INTERVIEW-PREP PACKET 2 OF 5 · PER-INTERVIEWER · BUSINESS-PROCESS LENS

Amazon NACF Supply Chain loop — Packet 2: Ryan Hoerger

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

Ryan Hoerger

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Wednesday, October 29, 2026 · 12:00 pm PDT · 55-minute video interview

SHADOW INTERVIEWER: SANDY HSIAO (A SECOND NOTE-TAKER — SAME PACE, SAME NUMBERS; EXPECT ONE QUESTION FROM THE SHADOW NEAR THE END)

Strategy in one line. The most quantitative interviewer on the panel gets the Elementum stories as primaries — ten, each with a real analytical spine (distribution, sample size, interval, counterfactual, sensitivity) and substituted worked numbers. Lead with S8 (measured lanes → p99 promise → ~$1.5M released, zero stockouts) or S6 (honest supply → shortage alert a lead-time early → $2M released under a two-signal rule); go deeper with S8a (the distribution analysis), S8b (the safety-stock formula redesign), S8c (the back-test) and S9 (1,000 lanes → 597); reach for the Walmart Marketplace delivery-experience stories (S19, S20, S21) and the new-market newsvendor design (E7) as the second and third examples. The two lane programs' dollars never meet. In his language: S6 positions safety stock where measured errors justify it; S8 positions the promise at p99 per lane. Every dollar keeps its basis word — record · user anchor · modeled · illustrative.

LP focus — my guess, not confirmed: Dive Deep · Are Right, A Lot · Learn & Be Curious — with a mentorship angle (his public About line names "a culture of mentorship"; expect one people question). Amazon does not disclose the assignment; the guesses are reasoned from the interviewer's title, the JD and his public profile in "Read me first" (public-profile inference, labeled as such) and should be held loosely — Insist on the Highest Standards remains a live alternate via the 99% bar in S8.

Shadow etiquette: address both; the shadow is usually a new interviewer being calibrated — same numbers, same pace; if the shadow asks the last question, answer it with the same discipline (phase asked, then stop).

Canon: amazon/amazon-l6-scm-master-packet.html (§1.2 roster, §4.1 ledger). Every number below is that packet's value; every dollar carries its basis label (record · user anchor · modeled · priced).

Contents NEW — existing sections kept; red NEW pills mark what this upgrade added; teal ELEVATED marks an existing story that gained sections

FRONT MATTER

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PRIMARY STORIES — EACH: LPS · 20-SECOND VERSION · PITCH · S/T · ACTION · RESULTS · L6 · FIVE FACETS · PROBING + FOUR-TABLE MATH · BEHAVIORAL PACK · THREE ANGLES · TRAP DRILLS

S6	Elementum — one real-time supply/demand view: shortage a lead-time early; 70% → 85%; $2M released (existing, elevated) ELEVATED	9
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BACK MATTER

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How to use this packet in 90 seconds. Page A is the cram sheet. Page B routes any question to a story, an angle, a 15-second opener and one number. Every primary has four behavioral questions, three angle cards and three trap drills — the trap drills are the statistical follow-ups this interviewer is most likely to throw. Anything tagged VERIFY-WITH-RAJESH is a detail that is not in the record: confirm it or leave it out; the one-page checklist near the back collects them all.

A · 5-minute cram sheet — Ryan Hoerger, Placement Analytics NEW

WHO IS IN THE ROOM (PUBLIC-PROFILE INFERENCE — FRAMING ONLY, NEVER QUOTED BACK)

- Manager, Transfer/Placement Analytics; Chicago Booth MBA (Siebel Scholar); analytics manager at Nike; Cornerstone Research before that — expert-grade statistical work. Shadow: Sandy Hsiao (same numbers, same pace).
- **Expect:** definitions and denominators ("how was it measured?"), n and intervals, the counterfactual ("how do you know it was you and not seasonality?"), and "what is your softest input?"
- **Style:** claim → number with its basis word → stop. Memo-like, not a wind-up.

THREE SENTENCES TO OPEN WITH

> "Every dollar I give you carries its basis — record, modeled or illustrative — in the same breath, and I will tell you the n behind every percentage."

> "The planning parameter was wrong before the policy was: the brochure lead time on 200 lanes, and what counted as supply — I measured the first lane by lane and defined the second before any screen existed."

> "I owned the definitions, the policy and the release rule; planners owned the forecast and the buffers, the data team built the distributions, Finance signed each release."

THE TEN STORIES IN THE ORDER I REACH FOR THEM — THE SINGLE NUMBER EACH ONE LANDS

#	STORY	REACH FOR IT WHEN HE ASKS…	ONE NUMBER (BASIS)
1	**S8**	Used data to change an inventory policy · went deeper than the request	580K units / ~$1.5M released, zero stockouts (record, one-time)
2	**S6**	Improved forecast accuracy · needed Finance and Planning buy-in	70% → 85%; $2M released (record, one-time)
3	**S8a**	Dug into the data · data contradicted belief	plan 18 d vs actual mean 21 d, p99 34 d (illustrative lane)
4	**S8c**	How did you validate? · how do you know it was not seasonality?	held-out quarter 99.1%, Wilson [98.7, 99.4] (illustrative inputs)
5	**S8b**	How do you set safety stock? · simplified · backbone	lane A 12,937 → 8,102 units; lane B +1,383 (illustrative)
6	**S9**	Prioritized 1,000 items · held a quality bar · pushed back upward	1,000 asked → 597 = 80% of spend; completeness 60% → 95% (record)
7	**S19**	A metric hid the problem · simplified a decision	fast-promise share 7.6% → 39.6% at the held 95% floor (engagement model)
8	**S20**	Took ownership of an un-owned number · built something to act faster	Promise Yield 41.5% = 43.6% × 95.2%; ~30 d → < 24 h (design)
9	**S21**	Why a program stalled · precise about what was mine	≈$3.7M/yr per +10 pts of enrolled GMV (modeled)
10	**E7**	Decision with little data · learned a new method	critical ratio 0.99 → z 2.33 → first buy 9,495 on a 6,000 mean (illustrative; design)

FIVE NUMBERS NEVER TO GET WRONG

- **580K units / ~$1.5M** — S8, record, one-time, zero stockouts; $2.59/unit is derived. Never merged with the $2M into a range; never per year.
- **$2M** — S6, record, one-time; a different program — never added to the $1.5M.
- **597 lanes = 80% of spend** out of 1,000 asked; completeness 60% → 95% — S9, record. Freight −10% ≈ $2M is six months, not annualized.
- **99% lead-time accuracy at the p99 promise on 200 lanes** — record; know which definition (coverage or ratio) before quoting a tolerance.
- **Six sites** — no regional qualifier; S19's ≈$13.3M/yr and S21's ≈$3.7M/yr are *modeled*, never record.

STATISTICAL REFLEXES (SAY THESE WITHOUT BEING ASKED)

- Give n and the interval: 22 late of 2,500 → 99.1%, Wilson [98.7, 99.4] — illustrative inputs.
- One held-out quarter is one draw; rolling origin is the upgrade. Zero stockouts bounds the rate: 0 in 1,950 lane-weeks → below 0.154% (rule of three).
- Name the softest input before he does.
- **NACF bridge:** inbound lead-time variability and the definition of supply sit under every WOC target and every placement promise — booklet INV-A (WOC), INV-B (safety stock), INV-C (placement).

B · Question → story router NEW

How to use. Find the LP or prompt on the left, open with the 15-second sentence, land the one number with its basis word, stop. "Q#" is the behavioral question inside that story's own pack (a separate 90-second STAR from a different moment of the story). Shaded cells are honest stretches — no formal people-development story in this packet, and I will not borrow another interviewer's.

IF ASKED ABOUT…	PRIMARY STORY · ANGLE (BEHAVIORAL Q#)	BACKUP	15-SECOND OPENER	ONE NUMBER (BASIS)
Customer Obsession	S19 · Q3 — pin accuracy at 95% on every cohort; let speed float	S8 · Q2 — the customer was the production line	"Fast-promise share can be juiced by bleeding accuracy, so I pinned accuracy at 95% on every cohort and let speed float."	**Fast-promise 7.6% → 39.6% at the held 95% floor (engagement model)**
Ownership	S20 · Q2 — promise accuracy was owned by nobody between Tuesdays	S8 · Q4 — a promise policy planning now owns	"The one number the customer feels — promise accuracy — was owned by nobody between Tuesdays, so I took it and gave every factor a named owner."	**7 outputs over 10 inputs; Promise Yield 41.5% = 43.6% × 95.2%**
Invent and Simplify	S8b · Q1 — a formula nobody could defend plus a week of pad → two readable terms	S19 · Q2 — one dial b = k·σ instead of per-lane learned buffers	"Safety stock was a formula nobody could defend plus a week of pad nobody could count; I replaced both with two terms a planner could read."	**Lane A 12,937 → 8,102 units (illustrative); k = 1.645**
Are Right, A Lot	S8c · Q1 — held a quarter out before releasing 580K units	S8a · Q2 — the scans said 21 days, the plan said 18	"A plan fitted to history always looks perfect on that history, so before I released anything I held a quarter out."	**Held-out 99.1% vs 96.5% for a bell-curve plan (illustrative inputs)**
Learn and Be Curious	E7 · Q1 — the newsvendor model from MIT coursework on a launch problem, labeled toolkit	S8c · Q4 — shadow-mode the drift flags before go-live	"I took the newsvendor model from my MIT MicroMasters coursework and applied it to a launch problem — and I label it as the toolkit."	**CR 0.99 → z 2.33 → first buy 9,495 on a 6,000 mean (illustrative)**
Hire and Develop the Best	No packet primary — honest stretch; closest: the S6 hand-over (planners run the alert and the measurement without me)	S8 · Q4 — planning re-applies the release rule on its own cadence	"I have no formal mentoring story here; the nearest true one is that planners now run the alert, and domain owners the gates, without me. VERIFY-WITH-RAJESH"	**Adoption 8×; planner effort −50% (record)**
Insist on the Highest Standards	S9 · Q2 — the ≥95% valid-ASN gate failed twice and was held twice	S8c · Q2 — 99% per lane, no pooling	"A cohort failed our ≥95% valid-ASN gate twice on a carrier timestamp format, and I held the gate both times."	**Completeness 60% → 95% behind a ≥95% gate (record)**
Think Big	E7 · Q2 — a launch chain designed as the template for market N+1	S6 — eight data domains, one view across six sites	"I designed the launch chain as a template for market N+1, not a one-off for market N."	**One chain · three books of record · five invariants (design, not run)**
Bias for Action	S20 · Q4 — an Andon that acts in 24 hours instead of a 30-day scorecard	S6 · Q3 — alert live in three months	"A 30-day rolling scorecard lets a degrading seller or carrier burn customers for weeks; I designed a control that acts inside 24 hours."	**~30 days → < 24 hours (design); false-trip bar < 5%**
Frugality	S8b · Q3 — less stock and better service	S8 — the release itself, with the watch armed first	"Less stock and better service — on one lane, 37% fewer units and seven late shipments in a hundred became one."	**580K units / ~$1.5M net, zero stockouts (record, one-time)**
Earn Trust	S6 · Q2 — the two-signal rule signed before the first release	S8c · Q3 / S21 · Q3 — say what the test cannot show; say what was mine	"Releasing $2M of safety stock is a risk Finance and Planning both carry, so the rule was theirs before it was mine."	**$2M released (record, one-time); both-errors rule**
Dive Deep	S8a · Q1 — 400 scans on the busiest lane: 21 days actual against 18 planned	S20 · Q1 — 97% national, 82% in the Southeast: the factor math names the cell	"Buffers kept growing and nobody could say why, so I pulled the shipment scans on our busiest lane — about 400 shipments, planned date against actual receipt."	**Mean 21 d vs plan 18 d; 240 of 400 (60%) slower than the brochure (illustrative lane)**
Have Backbone; Disagree and Commit	S9 · Q1 — told the Global Head that 1,000 lanes in two months means confident wrong data	S8b · Q2 — a rule that can add stock, not just release it	"The Global Head of Supply Chain wanted 1,000 lanes of inbound visibility in two months; I told him the schedule would deliver confident wrong data and showed him the spend curve instead."	**597 lanes = 80% of spend; ~400 lanes (~20%) deferred (record)**
Deliver Results	S6 · Q3 — a shortage visible a lead-time early, in three months	S9 — completeness 60% → 95%; freight −10% ≈ $2M in six months (record)	"Three months from kickoff, a planner could see a shortage a lead time before the line stopped."	**70% → 85% forecast accuracy; $2M one-time (record)**
Strive to be Earth's Best Employer	No packet primary — honest stretch; closest: S6 planner effort −50% (time redeployed, not headcount)	S8 — the drift flag and refresh cadence handed to planning	"I will not dress up a story I do not have; what is true is that the work moved off the planners' week into root-causing. VERIFY-WITH-RAJESH"	**Planner effort −50% (record)**
Success and Scale Bring Broad Responsibility	S9 · Q4 — carriers without EDI would have been priced out; the standard became the data, not the transport	S20 — an Andon with a < 5% false-trip bar so a seller is not penalized on noise	"Carriers without EDI would have been priced out of the standard, so I made the standard the data, not the transport."	**Adoption 8× (record); XML/portal floor**

COMMON PROMPTS THAT NAME NO LEADERSHIP PRINCIPLE — AND THE STATISTICAL PROMPTS HE IS MOST LIKELY TO ASK

COMMON PROMPT	PRIMARY · ANGLE	BACKUP	ONE NUMBER
Tell me about a time you failed	S4 (bench) — the in-transit promise rule I got wrong, fixed through the signed policy	S6 angle (iii) — MPN worklist gated longer than planned · S9 angle (iii) — the gate failed twice	**Lane σ ≈ 0.3 d; ~0.49-day window ≈ 95% (record promise math)**
…disagreed with your manager	S9 · Q1 — the Global Head, two levels above my sponsor (a client leader)	S8b · Q2 — release-only rule vs add-stock clause	**597 lanes = 80% of spend**
…influenced without authority	S8 — Finance signed each lane; none reported to me	S21 — contract design across seller, marketplace, carrier	**580K units / ~$1.5M (record)**
…reduced cost	S9 — freight −10% ≈ $2M in six months (record, not annualized)	S8b — inventory capital, 580K units / ~$1.5M	**$2M in six months; ≈$0.4M/yr doc labor modeled**
…dealt with ambiguity	S8 — a "variability report" request reframed as a parameter problem	E7 — a launch with no local history	**200 lanes scoped by site, lane and mode**
…simplified a process	S19 — one governed dial instead of per-lane learned buffers	S8b — two terms; S20 — 7 outputs over 10 inputs	**k = 1.645; 7.6% → 39.6%**
…decided with incomplete data	E7 — newsvendor on no history; the critical ratio sets the quantile	S8a — thin lanes print their interval	**CR 0.99; z 2.33; trigger ±20% over two weeks**
…missed a deadline	S6 — MPN worklist gated longer; re-sequenced date with the coverage number	S9 — gate held twice; schedule moved, not the bar	**Alert live in 3 months (record)**
…mentored someone	Honest stretch — S6 owners take over gates and root-cause reviews VERIFY-WITH-RAJESH	S8 — planning owns the refresh cadence	**Adoption 8× (record)**
…delivered under a tight timeline	S6 — three months to a live shortage alert	S9 — 1,000 in two months became 597 in month one	**Alert in 3 months; 597 lanes live month one**
…changed your mind with data	S8b angle (iii) — the textbook form was my starting point; one skewed lane showed it understates the tail VERIFY-WITH-RAJESH	S21 angle (iii) — the funnel said awareness; interviews showed an unpriced trade VERIFY-WITH-RAJESH	**normal p99 30.3 d vs ranked 34 d; release 8,009 vs 4,835 (illustrative lane)**
How would you set safety stock / service level?	S8b — two terms; critical ratio; lane A and lane B	S19 — one dial b = k·σ at a 95% floor	**SS 12,937 → 8,102 (illustrative)**
How do you validate / back-test a policy?	S8c — held-out quarter, paired replay, interval, rule of three	S21 — shadow gate before day one	**99.1% [98.7, 99.4]; 0 in 1,950 → < 0.154%**
MAPE vs bias · alert thresholds · false alarms	S6 — alert at lane p90; precision/recall; cost of a false alarm	S20 — Andon false-trip bar < 5%	**p90: precision 71.2%, recall 98.6% (illustrative lane)**
Lead-time variability — how do you measure it?	S8a — p50 / p90 / p99, empirical not bell-curve	S8 — per-lane distribution, drift flag	**plan 18 d vs mean 21 d, p99 34 d**
New market / placement with no history	E7 — newsvendor + risk pooling + re-forecast trigger; NACF INV-C	S19 — promise at the tail you can defend	**Q = 9,495 on a 6,000 mean (illustrative)**
Prioritize 1,000 items	S9 — spend Pareto, four-factor score, 597 first	S8a — which lanes are thin	**597 = 80% of spend**
Metrics architecture / WBR · promise policy · incentives	S20 · S19 · S21 respectively	S6 — metrics tree	**Promise Yield 41.5%; 7.6% → 39.6%; ≈$3.7M/yr per +10 pts (modeled)**
Your statistical background?	MIT SCM MicroMasters toolkit, labeled; S8a / S8c are the interval work	Never "I build models for a living"	**Toolkit, not job title**

Read me first

The interviewer's lens (inference from title, JD and public profile — the profile parts labeled)

Inference from the title. "Placement Analytics" is the analytical arm of inventory placement — deciding where inventory should sit in the network so that the promise holds at the lowest landed cost. A manager in that function lives on forecast error, lead-time variability, safety-stock policy, service levels and the data-quality underneath them. He will be the most quantitative interviewer on the panel and the most likely to ask "how was that measured?", "what was the denominator?", and "what is the softest input in your model?" He is also the one most likely to recognise a textbook formula being passed off as a program artifact — so the honesty labels matter more here than anywhere.

What that means for the answers. Ten Elementum primaries (the router and cram sheet say which to reach for). The first two are the Plan-process redesigns with the math visible: the shortage-alert / real-time supply-demand view (S6 — forecast accuracy 70 → 85%, $2M safety stock released under a two-signal rule) and the measured-lane lead-time program (S8 — promise at p99 on 200 lanes, 580K units / ~$1.5M released with zero stockouts). The other eight are the same facts cut for a statistician — the lead-time distribution (S8a), the safety-stock formula (S8b), the back-test (S8c), the 1,000 → 597 lane prioritization (S9), the Walmart Marketplace delivery promise, WBR and managed-shipping design (S19, S20, S21) and the new-market newsvendor design (E7). The lane programs end on *capital released on evidence with service held* — asset utilization and reliability in the JD's words. Keep the two programs' dollars apart in every sentence. A shadow (Sandy Hsiao) is on the call: same pace, same numbers, expect one question from the shadow near the end.

What his public profile adds (public-profile inference, labeled)

The profile (public-profile inference — directional, not verified biography). Manager, Transfer/Placement Analytics at Amazon (Jul 2025–), New York; before that Senior Business Intelligence Engineer at Amazon (Dec 2024–Jul 2025); analytics manager at Nike (2021–24); analyst at Cornerstone Research (2016–20) — an economics-consulting shop where the work product is expert-testimony-grade statistical and financial analysis; Chicago Booth MBA (Siebel Scholar); Duke. His own About line: "rigorous statistical and financial analysis… a blend of data analysis and writing skills… a culture of mentorship." **Quoted for framing only — never referenced in the room; none of it enters my stories.**

What that means for the answers. Expect *definition-level probing*: MAPE versus bias and which one steered; the grain (SKU-family? per lane?); the window and refresh; what "99% lead-time accuracy" tolerates in days. Three such definition Q&A are drilled into the S6/S8 probing below. Second, *answer style: crisp and memo-like* — claim, evidence, stop; he writes for a living, and a long wind-up reads as evasion. Third, speak *placement*: S6 is positioning safety stock where the measured errors justify it; S8 is positioning the promise at p99 per lane — his daily work is deciding where inventory sits versus where it is promised. Fourth, the mentorship line in his About suggests one people probe — answer it with planners grown into owners (S6's measurement handover; S8's drift flag owned by planning), not with org charts.

Three guessed LP focus areas, with rationale

> **Labeled guess.** Amazon assigns 2–3 Leadership Principles per interviewer and does not disclose them. These three are inferred from the title, the JD and the public profile above (public-profile inference); if the first question lands on a different LP — Insist on the Highest Standards via the 99% bar, or Frugality via the capital releases — the twist maps under each story carry the pivot.

GUESSED LP	WHY THIS INTERVIEWER WOULD CARRY IT	STORY TO OPEN · BACKUP
Dive Deep	An analytics manager tests whether the candidate goes to the data grain — measured lane distributions instead of brochure values; the definition of "supply" before any screen. S8 is the Dive Deep primary in the master map.	S8 (open) · S6 (the supply definition, the MAPE/bias measurement)
Are Right, A Lot	Placement decisions are judgment under uncertainty; he will probe how the candidate was right against the room's belief and how the decision was made safe (release rule, watch window, fat-tail exception).	S8 (measured vs brochure; p99 chosen by planning on the asymmetry) · backup: S6 (refusing unfirmed MRP orders as supply)
Learn & Be Curious (with a mentorship angle)	A Cornerstone-trained analyst with a Booth MBA who writes about rigor and mentorship (public-profile inference) will test whether the candidate learns at the definition level and grows people at the mechanism level: the MIT toolkit acquired to design what I had been implementing — labeled toolkit, never day job — and planners grown into owners of the measurement, the drift flag and the refresh cadence. Insist on the Highest Standards stays live as the 99% bar before any release and the two-signal rule before any buffer came out.	S8 (99% bar; drift flag owned by planning) · S6 (measurement handed to planners)

Answer strategy — business first, systems as enablers

- **Open with the process problem and its cost, not the system.** First sentence names the broken process (Plan, Source, Make, Deliver or Return), who it hurt (customer, Finance, the floor) and the baseline number. The platform appears only as the enabler of the redesign, in one clause.
- **Use the JD's four outcome words as the results frame:** reliability (did the promise hold?), responsiveness (how much earlier did we see it?), cost efficiency ($ with its basis label), asset utilization (capital, labor-hours, capacity released).
- **Action = operating-model redesign in gated phases** — Discovery → Plan & Design → Build & Test → Cutover → Hypercare, each spoken as entry bar · what I delivered · exit bar. Say the gate that fired, not the meeting that happened.
- **Every dollar carries its basis word in the same breath** — record · user anchor · modeled · priced. A record dollar is said flat; a modeled dollar is said with its inputs. Never promote a modeled number to record.
- **Program-lead voice.** I owned definitions, rules, gates and the signature chain; engineers, SIs and vendor teams configured and built to the spec I signed. The only platform personally configured in canon is Manhattan DOM.
- **The pitch ends on the primary metric.** Then stop. Silence after the pitch is the interviewer filling their STAR worksheet; their probes are the second half of the answer.
- **For Placement Analytics specifically:** lead with the planning parameter that was wrong (brochure lead time; unfirmed orders counted as supply), the decision it corrupted (buffer size, shortage response), and the rule that released capital safely. Show one formula only when asked, labeled as the MIT toolkit illustration — the data team built the distributions; planners sized the buffers; Finance signed.
- **Answer style for this room: crisp, memo-like.** Claim, evidence, stop. His background (Cornerstone rigor; a writing-heavy About line — public-profile inference) means a long wind-up reads as evasion; the one-breath basis label ("record", or "modeled, on illustrative inputs") does more credibility work here than any extra sentence.
- **Speak placement.** Frame S6 as positioning safety stock — buffer placed only where measured forecast and lead-time errors justify it — and S8 as positioning the promise at p99 per lane, recomputed on a drift flag. His

team decides where inventory sits versus where it is promised (public-profile inference); that is the exact shape of both stories.

- **Keep S6 and S8 apart:** $2M (S6, forecast-driven, two-signal rule) and ~$1.5M / 580K units (S8, lead-time-driven, 99% bar) are different programs — never add them, never swap their mechanisms. "Six sites" — never "six NA sites"; never paste S6's "3 months" onto the six-site ingest (that is S7, not in this packet).

Loop-consistency rule. One question, one story; never the same story to two interviewers — interviewers compare notes. Held for other packets (do not spend here): S11/S12 (Tripathy) · S2/S18 (Goforth) · S1/S3 (Lan) · S16/S15/S10/S14 (Josyula). S7 (CM BOM) is an unallocated master-packet backup — pointer only, and no BOM-explosion content. S9, S19, S20, S21 and E7 are used here as primaries: if the same story is also drafted for another interviewer, tell it in one room only. If pushed for "another example" on an LP, use the named backup in the question bank or the backup bench at the end of this packet, never a story allocated to another interviewer.

How to read the "L6 level evidence" block under each story. Amazon's L6 functional bar (user-supplied rubric): *Ambiguity* — strategy or network undefined; I select the stakeholders, set the goals, deliver independently. *Scope & influence* — team-level scope; influences one to two levels up. *Execution* — tactical and strategic; simplifies or extends; applies best practices; force-multiplies; mitigates long-term risk; makes the short-versus-long-term trade-off explicit. *Impact* — long-term team goals. *Process improvement* — operational-excellence best practices institutionalized. *Complex* — visible risks, roadblocks and constraints mitigated through expertise, judgment and influence to consensus. Overview themes the block is tuned to: designing and modeling the supply chain rather than working in it; developing new processes for others to own; root-cause mechanisms that resolve trending issues at scale; recommendations to leadership that outline competing metrics. The block sits right after Results so the evidence is on the page when the probing starts.

The stories — business-first, canon numbers

Ten Elementum primaries from the same six-site manufacturer, the Walmart Marketplace client work and a new-market design engagement. Each primary: LPs carried · 20-second version · 60-second pitch → Situation/Task → Action in gated phases → Results → L6 evidence → the five facets → Probing with the four-table math → behavioral pack (four separate 90-second answers) → three angles → three trap drills. S6 and S8 are the existing stories, elevated; the other eight are new to this packet (red NEW pills mark every new section).

S6 — Elementum — one real-time supply/demand view across eight data domains: a shortage visible a lead-time early, forecast accuracy 70% → 85%, $2M of safety stock released under a two-signal rule ELEVATED

LPs this story can serve here: Dive Deep (the definition of supply; MAPE/bias at the decision grain) · Earn Trust (the two-signal release rule signed with Planning and Finance) · Deliver Results (third — alert live in 3 months) · Think Big (one view over eight domains) · Are Right, A Lot (backup — refusing unfirmed orders as supply)
Process family: Plan — S&OP / shortage management / safety-stock policy; data-integrity program behind the plan

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Dive Deep** — I profiled every domain and defined supply honestly before any screen — released planned orders only, because unfirmed MRP orders breathe every run.
- **Earn Trust** — The $2M left the buffer only under a both-errors-improved rule Planning and Finance signed — the exception was written in before anyone asked.
- **Deliver Results** — Forecast accuracy 70% → 85% on scoped class-A SKUs; the shortage alert live in three months; planner effort −50%; adoption 8×.
- **Ownership** — Nobody owned the seam between eight data domains and the planners — I took the definitions, the data-integrity program and the delivery.

Honesty label. I owned the definitions, the data-integrity program and the delivery; the client's SAP teams and our integration engineers built the feeds; planners owned the forecast — I owned its measurement and the release rule. I have not owned a demand forecast end-to-end and won't claim it. The safety-stock formula is how I explain it now with the MIT toolkit — an illustration, not a program artifact.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. A six-site manufacturer — mixed make-to-order and make-to-stock — learned about a component shortage when the line stopped. S&OP ran off private spreadsheets, shipment data was 60% complete, and supplier, material, BOM, inventory, production, transportation, sales-order and planning data lived in SAP and site systems but never met. Leadership's question — which customer orders are at risk this quarter? — had no data path, so the plan responded with expedites instead of re-plans and buffers sized on noise. **Task.** Nobody owned the seam; I took the definitions, the data-integrity program and the delivery. **Action.** I defined supply honestly before any screen — on-hand plus production plan plus released planned orders only, at SKU × location × date — and refused to count unfirmed MRP orders because they change every run. Domains went live in waves sequenced by planner trust, each behind a published data-quality gate, with daily reconciliation to the SAP source and false alarms worked as defects. The shortage alert fires when days-of-cover falls to the lane's measured p90 lead time — the last moment a re-plan is cheaper than a flight. Safety stock came out only under a rule Planning and Finance signed: release where forecast error and lead-time error had both improved. **Result.** Primary metric: forecast accuracy on the scoped class-A SKUs 70% → 85%. $2M of safety stock released under the two-signal rule — record, one-time working capital — with ≈$0.8M a year of modeled recurring value around it; the shortage alert live in three months; planner effort −50%. That $2M is a different program from the lane program's $1.5M; I never add them.

20-SECOND VERSION NEW

Eight data domains, one honest definition of supply, a shortage alert that fires when days-of-cover falls to the lane’s p90 lead time — a lead-time early — and $2M of safety stock released only where forecast error and lead-time error had both improved.

NACF BRIDGE — ONE SENTENCE

This is the NACF WOC loop: define supply honestly, alert at the last moment a re-plan is cheaper than an expedite, and cut weeks of cover only where both error signals have improved — INV-A and INV-B in the inbound booklet.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — A strategic manufacturing customer with six sites: planners learned about a component shortage when the line stopped; S&OP ran off private spreadsheets; shipment data was 60% complete; eight data domains lived in SAP and site systems but never met. Leadership's question — "which customer orders are at risk this quarter?" — had no data path.

The Plan process underneath was running on an unsafe definition of supply (unfirmed MRP orders counted because they looked better), an unmeasured lead-time error, and a shortage signal that arrived at line-down. The cost: expedites instead of re-plans, buffers sized on noise, and planners spending ~40% of their week reconciling rather than root-causing.

T — Nobody owned the seam — no spec, no agreed grain, no definition of "supply," no data-quality bar. I chose all four, as owner of the definitions, the data-integrity program and the delivery. Goal: a real-time supply/demand view with honest definitions and a shortage visible a lead-time early — and planner trust, which was the real product.

Baselines on the table: forecast accuracy 70% on scoped class-A SKUs · shortage detected at line-down (target: a lead-time early) · shipment-data completeness 60% · planner effort on reconciliation and firefighting (≈40% of the week, illustrative share; the −50% is record).

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (4–6 wks)** — profiled every domain (completeness, latency, joinability) to my own gap templates; wrote the definitions memo — grain SKU × location × date; supply = on-hand + production plan + released planned orders only — and the metrics tree (OTIF, inventory $, MAPE/bias, lane lead-time accuracy, completeness, latency, adoption). Refused Planning's request to count unfirmed MRP orders as supply. *Exit bar:* signed gap findings per domain; scope frozen in writing; planners agree the supply definition before any screen.

- **Plan & Design (parallel)** — master-data foundation plan; MPN mapping worklist with its coverage gate; domain-wave sequence by trust; DQ scorecard spec; the 10-step onboarding playbook. *Exit bar:* master-data gates green; unmapped-MPN list under threshold with a named owner; wave order signed by domain owners — who could fail their own wave.
- **Build & Test (8–12 wks)** — domain waves behind published DQ gates — inventory and production first (numbers planners already believed), then purchasing/ASN, then transportation; shortage-alert rule specified (days-of-cover vs lane lead-time p90); daily reconciliation to MD04/COOIS/VA05/MB52; weekly root-cause loop with false alarms worked as defects. *Exit bar:* per-domain completeness ≥ gate; reconciliation passing daily; false alarms trending down; pilot planners using the alert.
- **Cutover & Go-Live (4 wks)** — views live for pilot planners with the at-risk list replacing the Monday spreadsheet; nobody mandated off the spreadsheet — rollback stayed free. *Exit bar:* planners answer "where is my product" from the platform in the live S&OP meeting.
- **Hypercare / Close (4 wks)** — exception workbench staffed; weekly root-cause reviews handed to domain owners; the safety-stock release rule signed with Planning and Finance — release only where forecast error AND lead-time error both improved. *Exit bar:* blockers/day under threshold two consecutive weeks; the release survives audit.

Influence and ownership. Mine: the definitions memo, the metrics tree, the DQ scorecard spec, the wave sequence, the release rule and the onboarding playbook. The client's SAP teams and our integration engineers built the feeds; planners owned the forecast — I owned its measurement (MAPE/bias per SKU family) and the release rule; domain owners held sign-off over their own gates. The MPN worklist gated longer than planned — escalated to the sponsor as a re-sequenced date with the coverage number attached, not a slip.

> **Designed the process; others own it now.** I designed the definitions, the DQ gates, the release rule and the onboarding playbook; domain owners now run the gates and the weekly root-cause reviews, and planners run the alert without me. **Trade-off put to leadership.** The recommendation to Planning and Finance outlined the competing metrics — working capital released vs service risk — and resolved it as a rule: release only where forecast error and lead-time error both improved.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — $2M safety stock released (record, one-time working capital, under the both-errors-improved rule signed with Planning and Finance) + ≈$0.8M/yr *modeled* recurring (carry ≈$0.2M at 10% · expedites pre-empted ≈$0.3M · planner capacity ≈$0.3M; range $0.5–1.2M) — never added to S8's $1.5M.
- **#** — eight data domains into one view across six sites; shortage-alert product live in 3 months; adoption 8× (record).
- **%** — forecast accuracy 70% → 85% on scoped class-A SKUs (record); planner effort −50% (record); shipment data 60% complete at the start.
- **Time** — the alert fires at days-of-cover ≤ lane lead-time p90 — a lead-time early instead of at line-down (record mechanism); views live in 3 months (the alert product; the six-site ingest is a separate 5-month program — never paste 3 months onto it).
- **Customer** — "which customer orders are at risk this quarter?" answered from the platform in the live S&OP meeting, not a spreadsheet (record exit bar).
- **Quality** — daily reconciliation to MD04/COOIS/VA05/MB52; false alarms classified weekly to root cause and burned down as defects (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Eight data domains, six sites and no shared definition of supply. I wrote the definitions memo (grain SKU × location × date; supply = on-hand + production plan + released planned orders only) and the metrics tree, and chose the domain owners who would sign their own gates.
Scope & influence	The client's SAP teams and our integration engineers built the feeds; planners owned the forecast; Planning and Finance signed the release rule; domain owners held sign-off over their gates — none reporting to me.
Execution trade-off + mechanism others now own	Working-capital release vs service risk: release only where forecast error AND lead-time error both improved. Mechanisms others own now: the DQ scorecard, the 10-step onboarding playbook and the weekly root-cause reviews handed to domain owners.
Impact (long-term)	$2M of safety stock released (record); forecast accuracy 70% → 85% on class-A; planner effort −50%; adoption 8×; the shortage alert fires a lead-time early instead of at line-down.
Process improvement (standard work institutionalized)	Domain waves behind published DQ gates; daily reconciliation to MD04/COOIS/VA05/MB52; false alarms classified weekly to root cause and burned down as defects.
Complexity & consensus	Nobody was mandated off the Monday spreadsheet — rollback stayed free until planners answered "where is my product" from the platform in the live S&OP meeting; the MPN worklist gated longer than planned and the gate held.
Root cause at scale	Unfirmed orders inflating supply was a definition defect, not a planner habit — fixed once in the supply definition and applied across all six sites.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: eight domains (supplier, material, BOM, inventory, production, transportation, sales-order, planning) across six sites; shipment data 60% complete at the start; MAPE and bias at the SKU-family grain on scoped class-A SKUs. Revealed: unfirmed MRP orders inflated supply; the lane's own lead time was unmeasured.
2 · Solution design	Options: count unfirmed orders or not · alert at brochure lead time or measured p90 · release buffer on accuracy alone or on both signals. Chosen: released planned orders only; alert at DoC ≤ lane p90; release only where both errors improved.
3 · Math & data logic	Math: DoC = on-hand ÷ daily demand = 1,200 ÷ 100 = 12 ≤ p90 14 → alert; SS = z·d·σ_L = 2.33 × 100 × 4 = 932 → 466 at σ_L 2; alert precision 71.2% / recall 98.6% at p90; both-errors rule false-release rate under noise 25% → 2.5% with a one-SE bar.
4 · Implementation	Implementation: definitions memo → data-quality gates per domain → waves sequenced by planner trust → daily reconciliation to SAP source → alert live in 3 months → release rule signed → false alarms classified weekly.
5 · Probe questions	Probe questions: MAPE vs bias · what "accuracy" means · why p90 for the alert and p99 for the promise · false alarms · whether "both improved" is signal or noise — see Trap Drills.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

> **Analytics — How was 70 → 85% measured, and on what?**
>
> Accuracy = 1 − MAPE on the scoped class-A SKUs, with bias published alongside it from month two, at the SKU-family grain the planners decide on. The planners owned the forecast; I owned the measurement and the input integrity — I have not owned a demand forecast end-to-end and won't claim it.

> **Analytics — Why exclude unfirmed MRP planned orders from supply? That understates it.**
>
> Deliberately. Unfirmed orders breathe every MRP run; counting them amplifies noise and inflates supply, which is exactly how the line-down surprises happened. Released planned orders only — accepted by planners before the first screen was built.

Analytics — The $2M — how do I know it is not a prettier average?

Because of the release rule: buffer came out only on SKU families where forecast error and lead-time error had both demonstrably improved, signed by Planning and Finance. That is why the release survived audit. It is one-time working capital; the recurring value around it is my model and labeled.

Analytics — Your expedite row is the softest. Defend it.

Agreed — about sixty expedites a year pre-empted at a $5K premium, both illustrative. The record is the mechanism: the alert fires at days-of-cover ≤ lane p90. To firm it I would pull the expedited-freight ledger for the six sites twelve months before and after the alert went live and count the premium directly.

Analytics — Why p90 for the alert threshold and p99 for the promise (S8)?

Different decisions. The alert is a re-plan trigger — p90 gives the planner a window before the tail bites. The promise is a commitment — it sits at the tail you can defend. Same measured distribution, two policies read off it.

Analytics — Define your two error metrics: MAPE versus bias. Which one steered?

MAPE is mean absolute percentage error — how wrong the forecast is on average, direction ignored; bias is the signed mean error — whether it runs systematically high or low. MAPE without bias hides a chronic over-forecast inside a decent average, and a chronic over-forecast is exactly how buffer quietly accumulates. So accuracy was reported as 1 − MAPE on the scoped class-A SKUs at the SKU-family grain, with bias published alongside from month two — and the release rule required forecast error and lead-time error both to improve, so neither metric could carry the decision alone.

Analytics — What are the precision and recall of an alert at the lane's p90?

On an illustrative lane — replenishment time normal with mean 10 and σ 3 days, so p90 = 13.8 — and days-of-cover states uniform across 0–120 days: of every 1,000 checks about 83 would run short without action; the p90 alert fires on about 115; 71.2% of alerts are real (precision) and 98.6% of shortages are caught (recall). Moving to p99 lifts recall to 99.9% and drops precision to 58.8%. Illustrative lane; the record is the rule, DoC ≤ lane p90.

Analytics — Two kinds of false alarm — how do you tell them apart?

Statistical and defect. A designed false positive is an alert on a lane that turned out fine — at p90 about 29% of alerts by construction; that is the price of the window. A defect false alarm is the data lying — a wrong on-hand, a missing receipt, an unfirmed order counted as supply. The record burns the second kind down weekly by root cause; I do not let the first kind masquerade as a data problem.

Analytics — Is "both errors improved" signal or noise?

Under no real improvement each metric is a coin flip, so both improve by chance 25% of the time — on 100 families, about 25 false releases. Requiring each to beat its own standard error cuts that to 2.5% (about 2.5 families); two standard errors, 0.05%. The record says "demonstrably improved"; how that was operationalized is a number I would confirm before I quote one. VERIFY-WITH-RAJESH

Analytics — Why does bias matter when MAPE is already 15%?

Because direction compounds. A +10% over-forecast on 100 units a day adds 70 units a week; in 13.3 weeks the excess alone equals the whole 932-unit buffer on the example lane. MAPE tells you how far off; bias tells you which way the stock is piling up.

Shadow / closing — What would you do differently?

Start the MPN-mapping worklist on day one — it is never "done" and it gated more than I planned. And define supply honestly before any screen, which held.

THE KEY MATH — THE NETTING WALK AT FINISHED-GOODS GRAIN, THE ALERT RULE AND THE TWO-SIGNAL RELEASE (ILLUSTRATIVE VALUES; RECORD PERCENTAGES)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Accuracy before / after	`1 − 0.30 = 70% → 1 − 0.15 = 85%`	record	70% → 85%
2	Honest supply	`supply(t) = on-hand + in-transit ≤ t + production ≤ t + RELEASED planned orders`	record rule	unfirmed MRP orders excluded
3	Projected balance (FG SKU)	`PAB(3) = 400 − 500 = −100; PAB(7) = 1,000 − 1,200 = −200; PAB(10) = 1,800 − 1,600 = +200`	illustrative FG example — timing shortage, not volume	fix window = day 9 − day 7 = 2 days
4	Alert rule	`DoC = 1,200 / 100 = 12 days ≤ lane p90 14 → alert; p90 = 9 → no alert`	record design; values illustrative	a lead-time early, no false alarm on the fast lane
5	Two-signal release	`SS = z·d·σ_L = 2.33 × 100 × 4 = 932 → σ_L 2: 466; release 466 units — only if BOTH errors improved`	toolkit illustration; the rule is record	release survives audit
6	Recurring value	`$2M × 10% = $0.2M · 60 × $5K = $0.3M · 2.4 FTE × $120K = $0.3M`	modeled (benchmarks labeled)	≈$0.8M/yr; $2M one-time record

MRP math at finished-goods grain only — no BOM explosion in this packet. The six-site ingest (S7) is a separate program; never paste this story's "3 months" onto it.

Where this lands at Amazon. Placement rests on two error signals — forecast error and lead-time error — measured at the decision grain. The transferable disciplines: define supply before the screen, alert at the last moment a re-plan is cheaper than an expedite, and release buffer only when both error signals improve. Reliability (shortage seen a lead-time early), cost efficiency (expedites pre-empted), asset utilization ($2M of working capital released on evidence).

FULL DERIVATION — FOUR-TABLE MATH, SOLUTION FLOW, SENSITIVITY, SPOKEN STATEMENT, IF-CHALLENGED NEW

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Customer orders at risk this quarter -> answered a lead-time early
+- Inventory $ (stock): $2M safety stock released (record)
|    +- forecast error 30% -> 15% (accuracy 70 -> 85%, record)
|    +- lead-time error (lane p90 measured, not brochure)
|    +- release rule: BOTH errors improved -> SS = z*sigma*sqrt(L) falls
+- Shortage response (flow): alert when DoC <= lane LT p90
|    +- fix window = LT p90 - DoC: days to re-plan instead of fly
+- Planner effort -50% (record) -> hours to root cause, not reconciliation
+- Data-integrity loop: daily recon -> defects burned down -> planner trust (+)
     -> adoption 8x -> more domains feed the peg -> errors fall further (+)
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
Forecast accuracy, class-A SKUs	70%	85%	—	record
Safety stock released (one-time)	—	$2M out	—	record
Carry on released $2M	tied up	free	$0.2M	modeled ($2M × 10% WACC; benchmark 8–12%)
Expedites avoided via the alert	~150 events/yr	~60 pre-empted	$0.3M	modeled (60 × $5K; premium benchmark $3–8K)
Planner effort on reconciliation	12 × 40%	12 × 20%	$0.3M	modeled (2.4 FTE × $120K; −50% is record)

DRIVER	BASELINE	AFTER	$/YR	BASIS
Total modeled recurring	—	—	**≈$0.8M**	around the $2M one-time record; never added to S8's $1.5M

If challenged. *"The $2M — measured how?"* — "It's the record: the release Finance signed, applied only on SKU families where forecast error and lead-time error had both demonstrably improved. That rule is why the release survived audit — capital came out on evidence, not on a better-looking average." *"150 shortage events?"* — "Illustrative volume — six sites at roughly 25 escalations a year. The record is the mechanism: the alert fires at days-of-cover ≤ lane p90, which is the last moment a re-plan is cheaper than a flight."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Forecast accuracy	`accuracy = 1 − MAPE = 1 − 0.30 = 70%`	scoped class-A SKUs (record)	70%
2	What the error math looks like	`MAPE = (30/100 + 10/120 + 20/80 + 10/100)/4 = (0.30 + 0.083 + 0.25 + 0.10)/4 = 18.3%`	4 illustrative weeks: A = 100, 120, 80, 100; F = 130, 110, 100, 90; |e| = 30, 10, 20, 10	18.3% on this sample; bias = (430 − 400)/400 = +7.5%
3	The supply lie	`supply counted unfirmed MRP planned orders — they change every run`	record problem	plans built on sand
4	Shortage discovery	`at line-down: ~150 expedite events/yr`	benchmark-modeled count	~150 fire-drills/yr
5	Planner burn	`12 planners × 40% = 4.8 FTE-equivalent on reconciliation`	record share	4.8 FTE reconciling, not root-causing

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Honest supply	`supply(t) = on-hand + in-transit ≤ t + production ≤ t + RELEASED planned orders only`	unfirmed MRP planned orders excluded (record rule)	walked below on ROUTER-100
2	FG netting walk — setup	`FG SKU ROUTER-100 at DC-East: on-hand 400; in-transit 600 lands day 5; production 800 day 9; demand 500 d3 / 700 d7 / 400 d10`	illustrative (v105 §5.12)	projected balance below
3	Day 3	`PAB(3) = 400 − 500 = −100`	only on-hand has arrived by day 3	dated shortage of 100 on day 3
4	Day 7	`PAB(7) = (400 + 600) − (500 + 700) = 1,000 − 1,200 = −200`	day-9 production is too late for day-7 demand	shortage of 200; fix window = 9 − 7 = 2 days
5	Day 10	`PAB(10) = (400 + 600 + 800) − (500 + 700 + 400) = 1,800 − 1,600 = +200`	all supply lands by day 10	+200 — the shortage is timing, not volume
6	Shortage alert rule	`DoC = 1,200 / 100 = 12 days; 12 ≤ p90 14 → alert now`	FG SKU: on-hand 1,200 at 100/day; lane p90 = 14 days (illustrative)	alert a lead-time early, not at line-down
7	Counter-case	`p90 = 9 days: 12 > 9 → no alert`	same SKU, faster lane (illustrative)	3-day slack — no false alarm

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
8	Two-signal release rule	`SS = z × d × σ_L = 2.33 × 100 × 4 = 932; with σ_L 4 → 2: 2.33 × 100 × 2 = 466; release 932 − 466 = 466 units`	release only if forecast error AND lead-time accuracy both improved; d = 100/day, z = 2.33 at 99% (illustrative, toolkit form)	466 units off this SKU
9	Accuracy after	`error 30% → 15% (halved); accuracy = 1 − 0.15 = 85%`	gated waves; false alarms worked as defects (record)	85%

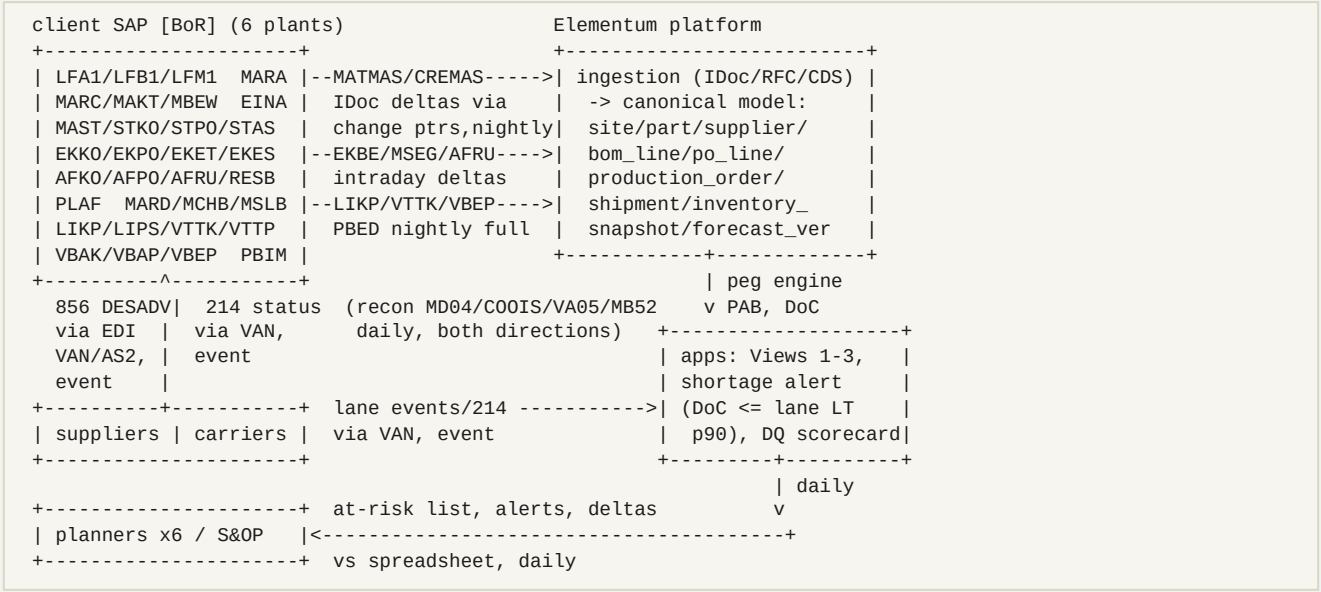
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Forecast accuracy, class-A SKUs	70%	85%	record
2	Shortage discovery	at line-down	a lead-time early (DoC ≤ p90 alert)	record design
3	Planner reconciliation share	12 × 40% of time	12 × 20% — halved	record (−50%)
4	Safety stock	sized on bad signals	$2M released one-time under the signed two-signal rule	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Carry on the release	`$2,000,000 × 10% = $200,000`	$2M release (record, one-time) × WACC (benchmark 8–12%)	≈$0.2M/yr
2	Expedites pre-empted	`150 × 40% = 60 caught early; 60 × $5,000 = $300,000`	events/yr modeled; premium benchmark $3–8K	≈$0.3M/yr
3	Planner capacity	`12 × 40% = 4.8 FTE; halved → 2.4 FTE; 2.4 × $120,000 = $288,000`	−50% is record; capacity redeployed, not headcount removed	≈$0.3M/yr
4	Total recurring	`0.2 + 0.3 + 0.3 = $0.8M`	modeled; never added to S8's $1.5M	≈$0.8M/yr
5	One-time	`$2M safety stock released, taken as stated`	record, under the signed two-signal rule	$2M

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Cost of capital (benchmark)	8% → $0.16M	10% → $0.20M	12% → $0.24M
Expedites avoided (illustrative)	45	60	75
Expedite premium (benchmark)	$3K → $0.14M	$5K → $0.30M	$8K → $0.60M
Loaded planner cost (benchmark)	$100K → $0.24M	$120K → $0.29M	$140K → $0.34M
Total modeled $/yr	≈$0.5M	≈$0.8M	≈$1.2M

ALERT-THRESHOLD DESIGN — PRECISION, RECALL AND THE COST OF A FALSE ALARM (EXTRA TABLE)

The record is the rule — alert when days-of-cover ≤ the lane's measured p90 lead time. Everything below is my after-the-fact design arithmetic on an illustrative lane, so the threshold can be defended in numbers. VERIFY-WITH-RAJESH how p90 was originally chosen.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The lane	`replenishment time L ~ Normal(μ = 10, σ = 3); p90 = 10 + 1.2816 × 3 = 13.84 ≈ 14 d`	illustrative lane (matches the p90 = 14 example)	θ = 13.84 d
2	Shortage if no action	`P(L > DoC) = 1 − Φ((DoC − 10)/3); DoC uniform on 0–120 d → ≈ 10/120 = 8.3% of checks`	states illustrative; ∫P(L > x)dx = E[L] = 10	83.3 per 1,000 checks
3	Alert rule (record)	`alert iff DoC ≤ θ: alerts per 1,000 = 1,000 × 13.84 ÷ 120 = 115`	record design; values illustrative	115 alerts / 1,000
4	Precision (p90)	`TP ÷ alerts = 82.15 ÷ 115.37 = 71.2%`	numerical integral of P(L > DoC) over DoC ≤ θ	71.2%
5	Recall (p90)	`TP ÷ shortages = 82.15 ÷ 83.34 = 98.6%`	same integral	98.6%
6	Same lane at p99	`θ = 16.98 d; alerts 141; precision 58.8%; recall 99.9%`	same method	more alerts, little recall gained
7	Cost of a false alarm vs a miss	`planner hour ≈ $120K ÷ 2,000 h = $60; a missed shortage ≈ $5,000 (master expedite premium)`	both illustrative (master)	$60 vs $5,000
8	Expected cost per 1,000 checks	`p75 $24,656; p90 115.372 × $60 + 1.1836 × $5,000 = $12,840; p95 $10,079; p99 $8,913`	alerts × c_fp + misses × c_miss	p90 $12,840 vs p99 $8,913
9	Cost-optimal threshold	`alert iff P(L > DoC) ≥ c_fp ÷ c_miss = 60 ÷ 5,000 = 1.2% → quantile 98.8% → θ = 16.8 d`	critical-ratio logic	θ* = 16.8 d if c_miss were $5,000
10	What p90 implies	`p90 ⇔ c_miss ÷ c_fp = 0.90 ÷ 0.10 = 9 → c_miss = 9 × $60 = $540; p99 ⇔ $5,940`	after-the-fact derivation — not how p90 was chosen	p90 prices a miss at $540
11	Minimum shipments to bound the p90	`n = 31: t(24)…t(31); n = 30 → t(24)…open`	exact order-statistic interval (95%)	n ≥ 31 per lane; else cohort p90

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The measured number is two million dollars of safety stock released, and that is record, signed with planning and Finance under the both-errors-improved rule. Around it I model roughly eight hundred thousand a year of recurring value: about two hundred thousand in carry cost on the capital that came out, about three hundred thousand in expedite premium the shortage alert pre-empts because it fires a lead-time early instead of at line-down, and about three hundred thousand of planner capacity from the fifty percent effort cut. The two million and the fifty percent are measured; the three recurring pieces are my model on benchmark rates, and I keep them separate from the one-and-a-half million in the lane program."

If challenged — the full answers.

The expedite figure is the softest

It rests on about sixty expedites a year pre-empted at a five thousand dollar premium, both illustrative; to firm it I would pull the expedited-freight ledger for the six sites for the twelve months before and after the alert went live and count the premium directly.

Planner capacity is not cash

Agreed, it is two point four FTE of time redeployed, not headcount removed; the fifty percent is measured, the forty percent share of the week is my assumption, and a two-week time study on the S&OP prep cycle would replace it.

Why not claim the whole two million every year

Because a release is a one-time working-capital event; the recurring value is the carry at eight to twelve percent, and I would rather say two hundred thousand a year honestly than two million a year wrongly.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S6

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you had to dig into the data to find the real problem.	Dive Deep	The supply definition — unfirmed orders inflating supply — found by profiling eight domains; MAPE and bias at the grain planners decide on.	Skip the SAP table names unless asked; systems are enablers.
Tell me about a time you released inventory / improved working capital.	Frugality / Deliver Results	$2M under the two-signal rule — record, one-time; the recurring ≈$0.8M labeled modeled.	Never add the $2M to S8's $1.5M; never claim $2M per year.
Tell me about a time you were right when the business disagreed.	Are Right, A Lot	Refusing unfirmed MRP orders as supply because it looked better; refusing a blanket safety-stock release — agreed as a rule instead.	Do not make Planning the villain; the rule did the refusing.
How do you get planners to trust a new signal?	Earn Trust	Waves sequenced by trust (inventory and production first); false alarms worked as defects; nobody mandated off the spreadsheet; domain owners could fail their own wave.	Do not claim adoption dollars; adoption 8× is the record.
How would you design a shortage / at-risk-order signal for the network?	Functional — Plan process design	Grain, honest supply, days-of-cover vs measured lane p90, DQ gate per feed, two-signal release — the §3.99 opening: book of record and metric first.	No BOM explosion; finished-goods grain only.
What's the softest number in your model?	Highest Standards / Earn Trust	The expedite row (60 × $5K); how I would firm it from the expedited-freight ledger.	Never present planner capacity as cash — it is time redeployed.

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 Tell me about a time you dug into the data to find the real problem. DIVE DEEP

Open (first two sentences): Leadership asked which customer orders were at risk, and the first thing I did was refuse to answer until we agreed what “supply” meant.

STAR (≈90 s): Situation: a six-site manufacturer found out about shortages when the line stopped; shipment data was 60% complete across eight domains. Task: nobody owned the seam, so I took definitions and data integrity. Action: profiled every domain for completeness, latency and joinability, and defined supply honestly — on-hand plus production plan plus released planned orders only; unfirmed MRP orders change every run and had been inflating supply. Result: the alert product live in three months; forecast accuracy 70 to 85% on the scoped class-A SKUs; planners accepted the definition before the first screen existed.

Land this number: 60% shipment-data completeness → daily reconciliation to SAP; 70% → 85% accuracy. · **Pivot to NACF:** NACF inbound “available” and “on-hand” need one signed definition before any placement model reads them. · **Do NOT say:** Do not say I owned the forecast; planners did — I owned the measurement and the input integrity.

Q2 Tell me about a time you needed buy-in from Finance and Planning on a risky change. EARN TRUST

Open (first two sentences): Releasing $2M of safety stock is a risk Finance and Planning both carry, so the rule was theirs before it was mine.

STAR (≈90 s): Situation: buffer had grown on noisy forecasts and unmeasured lead times. Task: take capital out without a stockout. Action: I proposed a two-signal rule — release only where forecast error and lead-time error had both demonstrably improved — and got Planning and Finance to sign it before the first release. Result: $2M released and it survived audit because the exception was in the rule; ≈$0.8M a year modeled around it; never added to the other program’s $1.5M.

Land this number: $2M released (record, one-time); both-errors rule. · **Pivot to NACF:** A WOC reduction at Amazon needs the same pre-signed exception: cut only where both signals improved. · **Do NOT say:** Do not add $2M and $1.5M, and do not call the $0.8M recurring a result — it is modeled.

Q3 Tell me about a time you delivered a result on a hard deadline. DELIVER RESULTS

Open (first two sentences): Three months from kickoff, a planner could see a shortage a lead time before the line stopped.

STAR (≈90 s): Situation: the plan responded with expedites, not re-plans. Task: ship a shortage alert fast without shipping wrong data. Action: sequenced the domains in waves by planner trust, each behind a published data-quality gate, with daily reconciliation to SAP and false alarms worked as defects. Result: alert live in three months, planner effort −50%, adoption 8×, forecast accuracy 70 to 85%. The six-site ingest is a separate five-month program — I never paste the three months onto it.

Land this number: Alert live in 3 months; planner effort −50%; adoption 8×. · **Pivot to NACF:** Same sequencing for NACF: trust-ordered waves, a quality gate per source. · **Do NOT say:** Do not stretch “3 months” to the full ingest.

Q4 Tell me about a time you owned something nobody owned. OWNERSHIP

Open (first two sentences): Eight domains, six sites, and the question “which customer orders are at risk?” had no data path and no owner.

STAR (≈90 s): Situation: S&OP ran off private spreadsheets and the seam between systems and planners belonged to nobody. Task: own it. Action: I wrote the definitions memo, the data-quality scorecard spec, the wave sequence and the release rule, and stayed through hypercare. The thing I would do differently: start the MPN-mapping worklist on day one — it is never done and it gated more than I planned. Result: the question was answered in the live S&OP meeting from the platform, not a spreadsheet.

Land this number: Eight domains into one view; alert in 3 months. · **Pivot to NACF:** Placement analytics lives on that seam too: forecast error and lead-time error from different systems. · **Do NOT say:** Do not claim the client’s SAP teams’ feeds were mine.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: "I would not build a screen until we agreed what supply was."

Headline metric: unfirmed MRP orders excluded; accuracy 70% → 85%

- MAPE with bias published from month two.
- Alert at DoC ≤ lane p90: precision 71% / recall 99% (illustrative lane).
- DoC 12 ≤ p90 14 → alert; p90 9 → no alert.
- False alarms: designed vs defect.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Earn Trust; Have Backbone

Open: "Planners wanted the unfirmed orders in supply because it looked better; I refused them."

Headline metric: released planned orders only; $2M under a signed rule

- Accepted by planners before the first screen.
- Release only if both errors improved.
- Finance and Planning signed.
- A coin-flip rule would false-release 25% of families — so "demonstrably."

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Ownership; Learn and Be Curious

Open: "The MPN-mapping worklist is never done — I should have started it on day one."

Headline metric: MPN mapping gated more than planned (record lesson)

- The lesson is in the record.
- Define supply honestly before any screen — that held.
- I did not own the forecast and say so.
- VERIFY-WITH-RAJESH the size of the delay MPN mapping caused.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 "Your alert fires at the lane's p90 — what are its precision and recall, and why not p99?"

The numbers: Illustrative lane N(10, 3): θ = 13.84; precision 71.2%, recall 98.6%; p99 precision 58.8%, recall 99.9%; p90 ⇔ miss costs 9× a false alarm.

Say exactly: "On an illustrative lane the p90 alert catches about 99 percent of shortages with about 71 percent precision; p99 buys about one more point of recall for twelve points of precision. A p90 threshold prices a miss at nine false alarms — at a sixty-dollar alert and a five-thousand-dollar miss the cost-optimal threshold would sit higher, which is why I would show planners the curve and let their own cost of an alert set the dial."

T2 "Both-errors-improved — what is the false-release rate if neither metric really improved?"

The numbers: Coin-flip: 25% of families; each beat by one SE: 2.52%; two SE: 0.052%; on 100 families: 25 / 2.5 / 0.05.

Say exactly: "If nothing improved, a bare both-up rule releases a quarter of the families by luck; requiring each metric to beat its own standard error takes it to about two and a half percent — which is why the rule said demonstrably improved, not merely improved."

T3 "Why does bias matter if your accuracy is 85%?"

The numbers: SS = 2.33 × 100 × 4 = 932; +10% bias × 100/day × 7 = 70/week; 932 ÷ 70 = 13.3 weeks.

Say exactly: "Accuracy is a magnitude; bias is a direction. A ten-percent over-forecast adds seventy units a week on a hundred-a-day SKU — in under fourteen weeks the excess alone equals a nine-hundred-unit buffer, which is how buffer quietly accumulates behind a decent MAPE."

S8 — Elementum — replacing brochure lead times with measured lane distributions: promise at p99 on 200 lanes, 99% lead-time accuracy, 580K units / ~$1.5M of safety stock released with zero stockouts ELEVATED

LPs this story can serve here: Dive Deep (primary — measured the distribution, not the average) · Are Right, A Lot (backup — measured vs brochure; p99 on the asymmetry) · Frugality (the release) · Insist on the Highest Standards (the 99% bar before any cut) · Earn Trust (Finance signed each lane; the exception written into the rule)
Process family: Plan / Source / Deliver seam — inbound lead-time policy, safety-stock sizing, promise policy; parameter governance with a drift flag

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Dive Deep** — I measured each lane's own distribution — mean, spread, tail — instead of accepting a carrier brochure or a network average.
- **Customer Obsession** — The customer was the production line: I put the cost of a line-down next to the carrying cost per lane and let planning choose p99 with the asymmetry in front of them.
- **Earn Trust** — Finance signed each lane; the release rule carried its own exception — fat tails kept their buffer — before anyone asked.
- **Ownership** — I designed the promise policy and release rule as a signed planning act that planning re-applies on its own refresh.

Honesty label. I owned the requirement (measured lanes, not brochures), the p99 promise policy and the release rule; the data team built the distributions; planners sized the buffers; Finance signed the release. Lane-cohort counts and week markers are not in the record. The safety-stock formula is the MIT inventory-theory toolkit as I explain it now — an illustration, not a program artifact. Depends on the six-site canonical model (S7 — pointer only).

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. The same six-site manufacturer planned inventory on carrier-brochure lead times — every buffer sized on a variability nobody had measured: too much on reliable lanes, too little on fat-tailed ones. The logistics manager asked for a "lead-time variability report." **Task.** I reframed it as a parameter and policy problem — nobody owned the seam between transportation reality and planning parameters, and I took it — scoped at 200 lanes by site, lane and mode. **Action.** I wrote the requirement to the data team: mean, spread and tail per lane from actual production cycle plus inbound transit; defined the promise at p99 cumulatively — the smallest day d with $P(L \le d) \ge 0.99$, not average-plus-padding — and showed each lane's asymmetry, line-down cost against carrying cost, and let planning choose the percentile; they chose p99. The release rule was written as a signed planning act with the exception inside it: fat-tail lanes keep their buffer until the source process moves; Finance agreed the rule in the same room, before any cut. I would not endorse a release until lead-time accuracy hit 99% on the 200-lane set; the stockout-and-expedite watch was armed first; every parameter got a drift flag and a refresh cadence so the promise never decays into a new brochure. **Result.** Primary metric: lead-time accuracy on 200 lanes — brochure values → 99% at the p99 promise. 580K units, about $1.5M of safety stock, released without a stockout — record, one-time working capital — with ≈$0.6M a year modeled around it. A different program from the $2M forecast release; never added.

20-SECOND VERSION NEW

Two hundred lanes planned on carrier brochures; I replaced them with measured per-lane distributions, set the promise at p99, gated any cut on 99% lead-time accuracy and released 580K units, about $1.5M, with zero stockouts.

NACF BRIDGE — ONE SENTENCE

Inbound lead-time variability is half of every WOC target and every placement promise at NACF — I would measure each vendor lane's own distribution and promise at the tail I can defend before releasing any weeks of cover.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The six-site manufacturer's logistics manager asked for a "lead-time variability report." The real problem: inventory planning ran on carrier-brochure lead times — every buffer sized on a variability nobody had measured; too much buffer on reliable lanes, too little on fat-tailed ones; the promise to the plan was folklore.

The risk cut both ways: keep planning on brochures and you hold too much on reliable lanes and stock out on fat-tailed ones; cut buffer without evidence and the stockout is yours. And an unmaintained promise decays into a new brochure — which is why every parameter needed a drift flag.

T — Nobody owned the seam between transportation reality and planning parameters; I took it. Goal: measured lanes replace catalog values; a promise at a defensible tail; buffer released only on evidence, with service held — scoped at 200 lanes by site, lane and mode.

Baselines on the table: lead-time parameters = carrier-brochure values, variability unmeasured · buffer capital sized on catalog values · no lead-time-accuracy measurement · target 99% lead-time accuracy on the 200-lane set before any release.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — reframed the report request as a parameter problem; scoped 200 lanes by site/lane/mode; wrote the success criteria — measured lanes replace catalog values; buffer released only on evidence; fat tails keep buffer; showed one right-skewed lane where the average hid the tail. *Exit bar:* planning and logistics agree the problem is the parameter, not the report.
- **Plan & Design** — requirement to the data team (mean, spread, tail per lane; production cycle + inbound transit vs actual); the p99 promise policy (cumulative, not μ + 2.33σ); the release rule as a signed planning act with the fat-tail exception inside it; the stockout-and-expedite watch; drift flag and refresh cadence. On p95 vs p99, showed the asymmetry per lane and left the choice to planning. *Exit bar:* policy signed by planning; Finance agrees the release rule in the room; data team accepts the spec.
- **Build & Test** — data team built the lane distributions and tracked lead-time accuracy per lane; manual reconciliation stayed alive. *Exit bar:* lead-time accuracy 99% on the 200-lane set — the bar I would not endorse a cut below.
- **Cutover & Go-Live** — planners executed the buffer release lane by lane under the signed rule; fat-tail lanes excluded until their source process moved; watch armed first. *Exit bar:* 580K units / ~$1.5M released; no stockout in the watch window.
- **Hypercare / Close** — read the watch; set drift flags per lane; handed the refresh cadence and the release rule to planning as an act they could re-apply. *Exit bar:* watch clean; the promise stays measured.

Influence and ownership. I owned the requirement, the promise policy and the release rule; the data team built the distributions and tracked accuracy; planners sized the buffers and executed; Finance signed each lane. When Finance pushed for a broader release, the rule they had signed did the refusing — fat-tail lanes keep buffer until the source process moves. The percentile fight never happened: I showed each lane's line-down cost against carrying cost and planning chose p99 — their choice surviving an audit is worth more than my choice winning a meeting.

Designed the process; others own it now. I designed the promise policy and the release rule as a signed planning act; planning now re-applies it on its own refresh cadence with a drift flag per lane. **Trade-off put to leadership.** The recommendation to Finance and Planning outlined the competing metrics — ~$1.5M of working capital out vs stockout

> exposure at the tail — and set the bar: p99, 99% lead-time accuracy, fat-tail lanes excluded until their source process moved.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — 580K units / ~$1.5M of safety stock released without a stockout (record, one-time working capital); ≈$0.6M/yr *modeled* recurring around it (carry ≈$0.3M at a benchmark 20% all-in rate + expedites avoided ≈$0.3M; range $0.3–1.4M) — a different program from S6's $2M; never added.
- **#** — 200 lanes measured by site, lane and mode; 580K units released (record); ≈$2.59/unit derived — low-value components.
- **%** — lead-time accuracy 99% on the 200-lane set at the p99 promise — the bar below which I would not endorse a cut (record).
- **Time** — release only after the 99% bar; watch window clean; refresh cadence and drift flag per lane handed to planning (record).
- **Customer** — service held — zero stockouts on released lanes through the watch window (record).
- **Quality** — release only under the signed rule; fat-tail lanes kept buffer until the source process moved; the promise stays measured, never a new brochure (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	A report request with no agreed parameter behind it. I reframed it as a parameter problem and wrote the success criteria myself: measured lanes replace catalog values; buffer released only on evidence; fat tails keep buffer.
Scope & influence	200 lanes by site, lane and mode; the data team built the distributions; planners sized and executed; Finance signed each lane — a planning act I designed and others executed.
Execution trade-off + mechanism others now own	Working capital out vs stockout risk: promise at p99 (cumulative), a 99% lead-time-accuracy bar below which I would not endorse a cut, fat-tail lanes excluded until their source process moved. Mechanism others own now: the release rule as a signed act, the refresh cadence and a drift flag per lane handed to planning.
Impact (long-term)	580K units / ~$1.5M of safety stock released without a stockout (record, one-time working capital); the promise stays measured, never a new brochure.
Process improvement (standard work institutionalized)	Lead-time accuracy tracked per lane; drift flags; the release rule written so planning can re-apply it without me.
Complexity & consensus	Finance pushed for a broader release — the rule they had signed did the refusing; the fat-tail exception was inside the rule, so it fired as policy, not as a judgment call.
Root cause at scale	Brochure lead times were the parameter error behind every excess buffer — fixing the parameter fixed all 200 lanes at once rather than one planner's buffer at a time.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: shipment durations per lane (ship-to-receipt) for 200 lanes by site, lane and mode; the brochure value beside each. Revealed: lanes with measured σ far below the brochure (release) and lanes above it (keep).
2 · Solution design	Options: brochure · average · p95 · p99 per lane · network-wide percentile. Chosen: promise at the lane's own p99, cut only behind a 99% accuracy gate, fat tails kept.
3 · Math & data logic	Math: SS = z·d·σ_L = 2.33 × 500 × 5 = 5,825 vs 2.33 × 500 × 1.2 = 1,398; release 4,427; p99 = min{d : P(L ≤ d) ≥ 0.99}; $1.5M ÷ 580K = $2.59; upper-bound σ at n = 50 → 1,675.
4 · Implementation	Implementation: reframe the request → scope 200 lanes → data team builds distributions → planning chooses p99 on the asymmetry → accuracy gate → lane-by-lane release with Finance signature → watch + drift flag.

5 · Probe questions	Probe questions: p99 vs p95 · skew and the normal shortcut · what “99% accuracy” means · whether the release is selection-biased · whether the $1.5M is measured — see Trap Drills.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

Analytics — Why p99 and not p95? That is a lot of buffer.

Per lane, the cost of a line-down on a fat-tailed lane dwarfed the carrying cost of the extra day. I showed the asymmetry lane by lane and let planning choose — they chose p99. On reliable lanes the measured σ was so far below the brochure that p99 still released buffer; on fat-tailed lanes the buffer stayed.

Analytics — How did you compute p99 on skewed lanes?

Cumulatively from the empirical distribution — the smallest d with P(L ≤ d) ≥ 0.99 — not μ + 2.33σ. On one right-skewed lane the normal shortcut understated the true p99 by about 3.4 days; that lane is why the definition is cumulative.

Analytics — Is the $1.5M measured?

The release is record: 580K units out, zero stockouts in the watch window, Finance signed each lane. The ≈$2.59/unit is the derived average. What is modeled is the recurring value around it — carry at 20% all-in (capital-only would be ≈$150K at 10%) and avoided expedites — each input labeled.

Analytics — The expedite number assumes what?

Ten thousand shipments a year on the 200 lanes and one late arrival in ten forcing an expedite before p99 — illustrative. I would pull the expedite ledger by lane for the year before and after and compare directly.

Analytics — What is lead-time accuracy, exactly?

1 − |actual − promised| / promised, tracked per lane against the p99 promise; the bar was 99% on the whole 200-lane set. Below it, no release — written into the act before anyone asked.

Analytics — At what grain did you measure lead time, and over what window?

At the lane grain, from actual ship-to-receipt durations — one empirical distribution per lane, never a network average. The record carries the per-lane distributions, the refresh cadence and the drift flag (p99 recomputed on refresh; the buffer decision re-opens if accuracy falls); it does not carry a quoted window length, and I will not invent one — what mattered design-wise was a window long enough to hold seasonality and recent enough that a dead lane's history could not vote, with the drift flag as the safety net either way.

Analytics — Why p99? What cost ratio does that imply?

A p99 promise is the critical ratio 0.99 = Cu ÷ (Cu + Co), i.e. a late day costs 99 times an extra day of cover. On the example lane an extra day of cover is 500 × $2.59 × 20% ÷ 365 = $0.71, so p99 is justified if a day of shortage costs more than about $70; p95 if more than $13. A line-down clears both bars by orders of magnitude — which is the asymmetry I showed planning lane by lane.

Analytics — Lanes released are lanes with a small measured σ. Isn’t that selection bias?

Yes — the winner’s curse. A lane picked because its measured σ looks small is partly a lucky draw. The standard error on σ is about σ ÷ √(2(n−1)): 10.1% at n = 50, 3.5% at n = 400. On the reliable lane (σ 1.2, brochure 5), sizing on the 95% upper bound at n = 50 gives σ = 1.44 and a buffer of 1,675 instead of 1,398 — 277 units more, a release of 4,150 instead of 4,427. That is the arithmetic; VERIFY-WITH-RAJESH what the program did on thin lanes.

Analytics — Define “99% lead-time accuracy” so I can test it.

The master’s definition is LTacc = 1 − |actual − promised| ÷ promised with a 99% bar on the 200-lane set. Read per shipment against a p99 promise it cannot be 99%: on the example lane the mean shipment is 21 days against a 34-day promise, |21 − 34| ÷ 34 = 38.2%, a score of 61.8%. The readings consistent with 99% are coverage — 99 of 100 shipments arrive within the promise — or the re-measured lane p99 within 1% of the promised p99, a tolerance of 0.34 day on a 34-day lane. On the ratio form 1% is 0.05 day on a 5-day lane and 0.3 day on a 30-day lane. I would say which reading before quoting a tolerance. VERIFY-WITH-RAJESH

Shadow / closing — How does the promise stay honest after you leave?

Drift flag per lane with a refresh cadence: p99 is recomputed on refresh, and the buffer decision re-opens if accuracy falls. The stockout-and-expedite watch became a standing report owned by planning.

THE KEY MATH — ONE RELIABLE LANE, ONE FAT-TAIL LANE, AND THE RECURRING VALUE (ILLUSTRATIVE UNITS; RECORD PERCENTAGES AND THE RECORD RELEASE PAIR)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Brochure-sized buffer	`SS = z·d·σ_L = 2.33 × 500 × 5 = 5,825 units`	illustrative; toolkit form	5,825 units of folklore
2	p99 defined cumulatively	`p99 = min{d : P(L ≤ d) ≥ 0.99}`	record definition	normal shortcut understates a skewed lane by 3.4 days
3	Reliable lane re-sized	`2.33 × 500 × 1.2 = 1,398; release = 5,825 − 1,398 = 4,427`	measured σ = 1.2 vs brochure 5 (illustrative)	4,427 units off this lane
4	Fat-tail lane	`2.33 × 500 × 6.5 = 7,573 — buffer KEPT`	measured σ = 6.5 > brochure 5 (illustrative)	kept until the source process moves
5	Accuracy gate	`LTacc = 1 − |actual − promised| / promised; bar 99% on 200 lanes`	record	99% held before any cut
6	The release	`$1,500,000 / 580,000 = $2.59/unit`	record pair; average derived	580K units / ~$1.5M, 0 stockouts
7	Recurring value	`$1.5M × 20% = $0.3M; 900 fewer late × 10% × $3K ≈ $0.27M`	modeled (benchmarks labeled)	≈$0.6M/yr; never added to S6's $2M

Depends on the six-site canonical model (S7) — six definitions cannot produce one σ. S7 is a pointer only in this packet; say "six sites," never "six NA sites."

Where this lands at Amazon. Inbound lead-time variability is half of every buffer and every promise in a placement model. Transferable: measure the lane's own distribution, promise at the tail you can defend, release capital only on evidence with the exception written into the rule, and give every parameter a drift flag. Reliability (zero stockouts), cost efficiency (expedites avoided), asset utilization (~$1.5M of working capital out).

FULL DERIVATION — FOUR-TABLE MATH, SOLUTION FLOW, SENSITIVITY, SPOKEN STATEMENT, IF-CHALLENGED NEW

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Service held at lower inventory: 580K units / ~$1.5M released (record)
+- Safety stock (stock) = z * d * sigma_L per lane
|    +- sigma_L: brochure folklore -> measured per lane (200 lanes)
|    +- release rule: 99% LT-accuracy bar + fat-tail lanes keep buffer; Finance
|         signed each lane (S6's two-signal rule is that program's own control)
|    +- fat-tail lanes: sigma_L worse than brochure -> buffer KEPT
+- Promise integrity (flow): p99 per lane, LT accuracy 99% (record)
|    +- cost asymmetry: line-down >> one day of holding -> p99 not p95
+- Drift loop: refresh cadence -> p99 recomputed -> drift flag ->
     buffer decision re-opened -> promise stays measured, never a new brochure
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
Lead-time accuracy, 200 lanes	brochure	99% at p99	—	record
Safety stock released (one-time)	brochure-sized	measured-sized	—	record: 580K units / ~$1.5M
Stockouts on released lanes	—	0 in watch	—	record

DRIVER	BASELINE	AFTER	$/YR	BASIS
Carry on released $1.5M	held	free	$0.3M	modeled ($1.5M × 20% all-in carry; benchmark 15–25%)
Expedites on late arrivals	~1,000 late × 10%	~100 × 10%	$0.3M	modeled (90 × $3K)
Total modeled recurring	—	—	**≈$0.6M**	around the ~$1.5M record; never added to S6's $2M

If challenged. *"Is the $1.5M measured?"* — "The release is record: 580K units out, no stockout in the watch window, Finance signed each lane. The ~$2.59/unit is the derived average on low-value components. What's modeled is the recurring value around it — carry and expedites — and each input is labeled." *"Why did you own the percentile choice?"* — "I didn't — I owned the frame. I showed each lane's asymmetry, line-down cost against carrying cost, and planning chose p99. Their choice surviving an audit is worth more than my choice winning a meeting."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Lead-time parameters	`carrier-brochure values; variability unmeasured on 200 lanes`	record	folklore, not data
2	One lane sized on folklore	`SS = z × d × σ_L = 2.33 × 500 × 5 = 5,825 units`	d = 500 units/day; brochure σ = 5 days; z = 2.33 at 99% (illustrative, toolkit form)	5,825 units of buffer
3	Tail blindness	`normal shortcut μ + 2.33σ understates a right-skewed lane's true p99 by 3.4 days`	illustrative from the record	the promise misses the tail
4	Late arrivals	`200 lanes × ~50 shipments = 10,000/yr; × 10% = 1,000 late/yr`	volumes illustrative	~1,000 late/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Measure per lane	`μ, σ, p50, p90, p99 from shipment history, 200 lanes`	production cycle + inbound transit vs actual (record)	measured distributions
2	p99 defined cumulatively	`p99 = min{d : P(L ≤ d) ≥ 0.99}`	not μ + 2.33σ unless the lane is normal (record definition)	the tail promise
3	Accuracy gate	`LTacc = 1 − |actual − promised| / promised; bar = 99% on the 200-lane set`	before endorsing any cut (record)	99% held
4	Reliable lane re-sized	`SS = 2.33 × 500 × 1.2 = 1,398; release = 5,825 − 1,398 = 4,427 units`	measured σ = 1.2 days vs brochure 5 (illustrative)	4,427 units off this lane
5	Fat-tail lane	`SS = 2.33 × 500 × 6.5 = 7,573 units — buffer KEPT`	measured σ = 6.5 > brochure 5 (illustrative)	kept until the source process moves
6	Drift flag	`p99 recomputed on refresh; buffer decision re-opened if accuracy falls`	record	the promise stays measured, never a new brochure

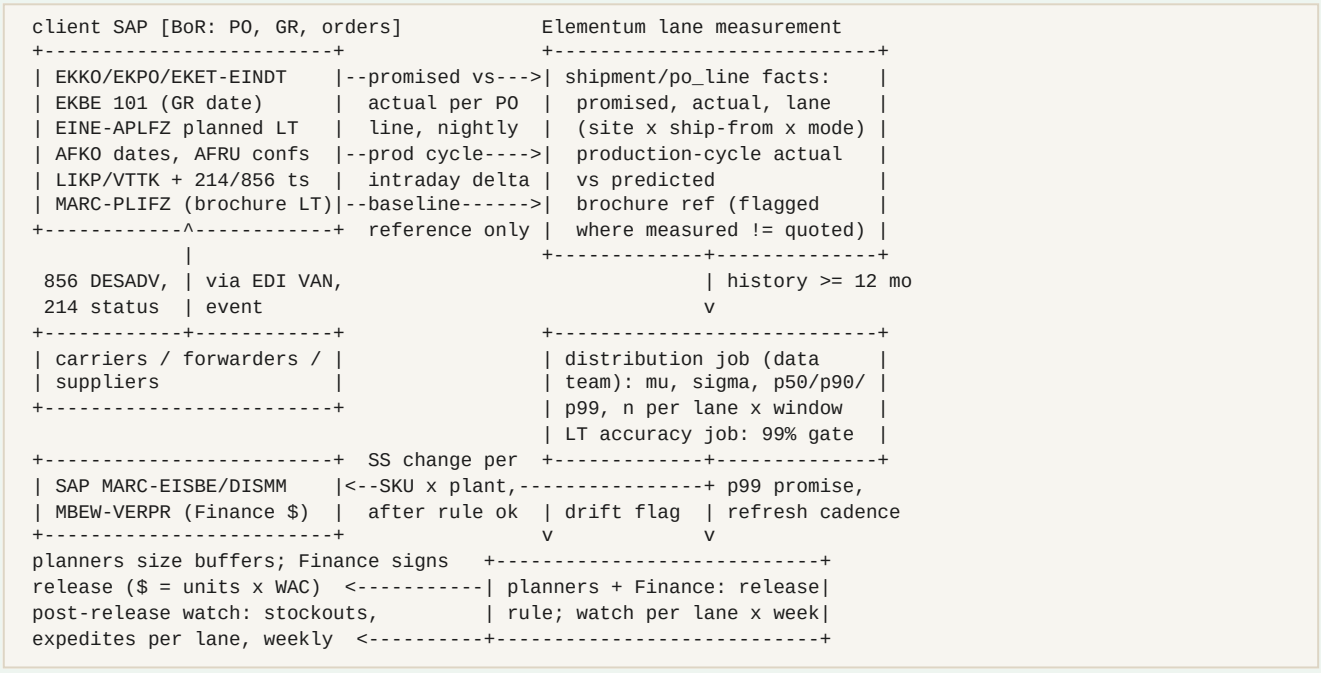
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Lead-time parameters	brochure folklore	measured distributions, 200 lanes	record
2	Lead-time accuracy	unmeasured	99% at p99	record
3	Inventory	brochure-sized	580K units / ~$1.5M released one-time	record pair
4	Stockouts on released lanes	—	0 in the watch window	record
5	Late arrivals	~10%	~1% at the 99% bar	record; volumes illustrative

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Unit value	`$1,500,000 / 580,000 = $2.59/unit`	derived from the record pair; low-value components	≈$2.59/unit
2	Carry on the release	`$1,500,000 × 20% = $300,000`	all-in carry (benchmark 15–25%)	≈$0.3M/yr
3	Late reduced	`10,000 × 10% = 1,000 before; 10,000 × 1% = 100 after; 1,000 − 100 = 900 fewer late`	volumes illustrative; the 99% is record	900 fewer late/yr
4	Expedites avoided	`900 × 10% = 90 expedited; 90 × $3,000 = $270,000`	expedited share and premium modeled	≈$0.3M/yr
5	Total recurring	`0.3 + 0.3 = $0.6M`	modeled; never added to S6's $2M	≈$0.6M/yr

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Carrying rate (benchmark)	15% → $0.23M	20% → $0.30M	25% → $0.38M
Late share before p99 (illustrative)	5% → 400 fewer	10% → 900 fewer	15% → 1,400 fewer

INPUT	LOW	BASE	HIGH
Share of late arrivals expedited (illustrative)	5%	10%	15%
Expedite premium (benchmark)	$2K → $0.04M	$3K → $0.27M	$5K → $1.05M
Total modeled $/yr	≈$0.3M	≈$0.6M	≈$1.4M

RELEASE-DECISION DESIGN — POINT ESTIMATE VS UPPER BOUND; WHAT "99% ACCURACY" CAN MEAN (EXTRA TABLE)

The record is the rule — measured per-lane distribution, p99 promise, 99% accuracy gate, fat tails kept, Finance signs. The statistics below are my design arithmetic on illustrative lanes. VERIFY-WITH-RAJESH what the program did on thin lanes and which accuracy definition it used.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Brochure buffer	$z \cdot d \cdot \sigma_L = 2.33 \times 500 \times 5 = 5{,}825$	brochure σ 5 (illustrative); z = 2.33 (toolkit)	5,825 units
2	Reliable lane, point estimate	$2.33 \times 500 \times 1.2 = 1{,}398$; release $5{,}825 - 1{,}398 = 4{,}427$	measured σ 1.2 (illustrative)	release 4,427
3	Standard error of $\hat{\sigma}$	$\sigma \div \sqrt{2(n-1)}$: $n = 50 \rightarrow 1.2 \times 0.1010 = 0.121$; $n = 400 \rightarrow 1.2 \times 0.0354 = 0.042$	large-sample approximation	10.1% (n = 50) · 3.5% (n = 400)
4	95% upper bound on σ	$n = 50$: $1.2 \times (1 + 1.96 \times 0.1010) = 1.44$; $n = 400$: $1.2 \times (1 + 1.96 \times 0.0354) = 1.28$	one lane, illustrative n	1.44 · 1.28
5	Buffer on the upper bound	$n = 50$: $2.33 \times 500 \times 1.438 = 1{,}675$; $n = 400$: 1,495	design arithmetic — not the record	1,675 · 1,495 units
6	Winner's-curse cost	$n = 50$: $1{,}675 - 1{,}398 = 277$ units × $2.59 = $717	$2.59 = $1.5M ÷ 580K (record pair)	277 units ≈ $717 held back
7	Implied cost of a late day (p99)	carry/day = 500 × $2.59 × 20% ÷ 365 = $0.71; Cu = 99 × $0.71 = $70	critical ratio 0.99 ÷ 0.01	p99 justified above $70 per shortage-day
8	Accuracy as a ratio, per shipment	$\lvert 21 - 34 \rvert \div 34 = 38.2\% \rightarrow$ score 61.8%	example lane (secondary source)	cannot be 99% — read as coverage or p99-match

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The measured result is five hundred eighty thousand units, about a million and a half dollars of safety stock, released at ninety-nine percent lead-time accuracy with zero stockouts on the watch; that is record and Finance signed it. What recurs around it I model at roughly six hundred thousand a year: about three hundred thousand in carrying cost on capital we no longer hold, and about three hundred thousand in expedites we no longer pay because the promise sits at the ninety-ninth percentile instead of a brochure number. The release and the accuracy are measured; the two recurring pieces are my model on benchmark carrying and expedite rates, and I never add this to the two million from the forecast program."

> **If challenged — the full answers.**
>
> **The expedite number is the softest**
>
> It assumes ten thousand shipments a year on the two hundred lanes and that one late arrival in ten forced an expedite before p99; to firm it I would pull the expedite ledger by lane for the year before and after the release and compare directly.
>
> **Twenty percent carrying cost is high**
>
> Fifteen to twenty-five is the textbook range and I use the middle; capital alone is eight to twelve, so if Finance wants capital only the carry is about a hundred and fifty thousand and I would say that instead.
>
> **Why is the million and a half not an annual saving**
>
> Because it is a one-time working-capital release, booked once as a signed planning act; what recurs is the carry and the avoided expedites, and I would rather be right at six hundred thousand a year than wrong at a million and a half.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S8

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you went deeper than the request.	Dive Deep	A "variability report" reframed as a promise-and-buffer policy; mean, spread and tail per lane from actuals; p99 defined cumulatively.	Do not claim to have built the distributions — the data team did, to my requirement.
Tell me about a time the data contradicted conventional wisdom.	Are Right, A Lot	Brochure values vs measured lanes; one right-skewed lane where the average hid the tail; planning chose p99 on the asymmetry.	S10 (Austria — "the noise said carriers") is the bar raiser's; do not use it here.
Tell me about a time you held a standard under pressure.	Insist on the Highest Standards	No release below 99% lead-time accuracy; Finance's push for a broader release answered with the rule they signed.	Do not frame Finance as the obstacle; the exception was in their rule.
How do you set safety stock / service levels?	Functional — placement policy	Two error signals (forecast, lead time), measured at the lane × SKU grain; policy by cost asymmetry; release by rule with a watch window; drift flags.	Never present the formula as a program artifact — MIT toolkit illustration.
What did you refuse to do?	Have Backbone	Averaging away a tail; endorsing a cut below 99%; a blanket release — all written into the act before anyone asked.	Keep the $2M (S6) out of this answer.
How did you make Finance comfortable releasing capital?	Earn Trust	Finance agreed the release rule in the same room as planning signed the policy — before any cut; signed each lane; the watch armed first.	No annualized $1.5M — it is a one-time release.

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 **Tell me about a time you went deeper than the request.** DIVE DEEP

Open (first two sentences): The ask was a “lead-time variability report.” I turned it into a parameter question: what number is the plan actually using, and how wrong is it?

STAR (≈90 s): Situation: a six-site manufacturer’s logistics manager wanted a report; inventory was sized on carrier-brochure lead times with variability unmeasured. Task: decide whether a report was the deliverable. Action: I reframed it as a parameter problem, scoped 200 lanes by site, lane and mode, wrote the success criteria — measured lanes replace brochures, and 99% lead-time accuracy before any cut — and had the data team build per-lane distributions. Result: planning chose a p99 promise and 580K units, about $1.5M, came out with zero stockouts.

Land this number: 200 lanes; 99% lead-time accuracy; 580K units / ~$1.5M, 0 stockouts. · **Pivot to NACF:** NACF inbound lead-time variability by vendor lane is the same “report” that is really a parameter. · **Do NOT say:** Do not say the data team’s distributions were mine; I owned the requirement, the policy and the release rule.

Q2 **Tell me about a time you put the customer’s cost ahead of your own team’s metric.** CUSTOMER OBSESSION

Open (first two sentences): The customer was the production line. A released unit of buffer that causes a line-down is worse than a unit never released.

STAR (≈90 s): Situation: carrying cost pushed to cut buffer; the line-down cost sat in another budget. Task: make the asymmetry visible. Action: I showed it lane by lane — an extra day of cover on the example lane costs about $0.71; a day of shortage costs far more — and let planning choose the quantile; they chose p99. On reliable lanes p99 still released buffer; on fat-tail lanes the buffer stayed. Result: zero stockouts across the released lanes.

Land this number: Zero stockouts (record); p99 on 200 lanes. · **Pivot to NACF:** Placement trades FC buffer cost against a missed promise the same way. · **Do NOT say:** Do not say planning was forced — say they chose with the asymmetry in front of them.

Q3 **Tell me about a time you got a finance partner to sign a risky decision.** EARN TRUST

Open (first two sentences): Finance signed each lane because the exception was written into the rule before anyone asked.

STAR (≈90 s): Situation: releasing safety stock is irreversible in practice. Task: a rule Finance could audit. Action: below 99% lead-time accuracy, no release; fat tails keep their buffer; Finance signs per lane; a stockout-and-expedite watch from day one. Result: $1.5M of working capital released, Finance-signed, no stockout in the watch window.

Land this number: 580K units / ~$1.5M (record, one-time). · **Pivot to NACF:** NACF WOC changes need a pre-signed exception clause too. · **Do NOT say:** Do not call the $0.6M recurring a result — it is modeled.

Q4 **Tell me about a process you designed that someone else now owns.** OWNERSHIP

Open (first two sentences): I designed the promise policy and the release rule as a signed planning act; planning re-applies it on its own refresh.

STAR (≈90 s): Situation: a one-off analysis decays into a new brochure. Task: leave a process, not a project. Action: drift flag per lane — p99 recomputed on refresh, the buffer decision re-opens if accuracy falls — and the watch became a standing report owned by planning. Result: the promise stays honest after I leave.

Land this number: Drift flag per lane; standing report owned by planning. · **Pivot to NACF:** Every NACF parameter I touch gets a drift flag and an owner. · **Do NOT say:** Do not say I run it today.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: "Planning to the slow day, not the average."

Headline metric: p99 per lane; 99% lead-time accuracy on 200 lanes

- Cumulative p99, not mean + 2.33σ — the normal shortcut understated one skewed lane by 3.4 days.
- Winner's curse on thin lanes: size on the upper bound.
- Accuracy definition: coverage or p99-match.
- Reliable lane release 4,427; fat tail kept.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Customer Obsession; Earn Trust

Open: "Finance signed each lane because fat tails kept their buffer."

Headline metric: 580K units / ~$1.5M, zero stockouts

- Asymmetry shown lane by lane.
- Planning chose p99.
- Exception in the rule before the ask.
- Release is one-time working capital.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Learn and Be Curious

Open: "I would put the interval on the page next to every p99."

Headline metric: p99 interval open-ended above at n < 330

- The record carries no window length — I do not invent one.
- S8a and S8c are the interval and validation work.
- Thin lanes: cohort pooling or a longer window.
- VERIFY-WITH-RAJESH the program's thin-lane rule.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 "Define 99% lead-time accuracy — what tolerance does it allow?"

The numbers: Ratio form: 1% × 5 d = 0.05 d; 1% × 30 d = 0.3 d. Per shipment vs a p99 promise: |21 − 34| ÷ 34 = 38.2% → 61.8%. Coverage form: 1 shipment in 100 beyond the promise.

Say exactly: "Written as one minus the absolute miss over the promise, one percent is a twentieth of a day on a five-day lane and a third of a day on a thirty-day lane — but against a p99 promise the average shipment misses by about thirty-eight percent, so I read ninety-nine as coverage: ninety-nine of a hundred shipments inside the promise. I would confirm which definition the record uses before I quote a tolerance."

T2 "Why p99 and not p95 — what is that buffer worth?"

The numbers: Carry per day of cover = 500 × $2.59 × 20% ÷ 365 = $0.71; p99 ⇔ a shortage-day worth > 99 × = $70; p95 ⇔ > $13.

Say exactly: "A p99 promise says a shortage day costs ninety-nine carrying days — about seventy dollars on that lane — and a line-down clears that by orders of magnitude; I showed planning the asymmetry lane by lane and they chose p99."

T3 "You released the lanes with the smallest measured σ. Isn't that a selection effect?"

The numbers: σ̂ standard error = σ ÷ √(2(n−1)) = 10.1% at n = 50; upper-bound σ 1.44; buffer 1,675 vs 1,398; 277 units held back.

Say exactly: "Yes, partly — a lane picked for a small measured spread is partly a lucky draw; on a fifty-shipment lane sizing on the upper bound holds back about two hundred seventy-seven units of the four thousand four hundred I would otherwise release. I would show that adjustment next to every release."

S8a — Elementum — actual-vs-planned lead-time distribution analysis: the brochure was wrong three ways — 3 days short on the mean, twice-too-wide on the spread, 16 days short on the tail NEW

LPs this story serves here: Dive Deep (primary — plan vs actual at the shipment grain) · Are Right, A Lot (rank, don't fit) · Insist on the Highest Standards (a "99%" plan that held 96.5% is not a 99% plan) · Earn Trust (the carrier's own scans, not my opinion)
Process family: Plan / Source seam — inbound lead-time parameter diagnosis before any buffer or policy change

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Dive Deep** — I compared plan to actual at the shipment grain on three axes — location, spread, tail — before changing a single parameter; 240 of 400 shipments ran slower than the brochure.
- **Are Right, A Lot** — I ranked the shipments instead of fitting a bell curve; the curve's "99%" plan held 96.5% — 14 of 400 shipments ran past it — and the ranked p99 held 99%.
- **Insist on the Highest Standards** — I refused to publish a p99 that carried its own error: a plan statistic had to be distribution-free and come with its sampling uncertainty.
- **Earn Trust** — I showed planners and commercial the carrier's own scans on one lane — brochure 18, mean 21, p99 34 — so the argument was the data, not my say-so.

Honesty label. I owned the question, the comparison design and the plan statistic; the data team built the per-lane distributions to my requirement. The 18 / 21 / σ 4 / p99 34 / n≈400 lane is the loop packet's example lane (secondary source — the master packet carries "18d brochure" in its crosswalk and "lane-cohort counts are not in the record"); the ERP default σ of 8 days and the 7-day manual pad are that packet's illustrative inputs. Say "illustrative inputs, real outcomes": the record is 200 lanes, 99% lead-time accuracy at the p99 promise, 580K units / ~$1.5M released. VERIFY-WITH-RAJESH that the example-lane numbers are the ones you are comfortable quoting.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. The six-site manufacturer's safety stock kept growing and nobody could say why. Planning ran on carrier-brochure lead times — on my example lane, 18 days — plus a manual week of pad that every planner added. Task. Before touching a parameter I owned one question: how, exactly, was the planned lead time wrong? Action. I pulled about a year of shipment history on the lane — roughly 400 shipments — and compared plan to actual three ways. Location: the actual mean was 21 days, three days longer than the brochure, fifteen standard errors — not noise — and 240 of 400 shipments, 60%, ran slower than the number in the system. Spread: the ERP's default assumed an 8-day standard deviation; measured, it was 4 — wrong in the opposite direction, which is why planners had padded by hand. Tail: a bell curve from that mean and spread puts the 99th percentile at 30.3 days, but 14 of the 400 shipments ran longer — a "99%" plan that actually held 96.5%. Ranking the shipments puts it at 34. Result. Primary metric: shipments late against the plan on that lane, 60% → 1%, planned lead time 18 → 34 days; at program level, lead-time accuracy 99% on 200 lanes — record. The example lane is the pattern, not the headline.

20-SECOND VERSION NEW

The planned lead time was wrong in three directions at once — the mean was 3 days short, the ERP's spread was twice the real one, and the tail was 16 days out; I measured all three on the carrier's own scans and planned to the ranked p99, not a bell curve.

NACF BRIDGE — ONE SENTENCE

Every NACF inbound plan is a promised date; I would run the same plan-vs-actual on appointment and ASN dates versus receive scans, by vendor lane, before anyone resizes a buffer — location, spread, tail.

SITUATION / TASK — THE PROBLEM SPACE

S — Elementum, the same six-site manufacturer. Buffers kept growing, service was still shaky and nobody could explain why. Inventory planning ran on carrier-brochure lead times (18 days on the example lane) and every planner added an informal week of pad. The risk of doing nothing: working capital grows every quarter and every stockout triggers more padding — a wrong number plus panic.

T — Own the diagnosis: characterize how the planned lead time was wrong — bias, spread, tail — with the uncertainty on each statement, so the later policy (S8) and the formula redesign (S8b) stand on measured facts. Scope: 200 lanes by site, lane and mode; the example lane is the busy one.

Baselines on the table: planned lead time = brochure (18 d on the example lane) · actual mean 21 d · 60% of shipments slower than plan · ERP default σ_LT = 8 d (never measured) · manual pad 7 d × 500 units/day = 3,500 units on the lane · no lead-time-accuracy measurement existed.

ACTION — FIVE GATED PHASES

- **Discovery** — agreed with planning and logistics that the question was “how is the parameter wrong,” not “write a variability report”; chose the example lane because it had the history (≈8 shipments a week for a year). Exit bar: the comparison design — three tests, one lane — accepted before any number was shown.
- **Plan & Design** — requirement to the data team: promised date (PO-line schedule) vs actual receipt per shipment, keyed by lane = site × ship-from × mode, 12+ months so every season is in the sample; outputs mean, σ, p50 / p90 / p99, n and the late share against the plan in force. Exit bar: data team accepts the spec; planning agrees which plan statistic is on trial (mean, brochure + pad, normal p99, ranked p99).
- **Build & Test** — data team built the lane file; I owned the comparison: bias with its standard error, late share with a binomial interval, σ against the ERP default, then the four candidate plans scored on the same 400 shipments. Exit bar: the candidate table reproduces on a second pull; the ranked p99 reproduces the “99%” by construction and its sampling interval is printed beside it.
- **Cutover & Go-Live** — the finding went to planners as one lane, three numbers — brochure 18, mean 21, p99 34 — then the same three tests per lane for all 200; the plan statistic moved to the ranked p99 under the release rule (S8) and the formula (S8b). Exit bar: planning signs the plan statistic; the old brochure stays in the master as a labeled reference only.
- **Hypercare / Close** — drift flag per lane (p99 recomputed on refresh; review if the tail or the spread moves) and a standing plan-vs-actual page owned by planning. Exit bar: the page refreshes without me; thin lanes carry their interval on the page so nobody reads a max as a quantile.

Influence and ownership. I owned the question, the comparison and the plan statistic; the data team built the distributions; planners owned the parameters and the buffers; Finance and planning signed the policy that used it. The conflict was commercial: quoting 34 days where the brochure said 18 — answered with the number that decided it: a fast promise missed 60% of the time costs expedites and trust; the honest date costs only optics, and where speed truly mattered we fixed the lane, not the number.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **%** — shipments late against the plan on the example lane 60% (brochure) → 1% (ranked p99); program level: lead-time accuracy 99% on 200 lanes at the p99 promise (record).
- **Time** — planned lead time on the example lane 18 → 34 days: the plan now says what 99 of 100 shipments do.
- **#** — 200 lanes measured by site, lane and mode; 580K units / ~$1.5M released later under the signed rule (record, one-time — S8).
- **$** — on the example lane only: ≈$9.0K a year modeled (carry on the 3,500-unit pad + expedites on late arrivals) — an illustrative lane, never scaled by 200 and never added to S8’s ≈$0.6M or S6’s $2M.
- **Quality** — every plan statistic reported with n and an interval; the max of a thin lane is never presented as its p99.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	No agreed definition of “how wrong is the planned lead time”; I defined it as three tests — location, spread, tail — and the unit of analysis (the shipment on a lane).
Scope & influence	Planners, logistics, commercial, Finance and a data team that did not report to me; the commercial pushback on 34 vs 18 resolved by the 60%-late number.
Execution	Chose a distribution-free plan statistic over a convenient formula and carried its interval; sequenced diagnosis (this story) before policy (S8) before formula (S8b).
Impact	99% lead-time accuracy on 200 lanes (record); the diagnosis is what made 580K units / ~$1.5M releasable without a stockout.
Process improvement	A standing plan-vs-actual page and drift flag per lane — the parameter can no longer decay into a new brochure.
Complexity	Skewed tails, thin lanes, seasonality inside a 12-month window, and a normal-fit shortcut that looks right and holds 96.5%.
Root cause	The root cause was the planning parameter, not the carrier: 60% of arrivals were “late” against a number the carrier never measured.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: shipment × lane (site × ship-from × mode), promised date (PO-line schedule) vs actual receipt, 12+ months; joins PO line → goods receipt → lane key; cuts by lane, mode, site, quarter. Revealed: bias +3 d (t = 15), 60% of shipments slower than plan, ERP σ 8 vs measured 4, normal-fit p99 holds 96.5%.
2 · Solution design	Options: plan to the mean · brochure + planner pad · normal-fit p99 · ranked (empirical) p99. Chosen: ranked p99 — distribution-free, defensible to the tail the business cares about; why: it is the only candidate whose coverage does not depend on an assumed shape.
3 · Math & data logic	Math: bias = 21 − 18 = +3.0 d; SE = 4/√400 = 0.20; t = 15.0 · late share 240/400 = 60% (Wilson 95% [55.1%, 64.7%]) · normal p99 = 21 + 2.33 × 4 = 30.32, 14/400 beyond → 96.5% · ranked p99 = t(⌈0.99 × 400⌉) = t(396) = 34 d · order-statistic 95% interval t(392)…t(400).
4 · Implementation	Implementation: spec → data team → lane file → comparison → planning sign-off → plan statistic moves to the ranked p99 under the S8 release rule; drift flag + refresh cadence; thin lanes print their interval. Risk: thin lanes — handled by cohort pooling or a longer window (VERIFY-WITH-RAJESH what was actually done).
5 · Probe questions	Probe questions: n and representativeness · +3 d noise vs signal · interval on a p99 from 4 observations · seasonality · thin lanes · why rank not fit — all answered with numbers below and in the Trap Drills.

PROBING — CHALLENGE Q&A TUNED TO A STATISTICIAN

> **Analytics — n = 400 on one lane. Is that representative of 200 lanes?**
>
> No, and I say so. It is the busiest lane, about eight shipments a week. The master’s illustrative volume is about 10,000 shipments a year across 200 lanes — roughly 50 on an average lane — so the example shows the mechanism and the same three tests ran per lane. At n = 50 the order-statistic interval for the p99 is open-ended above: t(48) up to “unbounded” — the maximum is only a lower bound, which is why thin lanes need a longer window or cohort pooling. VERIFY-WITH-RAJESH what the program did for thin lanes.

> **Analytics — Is a three-day difference statistically meaningful or just practically meaningful?**
>
> Both. Standard error of the mean is 4 ÷ √400 = 0.20 day, so the 95% interval on the actual mean is 21 ± 0.39 — the brochure sits fifteen standard errors away. Practically, 3 days × 500 units a day is 1,500 units of exposure per cycle that the plan did not know about.

Analytics — Your p99 is the 396th of 400. What is the confidence interval?

Wide, and I print it: with n = 400 the 95% interval for the true p99 runs from the 392nd-ranked shipment (between 31 and 34 days on this lane) to the slowest one (35 or more). In-sample coverage of the ranked plan is 99% by construction; the honest sampling interval on that coverage is 97.5%–99.6% — which is why I validated on a held-out quarter (S8c) and put a drift flag on every lane.

Analytics — Why not just mean + 2.33σ?

Because lead-time tails are right-skewed and the shortcut understates them. On this lane it gives 30.32 days; 14 of 400 shipments, 3.5%, ran longer, so a “99%” plan delivered 96.5%. On the whiteboard sample — 20 shipments, mean 12.5, σ 2.64 — the normal p99 is 12.5 + 2.33 × 2.64 = 18.65 against a ranked 22: about 3.4 days short, the canon figure.

Analytics — What about seasonality in the window?

The window was twelve months or more, so every quarter is inside the sample and the p99 prices the slow season rather than ignoring it. The test I would add is the quarterly cut: the +3-day bias should carry the same sign in all four quarters before I call it a parameter error rather than one bad season. The record does not carry that cut. VERIFY-WITH-RAJESH

Shadow / closing — What would you do differently?

Lead with the thin lane, not the busy one — show the interval first. A p99 from fifty shipments is a maximum with a label on it, and the plan page should say so.

BUSINESS IMPACT MODEL — WHAT THE WRONG PLANNED LEAD TIME COST (EXAMPLE LANE, MODELED)

```
Planned lead time wrong (brochure 18 d)
+- Location error: mean 21 vs plan 18  -> 60% of shipments late vs plan (240 / 400)
+- Spread error: ERP sigma 8 vs measured 4 -> wrong the OTHER way -> planners pad by hand
+- Tail error: normal p99 30.3 holds 96.5%; ranked p99 = 34 holds 99%
+- Consequence A (stock): 7-day pad x 500 units = 3,500 units held to cover the brochure
+- Consequence B (flow): late arrivals vs the planning date -> expedites
+- Fix: plan statistic = ranked p99 (distribution-free) + interval + drift flag
```

DRIVER	BASELINE	AFTER	$/YR (LANE)	BASIS
Planned lead time	18 d (brochure)	34 d (ranked p99)	—	loop-packet example lane (secondary)
Shipments late vs plan	28 of 400 vs the 25-d planning date (7%); 240 of 400 vs the 18-d brochure	4 of 400 (1%)	—	loop-packet example lane
Carry on the manual pad	3,500 units × $2.59 = $9,065 held	pad not needed	$1,813	modeled (20% all-in carry; benchmark 15–25%)
Expedites on late arrivals	28 × 10% × $3,000	4 × 10% × $3,000	$7,200	modeled (10% share and $3K premium illustrative)
Total, this lane	—	—	**≈$9.0K**	illustrative lane — never scaled ×200, never added to S6/S8

If challenged. *“Multiply by 200.”* — “No: lanes differ by two orders of magnitude in volume; the program-level recurring value is S8’s ≈$0.6M on its own inputs, and this lane is the mechanism, not the sample.”

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: how the plan was wrong (example lane, n ≈ 400).

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Sample	`n ≈ 400 (≈ 8 shipments/wk ≈ one year)`	loop-packet example lane (secondary) — a busy lane; average lane ≈ 50 (master, illustrative)	n = 400
2	Location error	`mean − brochure = 21 − 18 = +3.0 d (3 ÷ 18 = +16.7%)`	brochure 18, measured mean 21 (example lane)	+3.0 d
3	Is it noise?	`SE = σ/√n = 4/√400 = 0.20 d; t = 3.0/0.20 = 15.0; 95% CI of mean = 21 ± 1.96 × 0.20 = [20.61, 21.39]`	measured σ = 4 d	t = 15.0 — not noise
4	Late vs brochure	`240 / 400 = 60.0%; Wilson 95% [55.1%, 64.7%]`	counts of shipments slower than 18 d	60% late
5	Spread error	`ERP σ = 8 vs measured 4 → ratio 2.0, variance ratio 4.0; CI of σ ≈ 4 ± 1.96 × 4/√(2 × 399) = [3.72, 4.28]`	ERP default 8 d (illustrative); large-sample approximation	8 is far outside the interval
6	Tail error — bell curve	`21 + 2.33 × 4 = 30.32 d; 14 / 400 = 3.5% beyond → coverage 96.5%`	z = 2.33 (99%, toolkit label)	a “99%” plan that holds 96.5%
7	Tail error — vs brochure	`ranked p99 − brochure = 34 − 18 = 16 d`	ranked p99 = t(396)	16 d short on the tail
8	The manual patch	`7 d × 500 units/day = 3,500 units`	pad and demand rate illustrative	3,500 units of pad

Table 2 — the fix: four candidate plans scored on the same 400 shipments.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Plan = mean	`21 d; shipments slower: 160 of 400`	counts from the lane file (loop packet)	serves 60.0%
2	Plan = brochure + planner pad	`18 + 7 = 25 d; slower: 28 of 400`	pad illustrative	serves 93.0%
3	Plan = normal-fit p99	`21 + 2.33 × 4 = 30.32 d; slower: 14 of 400`	assumes a bell curve	serves 96.5% — not 99%
4	Plan = ranked p99	`position ⌈0.99 × 400⌉ = 396 → t(396) = 34 d; slower: 4 of 400`	distribution-free (record definition)	serves 99.0% in-sample
5	Interval on the ranked plan’s coverage	`4/400 late → Wilson [0.39%, 2.54%] late → coverage [97.5%, 99.6%]`	binomial; n = 400	in-sample 99% is not a guarantee
6	Interval on the p99 itself	`95% interval t(392)…t(400) (coverage 96.1%); n = 50 → t(48)…open; n = 100 → t(97)…open; n = 1,000 → t(983)…t(996)`	exact binomial order-statistic interval	thin lanes cannot bound a p99 from above
7	Whiteboard sample (n = 20)	`mean 12.5, σ 2.64; ranked p50 = 12, p90 = 14, p99 = 22; normal p99 = 12.5 + 2.33 × 2.64 = 18.65; gap = 22 − 18.65 = 3.35 ≈ 3.4 d`	values 9,10,11×5,12×6,13×3,14×2,15,22 (illustrative)	normal understates by ≈3.4 d (canon)

Table 3 — after: the metric, before → after.

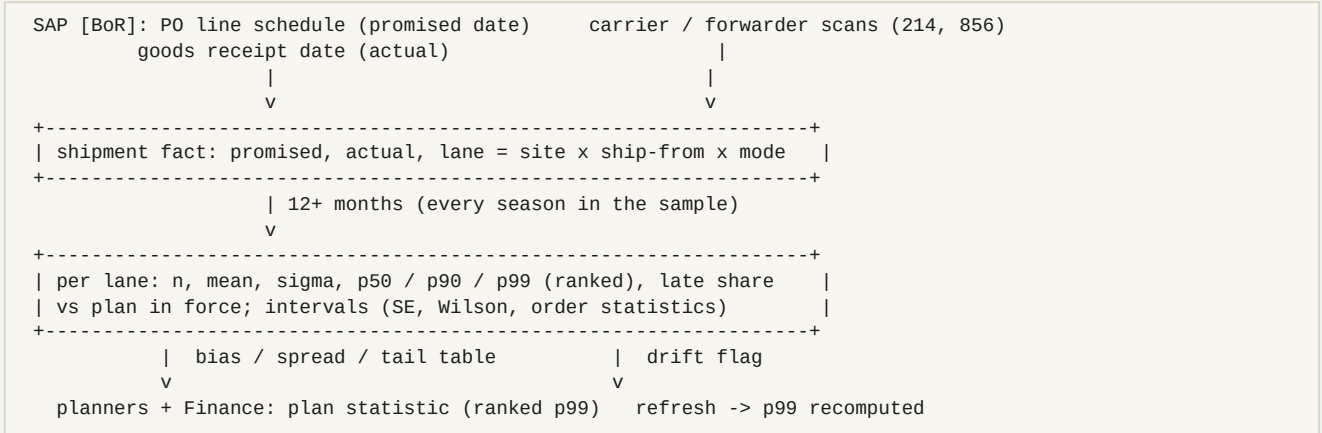
STEP	METRIC	BEFORE	AFTER	BASIS
1	Planned lead time (example lane)	18 d (brochure)	34 d (ranked p99)	example lane (secondary)
2	Shipments late against the plan	60% vs brochure (28 of 400 = 7% vs the 25-d planning date)	1% (4 of 400)	example lane
3	Lead-time cover in the plan (plan − mean)	18 − 21 = −3 d (plan below the mean)	34 − 21 = +13 d	derived; feeds S8b
4	Spread used in planning	8 d (ERP default, unmeasured)	4 d measured, per lane	illustrative default
5	Lead-time accuracy, 200 lanes	unmeasured	99% at the p99 promise	record

Table 4 — the dollars, every multiplication shown (example lane, modeled).

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Pad inventory	`7 d × 500 units/day = 3,500 units; 3,500 × $2.59 = $9,065`	$2.59 = $1.5M / 580K (derived record pair)	$9,065 of capital on the pad
2	Carry on the pad	`$9,065 × 20% = $1,813/yr`	benchmark 15–25% all-in	≈$1,813/yr
3	Expedites before	`28 late × 10% × $3,000 = $8,400`	late vs the 25-d planning date; share and premium illustrative	$8,400/yr
4	Expedites after	`4 late × 10% × $3,000 = $1,200`	late vs the 34-d plan	$1,200/yr
5	Expedites avoided	`$8,400 − $1,200 = $7,200`	row 3 − row 4	$7,200/yr
6	Lane total	`$1,813 + $7,200 = $9,013`	modeled, one illustrative lane	≈$9.0K/yr — never × 200

Dollar basis labels: the lane statistics are the loop packet's example lane (secondary); every dollar is modeled on illustrative inputs; the record is the 99% / 200 lanes / 580K units / ~$1.5M quartet in S8.

SOLUTION FLOW — THE PLAN-VS-ACTUAL ANALYSIS PIPELINE



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Shipments on the lane (n) — p99 interval	n = 50 → t(48)…open	n = 400 → t(392)… t(400)	n = 1,000 → t(983)… t(996)
Carry rate on the pad	15% → $1,360	20% → $1,813	25% → $2,266
Expedite share of late arrivals	5% → $3,600 avoided	10% → $7,200	15% → $10,800
Lane total (modeled, one lane)	≈$5.0K	≈$9.0K	≈$13.1K

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

“On one busy lane the brochure said eighteen days; the actual mean was twenty-one, a three-day gap that is fifteen standard errors wide, and sixty percent of shipments ran slower than the number in the system. The system’s default spread was eight days; measured, it was four. A bell curve would have promised a ninety-ninth percentile of thirty point three, and fourteen of four hundred shipments ran past it; ranking the shipments gives thirty-four, which held ninety-nine percent. The record is ninety-nine percent lead-time accuracy on two hundred lanes. The lane is an example and the dollars around it are my model on illustrative inputs — about nine thousand a year on that one lane, and I never multiply it by two hundred.”

If challenged — the full answers

The lane is the example, not the data. “The program-level result is the record — 200 lanes, 99%, 580K units, about $1.5M. The lane shows how I got there; its counts are the loop-packet example and I say so.” The interval is wide. “Yes. With 400 shipments the p99 rests on the top four; I print the interval and validate out of sample in S8c. With 50 shipments I would not call a maximum a p99.” The dollars are small. “One lane, modeled, on purpose — I would rather show the arithmetic than a program-wide number I can’t derive.”

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 **Tell me about a time you dug into the data to find the real problem.** DIVE DEEP

Open (first two sentences): Buffers kept growing and nobody could say why, so I stopped collecting opinions and pulled the shipment scans on our busiest lane. A year of history, about 400 shipments, planned date against actual receipt.

STAR (≈90 s): Situation: the plan said 18 days on that lane and every planner padded a week by hand. Task: find how the number was wrong before anyone changed a parameter. Action: I compared plan to actual three ways — the mean was 21 days, three days late and fifteen standard errors from the brochure; the ERP’s assumed spread was 8 days against a measured 4; and the tail ran to 34 days. I showed it as the carrier’s own data on one lane. Result: planners stopped debating whose buffer was right and agreed the parameter was the problem; that became the p99 promise and the 580K-unit release.

Land this number: 60% of shipments slower than the brochure (240 of 400); mean 21 vs 18 — t = 15. · **Pivot to NACF:** NACF inbound runs on promised dates — ASN ETAs and appointment slots; the same plan-vs-actual by vendor lane is the first diagnostic I would run. · **Do NOT say:** Do not say I built the distributions — the data team did, to my requirement; do not say “400 on every lane.”

Q2 **Tell me about a time the data contradicted what everyone believed.** ARE RIGHT, A LOT

Open (first two sentences): The belief was that the carrier was missing its transit times. The scans said the carrier was delivering in about 21 days and we were planning on 18.

STAR (≈90 s): Situation: service was shaky and the room's theory was carrier performance — re-contract, escalate. Task: decide whether this was carrier management or planning. Action: I did not argue; I compared actual to the planned number and to the ERP's assumed spread; the options ranked themselves — the carrier was consistent, our parameter was wrong. Result: the fix moved to planning parameters, not carrier contracts, and the lane held 99% at the ranked p99.

Land this number: Mean 21 d vs plan 18 d; p99 34 d. · **Pivot to NACF:** In NACF terms: before blaming a carrier or a vendor for dock chaos, check whether the inbound plan's lead-time parameter was ever right. · **Do NOT say:** Do not imply the carriers were perfect — say the distribution had a tail; the point is the plan ignored it.

Q3 **Tell me about a time you refused to accept a metric that looked good enough.**
INSIST ON THE HIGHEST STANDARDS

Open (first two sentences): A bell-curve p99 looked right — 30.3 days — and it was a "99%" plan that held 96.5%. I would not ship a 99% label on a 96.5% plan.

STAR (≈90 s): Situation: the standard shortcut, mean plus 2.33 standard deviations, was what the planning system and most analysts would use. Task: choose the plan statistic. Action: I scored four candidate plans on the same 400 shipments — mean 60%, brochure plus pad 93%, normal p99 96.5%, ranked p99 99% — and printed the sampling interval on the winner so nobody mistook in-sample for guaranteed. Result: the ranked p99 became the statistic; the interval is why I also ran a held-out quarter.

Land this number: 96.5% vs 99% — 14 vs 4 of 400. · **Pivot to NACF:** For NACF placement and WOC targets, the same discipline: name the service level, then check what the distribution actually delivers. · **Do NOT say:** Do not call 99% "guaranteed"; say in-sample by construction, interval 97.5–99.6.

Q4 **Tell me about a time you persuaded a skeptical group with data.** EARN TRUST

Open (first two sentences): Commercial hated the number — 34 days where the brochure said 18. I did not defend it; I showed them what 18 had cost.

STAR (≈90 s): Situation: commercial teams quote what the system says, and a longer date reads as slower. Task: get a defensible promise adopted without an organizational fight. Action: one lane, three numbers — 18 brochure, 21 mean, 34 p99 — and the price of the status quo: a fast date missed 60% of the time costs expedites and trust, the honest date costs only optics. Where speed genuinely mattered I offered to fix the lane, not the number. Result: planning signed the plan statistic and commercial stopped quoting the brochure.

Land this number: 60% late at the quoted date vs 1% at the plan. · **Pivot to NACF:** Amazon's promise engine faces the same trade at scale; I would lead with the cost of the missed promise, not the length of the date. · **Do NOT say:** Do not say commercial was wrong to want speed — say the lever was the lane, not the label.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: “The brochure was wrong three different ways.”

Headline metric: t = 15 on the 3-day bias; 60% of shipments slower than plan

- Location: +3 d, SE 0.20 d, 15 standard errors.
- Spread: ERP σ 8 vs measured 4 — wrong the other way, hence the pad.
- Tail: normal p99 30.3 holds 96.5%; ranked p99 34 holds 99%.
- One lane, three numbers, the carrier’s own scans.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Have Backbone; Earn Trust

Open: “Commercial wanted to keep quoting 18; I would not put an 18 on a promise that failed 60% of the time.”

Headline metric: 60% late at the brochure date vs 1% at p99

- The objection was real: 34 reads slower than 18.
- I priced the status quo — expedites and trust — instead of arguing.
- Offered the lane fix where speed mattered.
- Planning signed the statistic; the brochure became a labeled reference.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Learn and Be Curious

Open: “I led with the busiest lane because it had the data — and that was my analytical shortcut.”

Headline metric: n = 400 busy lane vs ≈ 50 on an average lane; p99 interval open-ended above at n = 50

- Honest limit: a thin lane’s max is not its p99.
- I would now show the interval first and pool thin lanes by cohort.
- The held-out quarter (S8c) and the drift flag were the backstops.
- VERIFY-WITH-RAJESH what the program actually did on thin lanes.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 “Your p99 rests on four observations. How can that be a plan?”

The numbers: n = 400 → 95% interval t(392)…t(400); coverage interval 97.5%–99.6%; n = 50 → open-ended above; held-out quarter 99.1% (loop packet; S8c).

Say exactly: “It rests on the top four, and I print the interval — ninety-seven and a half to ninety-nine point six on coverage. That is why I validated on a held-out quarter and put a drift flag on every lane; on a thin lane I would not call a maximum a ninety-ninth percentile.”

T2 “How do you know the three-day gap is not one bad quarter?”

The numbers: SE 0.20 d; t = 15.0; window ≥ 12 months; the test I would add: same sign in all four quarterly cuts.

Say exactly: “Fifteen standard errors across a twelve-month window is not noise, but one bad quarter could still dominate it — the check is the quarterly cut, same sign in all four. The record doesn’t carry that cut and I would not claim it.”

T3 “What was the counterfactual — what would the plan have delivered if you had done nothing?”

The numbers: Brochure plan late 240/400 = 60%; brochure + pad 28/400 = 7% (93% served); ranked p99 4/400 = 1%.

Say exactly: “On the same 400 shipments the brochure plan was late sixty percent of the time and the brochure-plus-pad plan seven percent; the ranked p99 plan one percent — that is the counterfactual, shipment by shipment, with the interval.”

S8b — Elementum — safety-stock formula redesign: one formula nobody could defend plus a week of pad → two honest terms; one lane released 4,835 units, one lane took 1,383 in, 580K net NEW

LPs this story serves here: Invent and Simplify (primary — two terms replace a formula and a pad) · Have Backbone; Disagree and Commit (a rule that adds stock where the tail is fat) · Frugality (less stock, better service) · Ownership (a release rule someone owns after I leave)
Process family: Plan / Source seam — inventory-policy formula and release rule on top of the measured lane distributions (S8a → S8 → S8b)

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Invent and Simplify** — I replaced "$z \cdot \sqrt{(LT \cdot \sigma_d^2 + d^2 \cdot \sigma_{LT}^2)}$ plus a week of pad" with two terms a planner can read — lead-time cover to the measured p99 and demand cover over the mean lead time.
- **Have Backbone; Disagree and Commit** — The rule I wrote cuts both ways — on a lane whose measured tail is fatter than the assumption it adds stock — and I kept it two-sided when a one-way cut would have been easier to sell.
- **Frugality** — Lane A went 12,937 → 8,102 units while late arrivals fell from 7-in-100 to 1-in-100: less stock and better service from one honest number.
- **Ownership** — The release was a signed planning act with a drift flag and a standing stockout-and-expedite report planning owns — the formula does not need me to keep working.

Honesty label. I owned the formula redesign, the two worked lanes and the release rule; planners owned the parameters and Finance signed each lane. Lane A (d = 500, σ_d = 150, brochure 18 d, ERP σ_{LT} = 8 d, measured mean 21 / σ 4 / p99 34) and Lane B (d = 200, σ_d = 80, brochure 24, measured mean 30 / σ 11 / p99 58) are the loop packet's worked lanes; the roll-up split — about 150 lanes releasing an average of 4,500 units, about 50 lanes adding an average of 1,900 — is the loop packet's illustrative decomposition. The record is the net: 580K units / ~$1.5M, zero stockouts.
VERIFY-WITH-RAJESH — the master and this packet's S8 say fat-tailed lanes' buffer was "kept"; the loop packet says about 50 lanes took stock in. Say "kept, and on the worst lanes raised" only if that is what happened.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. A six-site manufacturer's inventory buffers were set by a textbook formula fed an assumed lead-time swing of eight days that nobody had measured, plus a week of pad that every planner added by hand. Task. Redesign the buffer so it said what the lanes actually did, and write the rule for when capital could come out. Action. I took the formula apart: on the example lane, 97.5% of its variance came from that unmeasured eight-day swing — the formula was mostly measuring a guess. I replaced it with two terms: lead-time cover — units per day times the distance from the average lead time to the measured ninety-ninth percentile — and demand cover over the average lead time. The pad went to zero because the plan now held 99%. The rule cut both ways: release only where the measured spread was inside the assumption, and where the tail was fatter, keep or add. Result. Primary metric: safety stock on the example lane 12,937 → 8,102 units, 37% less, with late arrivals 7-in-100 → 1-in-100; the program released 580K units, about $1.5 million, with zero stockouts — record, one-time working capital.

20-SECOND VERSION NEW

I replaced a formula that was 97% an unmeasured guess, plus a hand-added pad, with two honest terms — cover to the measured p99 and demand cover — and a two-sided release rule: 580K units out net, zero stockouts.

NACF BRIDGE — ONE SENTENCE

The NACF version is the inbound booklet's INV-B: $SS = z \cdot \sigma \cdot \sqrt{LT}$ — I would show which of the two variance terms dominates before anyone moves a buffer, and I would write the release rule so it can raise a buffer as well as cut it.

SITUATION / TASK — THE PROBLEM SPACE

S — The same six-site manufacturer. Safety stock was the textbook form, $z \times \sqrt{(LT \cdot \sigma_d^2 + d^2 \cdot \sigma_{LT}^2)}$, with the brochure lead time and a σ_{LT} of 8 days that was the ERP's default, plus an informal manual pad of about a week per lane. Nobody could say which part of the buffer was protecting against demand and which against transit — and on the loud lanes the answer was always "add more."

T — Own the formula and the release rule: say what each unit of buffer is for, size it from the measured distribution (S8a), and release capital only where the measured evidence supported it — with the exception written into the rule before anyone asked.

Baselines on the table: Lane A buffer 12,937 units (25.9 days of demand) · pad 3,500 units uncounted in the ERP · 7-in-100 shipments late against the plan in force · no statement of what service level the buffer bought.

ACTION — FIVE GATED PHASES

- **Discovery** — took the ERP formula apart on one lane: the demand part contributed 405,000 to the variance, the lead-time part 16,000,000 — 97.5% of it an assumed number. Exit bar: planning and Finance agree the buffer is a parameter question, not a service-level argument.
- **Plan & Design** — two terms — lead-time cover $d \times (p99 - mean)$ using the ranked p99 from S8a, and demand cover $z \times \sigma_d \times \sqrt{(mean\ lead\ time)}$ — pad to zero; release rule: release only where the measured spread and tail are inside the assumption and lane accuracy holds; where they are not, the buffer stays or rises. Exit bar: Planning and Finance sign the rule's text, including the add-stock clause.
- **Build & Test** — the data team computed every lane; I recomputed two lanes by hand and ran the normal-formula counterfactual beside them. Exit bar: Lane A and Lane B reproduce to the unit, and the counterfactual is on the page.
- **Cutover & Go-Live** — released lane by lane under the signed rule with Finance signing each lane; stockout-and-expedite watch from day one. Exit bar: no lane released without its lane accuracy and signature. VERIFY-WITH-RAJESH wave structure and lane counts per wave are not in the record.
- **Hypercare / Close** — drift flag per lane — p99 recomputed on refresh, buffer decision re-opens if accuracy falls — and the watch report handed to planning. Exit bar: planning re-applies the rule on its own refresh without me.

Influence and ownership. I owned the formula, the rule and the two worked lanes; the data team built the distributions; Finance signed each lane; planners owned the parameters. The temptation was a blanket cut because a one-way release is the easiest thing to sell to Finance — Lane B is the page that shows why not: a blanket cut of the same size would leave about half of what the measured tail needs.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **$** — 580K units / ~$1.5M safety stock released, net (record, one-time working capital; ≈$2.59/unit derived); ≈$0.3M/yr carry on it at 20% (modeled) — inside S8's ≈$0.6M, never added to S6's $2M.
- **#** — Lane A 12,937 → 8,102 units (−4,835); Lane B 5,238 → 6,621 (+1,383); roll-up 150 × 4,500 − 50 × 1,900 = 580,000 (loop-packet decomposition, illustrative).
- **%** — late arrivals 7-in-100 → 1-in-100 on Lane A (loop-packet example lane); lane accuracy 99% on 200 lanes (record).
- **Quality** — zero stockouts in the watch window (record); the rule is two-sided and written; the pad is gone.
- **Customer** — the production line did not stop on a released lane; the planners' pad stopped being a second, hidden policy.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	No statement of what each unit of buffer protected against; I defined the two terms and the service level each one buys.
Scope & influence	Planning, Finance, the data team and six sites; Finance signed lane by lane, planning owns the report.
Execution	Hand-recomputed two lanes, ran the normal-formula counterfactual, wrote the add-stock clause into the rule.
Impact	580K units / ~$1.5M out net, zero stockouts; Lane A 37% less stock with fewer late arrivals.
Process improvement	A written, two-sided release rule and a drift flag replaced the pad as the de-facto policy.
Complexity	Skewed tails, two lanes moving in opposite directions, and a service-level definition (cycle service vs fill rate) that has to be named.
Root cause	97.5% of the old formula's variance was a default σ_LT nobody measured — the root cause was the parameter, not the planners.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: lane × (d, σ_d, brochure LT, measured LT mean / σ / ranked p99) — Lane A and B; variance decomposition of the ERP formula: 97.5% lead-time part. Revealed: a guessed σ_LT dominated the buffer.
2 · Solution design	Options: keep the ERP form with a measured σ_LT · normal-fit p99 · ranked-p99 cover + demand cover. Chosen: two terms on the ranked p99 — it does not assume the lead-time tail is a bell curve. Release rule: two-sided, signed.
3 · Math & data logic	Math: 9,437 + 3,500 pad = 12,937; after = 500 × (34 − 21) + 2.33 × 150 × √21 = 6,500 + 1,602 = 8,102; Lane B +1,383; z-table for the demand term; sensitivity to the assumed σ_LT.
4 · Implementation	Implementation: data team per lane → hand-check two lanes → signed rule → lane-by-lane release with Finance signatures → watch report and drift flag handed to planning.
5 · Probe questions	Probe questions: independence and normality of the demand term · cycle-service vs fill-rate · why z = 2.33 · blanket-cut counterfactual · what happens on a thin lane — see Trap Drills.

PROBING — CHALLENGE Q&A TUNED TO A STATISTICIAN

> **Analytics — Your formula adds a demand term and a lead-time term. Are they independent?**
>
> The ERP form assumes demand per day and lead time are independent; I kept that for the demand term only, and it is 19.8% of the after-buffer (1,602 of 8,102). The lead-time term — 80.2% — is empirical, not a bell curve. A fifty-percent error in the demand term moves the total by about ±10%. If demand surges when transit is slow the two correlate and the true figure is higher; I would test that on the lane history before calling it closed.

> **Analytics — Why z = 2.33?**
>
> Because the signed service level was 99% and Φ(2.33) = 0.9901. The price of more is steep and nonlinear: the demand cover on Lane A is 1,131 at 95%, 1,602 at 99% and 2,124 at 99.9% — +522 units (32.6%) for the last nine-tenths of a point.

> **Analytics — Is 99% a fill rate or a cycle service level?**
>
> A cycle service level — the probability a replenishment cycle ends without a stockout, measured as lane accuracy against the p99 promise. Fill rate — the share of units shipped from stock — is a different, much higher number on the same buffer; I do not mix the two and I say which one I mean.

Analytics — Why release 4,835 units and not 8,009, which is what the textbook form with the measured σ would release?

Because the textbook form with the measured σ_LT = 4 gives 4,928 units after, i.e. a release of 8,009, and it plans to 30.3 days when 3.5% of shipments run past that. The extra 3,174 units is exactly the tail the bell curve cannot see — that is the S8a result paid for in units.

Shadow / closing — What would you do differently?

Put Lane B on the first page. The two-sided result — some lanes need more stock — was the part that earned Finance's trust, and it should have been the opening exhibit, not the second slide.

BUSINESS IMPACT MODEL — WHAT THE FORMULA REDESIGN MOVED (LANE A, LANE B, PROGRAM)

```
Safety stock = formula nobody could defend + a hand-added pad
+- Lead-time part = 97.5% of the variance, driven by an unmeasured sigma_LT = 8
+- Pad = 7 days x demand, uncounted, invisible to Finance
+- Fix: lead-time cover to ranked p99  +  demand cover over mean LT  ;  pad -> 0
+- Rule: release where measured spread/tail is inside the assumption; keep or add where it is not
+- Result: Lane A -4,835 units, Lane B +1,383 units, program 580K net, zero stockouts
```

DRIVER	BASELINE	AFTER	$ / UNITS	BASIS
Lane A buffer	12,937 units	8,102 units	−4,835 ≈ $12,523	loop-packet lane (illustrative); $2.59/unit derived
Lane B buffer	5,238 units	6,621 units	+1,383 (stock in)	loop-packet lane (illustrative)
Program net release	—	580,000 units	≈$1.5M	record (one-time)
Carry on the net release	—	—	≈$300,440/yr	modeled (20% all-in; benchmark 15–25%)

If challenged. *"Your $1.5M is a decomposition of two example lanes."* — "No: the $1.5M is the record — 580K units net, Finance-signed. The two lanes are how the rule behaves; the 150-and-50 split is the loop packet's illustration and I label it."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: why the old buffer was wrong (Lane A).

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	ERP demand part	$LT_{brochure} \times \sigma_d^2 = 18 \times 150^2 = 18 \times 22{,}500 = 405{,}000$	brochure 18 d; σ_d 150 (illustrative lane)	405,000
2	ERP lead-time part	$d^2 \times \sigma_{LT}^2 = 500^2 \times 8^2 = 250{,}000 \times 64 = 16{,}000{,}000$	σ_LT = 8 d — ERP default, never measured	16,000,000
3	Combine	$\sqrt{(405{,}000 + 16{,}000{,}000)} = \sqrt{16{,}405{,}000} = 4{,}050$	sum under the root	4,050
4	Service factor	$2.33 \times 4{,}050 = 9{,}437$	z = 2.33 (99%, toolkit label)	9,437 units
5	Variance share of the guess	$16{,}000{,}000 \div 16{,}405{,}000 = 97.5\%$	derived	97.5% unmeasured
6	Manual pad	$7 \text{ days} \times 500 \text{ units/day} = 3{,}500$	pad ≈ a week (planner habit; illustrative)	3,500 units
7	Total before	$9{,}437 + 3{,}500 = 12{,}937$ (= 25.9 days of demand)	rows 4 + 6	12,937 units

Table 2 — the fix, values substituted (Lane A, the counterfactual, Lane B).

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Lead-time cover	`d × (p99 − mean) = 500 × (34 − 21) = 6,500`	ranked p99 = 34 (S8a); mean 21	6,500 units
2	Demand cover	`z × σ_d × √LT_mean = 2.33 × 150 × √21 = 2.33 × 150 × 4.583 = 1,602`	σ_d = 150; mean LT 21	1,602 units
3	Pad	`plan holds 99% → no manual pad needed`	S8a	0
4	Total after	`6,500 + 1,602 = 8,102`	rows 1 + 2	8,102 units
5	Release, Lane A	`12,937 − 8,102 = 4,835 (37.4%)`	before − after	4,835 units
6	Textbook-form counterfactual	`2.33 × √(21 × 150² + 500² × 4²) = 2.33 × √4,472,500 = 4,928`	normal form, measured σ_LT = 4	4,928 units — plans to 30.3 d; serves 96.5%
7	Lane B before	`2.33 × √(24 × 80² + 200² × 8²) + 7 × 200 = 2.33 × √2,713,600 + 1,400 = 3,838 + 1,400 = 5,238`	d 200, σ_d 80, brochure 24, ERP σ_LT 8	5,238 units
8	Lane B after	`200 × (58 − 30) + 2.33 × 80 × √30 = 5,600 + 1,021 = 6,621`	measured mean 30, σ 11, p99 58	6,621 units (+1,383)
9	Blanket-cut counterfactual on B	`Lane A's cut 37.4% applied to B: 5,238 × (1 − 0.3737) ≈ 3,280; ÷ 6,621 = 49.5%`	illustrative counterfactual (a blanket rule)	leaves 49.5% of the measured need — short 3,341 units ≈ 16.7 days of demand

Table 3 — after: the metrics, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Lane A buffer	12,937 units (25.9 days of demand)	8,102 units (16.2 days)	loop-packet lane
2	Lane A late arrivals	7 in 100 (brochure + pad plan)	1 in 100 (ranked p99 plan)	S8a (28/400 → 4/400)
3	Lane B buffer	5,238 units	6,621 units	loop-packet lane
4	Program net safety stock	base	−580,000 units	record
5	Stockouts in the watch window	—	0	record
6	Owner of the buffer parameter	ERP default + planner habit	signed rule + drift flag, planning	design record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Lane A release value	`4,835 × $2.59 = $12,523`	$2.59 = $1.5M ÷ 580K (record pair)	≈$12,523 one-time
2	Lane A carry saved	`$12,523 × 20% = $2,505`	20% all-in carry (benchmark 15–25%)	≈$2,505/yr (overlaps S8a's pad carry — do not add)
3	Roll-up, units out	`150 lanes × 4,500 = 675,000`	loop-packet decomposition (illustrative)	675,000 units
4	Roll-up, units in	`50 lanes × 1,900 = 95,000`	loop-packet decomposition (illustrative)	95,000 units
5	Net	`675,000 − 95,000 = 580,000`	ties to the record	580,000 units

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
6	Net value	`580,000 × $2.59 = $1,502,200`	record pair	≈$1.5M one-time (record)
7	Carry on the net	`$1,502,200 × 20% = $300,440`	modeled	≈$300,440/yr (part of S8's ≈$0.6M)

SOLUTION FLOW — FORMULA, RULE, WATCH

```
ERP: PO lines + goods receipts --> S8a lane distribution (n, mean, sigma, ranked p99)
                                    |
                                    v
per lane:  LT cover = d x (p99 - mean_LT)     demand cover = z x sigma_d x sqrt(mean_LT)
                        \                       /
                         v                     v
                    buffer = LT cover + demand cover          (pad = 0)
                                    |
       measured spread/tail inside assumption AND lane accuracy 99%?
          yes -> release (Finance signs the lane)      no -> keep or raise buffer
                                    |
            refresh: p99 recomputed; accuracy falls -> decision re-opens (drift flag)
```

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
ERP σ_LT assumed (Lane A release)	σ = 6 → release 2,544	σ = 8 → release 4,835	σ = 10 → release 7,142
Service level (demand cover, Lane A)	95% → 1,131	99% → 1,602	99.9% → 2,124
Measured p99 on Lane A	30 d → lead-time cover 4,500	34 d → 6,500	38 d → 8,500
Carry rate on the 580K-unit release	15% → $225,330/yr	20% → $300,440/yr	25% → $375,550/yr

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The old buffer was a textbook formula fed an eight-day lead-time swing nobody had measured — ninety-seven percent of its variance was that guess — plus a week of hand-added pad. I replaced it with two terms a planner can read: cover to the measured ninety-ninth percentile, and demand cover. On the example lane that took twelve thousand nine hundred units to about eight thousand one hundred with late arrivals falling from seven in a hundred to one; on a lane with a fatter tail it added about fourteen hundred units, because the rule cuts both ways. The program record is five hundred eighty thousand units, about one and a half million dollars, released net with zero stockouts — the lane arithmetic is the loop packet's example and I label it."

If challenged — the full answers

The split is illustrative. "The 150-and-50 decomposition is how I explain the net; the record is 580K units, about $1.5M, zero stockouts." The fat-tail lanes — kept or raised? "The rule keeps or raises where the measured tail is fatter than the assumption; whether a given lane saw stock go in is a lane-level fact I would pull from the release log rather than assert." The normal form is simpler. "And it plans to 30.3 days where 3.5% of shipments run past — I would rather defend 4,835 units than 8,009."

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 **Tell me about a time you simplified something complicated.** INVENT AND SIMPLIFY

Open (first two sentences): Safety stock on our lanes was a formula nobody could defend plus a week of pad nobody could count. I replaced both with two terms a planner could read.

STAR (≈90 s): Situation: the ERP buffer was z times a root of two variances, with an eight-day lead-time swing nobody had measured — 98% of the formula was that guess — and planners added a week on top. Task: say what each unit of buffer was for. Action: I split it into lead-time cover to the measured p99 and demand cover over the mean lead time, set the pad to zero, and wrote the release rule. Result: Lane A went from 12,937 to 8,102 units with late arrivals falling from seven in a hundred to one, and the program released 580K units with zero stockouts.

Land this number: 12,937 → 8,102 units; 97.5% of the old variance was one unmeasured input. · **Pivot to NACF:** INV-B is the same two-variance structure; I would show a planner which term dominates before touching a number. · **Do NOT say:** Do not say I replaced the ERP; say I changed two parameters and the rule that governs them.

Q2 **Tell me about a time you held your position when it was the harder path.** HAVE BACKBONE; DISAGREE AND COMMIT

Open (first two sentences): The easy rule to sell was a release-only rule. I wrote one that could also add stock — and defended it.

STAR (≈90 s): Situation: the obvious design was to cut buffers and report the release. Task: write a rule that would survive an audit. Action: I wrote it two-sided — release where the measured spread was inside the assumption, keep or add where it was not — and showed the alternative on Lane B: a blanket cut of Lane A's size would leave 50% of what the measured tail needs. Finance and Planning signed it with the add-stock clause. Result: 580K net, zero stockouts, and the credibility to do the next release. VERIFY-WITH-RAJESH whether a specific person asked for a one-way cut.

Land this number: Lane B +1,383 units; blanket cut leaves 49.5% of need. · **Pivot to NACF:** In placement, the same discipline: a rule that can only move one way is a target, not a policy. · **Do NOT say:** Do not name or imply a person pushed back unless it is true; say "the easy design was."

Q3 **Tell me about a time you did more with less.** FRUGALITY

Open (first two sentences): Less stock and better service — on one lane, 37% fewer units and seven late shipments in a hundred became one.

STAR (≈90 s): Situation: working capital sat in buffers that were partly folklore. Task: take capital out without moving the service line. Action: replaced the pad with a measured p99 plan and kept the buffer wherever the tail demanded it. Result: Lane A 4,835 units out (≈$12,523) with fewer late arrivals; program 580K units, about $1.5M, zero stockouts.

Land this number: 580K units / ~$1.5M net, zero stockouts. · **Pivot to NACF:** WOC targets are the NACF equivalent — fewer weeks of cover only where the measured distribution supports it. · **Do NOT say:** Do not claim recurring savings as the headline; the $1.5M is one-time working capital.

Q4 **Tell me about something you owned beyond the delivery date.** OWNERSHIP

Open (first two sentences): A release is not done when the units leave the shelf. It is done when the rule runs without you.

STAR (≈90 s): Situation: a buffer cut that nobody re-checks decays into a new brochure. Task: make the release self-correcting. Action: drift flag per lane — p99 recomputed on refresh, the buffer decision re-opens if accuracy falls — and the stockout-and-expedite watch handed to planning as a standing report. Result: planning re-applies the rule on its own refresh; the watch showed zero stockouts.

Land this number: Zero stockouts in the watch window (record). · **Pivot to NACF:** NACF inbound needs the same drift flag on every lead-time and WOC parameter. · **Do NOT say:** Do not say I ran the watch; say planning owns it.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: "97.5% of the old formula's variance was a number nobody measured."

Headline metric: variance share 97.5%; 12,937 → 8,102 units

- Demand part 405,000 vs lead-time part 16,000,000.
- Two terms replace it: cover to p99 + demand cover.
- Textbook form with measured σ would release 8,009 — and serve 96.5%.
- Lane A −4,835, Lane B +1,383.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Have Backbone; Disagree and Commit

Open: "I wrote the rule so it could add stock — that is why Finance signed it."

Headline metric: blanket cut leaves 49.5% of Lane B's measured need

- A one-way cut is the easy sell and the wrong rule.
- Lane B is the exhibit.
- Signed with the add-stock clause.
- VERIFY-WITH-RAJESH the actual stakeholder dynamics.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Learn and Be Curious

Open: "The textbook form was my starting point; one skewed lane showed it understates the tail by 3.4 days."

Headline metric: normal 30.3 d vs ranked 34 d; 8,009 vs 4,835 release

- The cumulative definition came from a skewed lane.
- I would now show the demand-supply correlation test on lane history.
- Cycle service ≠ fill rate — name which one.
- VERIFY-WITH-RAJESH that this was my own starting point.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 "Your safety-stock formula assumes normality and independence. Why trust it?"

The numbers: Demand term = 1,602 of 8,102 = 19.8%; ±50% error → ±10% of the total; lead-time term empirical (ranked p99).

Say exactly: "Only the smaller term assumes a bell curve — about a fifth of the buffer; the lead-time term, four-fifths, is the measured ranked p99 with no distribution assumed, and a fifty-percent error in the demand term moves the total by about ten percent."

T2 "Is 99% a fill rate or a cycle service level, and what does 99.9% cost?"

The numbers: Demand cover at 99% = 1,602; at 99.9% = 2,124 (+522, +32.6%); z 2.33 → 3.09.

Say exactly: "Ninety-nine percent is a cycle service level — probability of no stockout per replenishment — not a fill rate; going to 99.9 adds about five hundred units on the demand term alone, a third more, and I would only buy it where a stockout costs more than that."

T3 "Why not just cut everything by the same percentage — it is simpler and Finance would like it?"

The numbers: Lane A cut 37.4%; applied to Lane B: 3,280 vs need 6,621 → 49.5%; short 3,341 units = 16.7 days of demand.

Say exactly: "A blanket cut of Lane A's size would leave Lane B with half the buffer its measured tail needs — about seventeen days of demand uncovered — which is a line-down waiting to happen; the two-sided rule is what let Finance sign the release."

S8c — Elementum — validation and back-test design: how I proved the p99 plan held out of sample and was not a quiet quarter — held-out quarter 99.1%, zero stockouts, and what that can and cannot show NEW

LPs this story serves here: Are Right, A Lot (primary — test my own decision out of time) · Insist on the Highest Standards (per-lane gate, no pooling) · Earn Trust (the limits of the test on the page) · Learn and Be Curious (shadow-mode the drift flags first)
Process family: Plan / Source seam — validation and monitoring design for a parameter change before capital moves

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Are Right, A Lot** — I did not trust in-sample 99%: I trained on twelve months, scored a later quarter the plan had never seen, and read the result against an interval — 99.1% [98.7%, 99.4%].
- **Insist on the Highest Standards** — The bar was per lane, 99%, before any buffer released — no pooling to hide a bad lane inside a good average.
- **Earn Trust** — I put what the test cannot show on the page — a lane has about thirteen shipments a quarter — so Finance signed knowing the limits.
- **Learn and Be Curious** — My own lesson is shadow-mode: run the drift flags against history before they go live.

Honesty label. Record: validated on a held-out quarter at 99.1% (loop packet, secondary source); zero stockouts in the watch window; Finance signed each lane; drift flag per lane. The held-out sample size (2,500 shipments = the master's illustrative ~10,000 a year ÷ 4), the old-plan late share (7%, the loop-packet example lane applied program-wide), the 150 releasing lanes and the 13-week window are illustrative inputs for the arithmetic — not the record. VERIFY-WITH-RAJESH (a) the length of the watch window, (b) which quarter was held out and whether it was a peak or a quiet quarter, (c) whether any contemporaneous control was run, (d) whether one held-out quarter or several were used.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. We had replaced brochure lead times with measured p99 plans and were about to take capital out of 150-odd lanes. A plan fitted to a year of history will look perfect on that year — that is not evidence. Task. Prove the plan would hold on shipments it had never seen, and that the result was not a quiet season. Action. I trained each lane on twelve months and scored the next quarter out of time. At program level that is about 2,500 shipments; 22 ran past the plan — 99.1% held, with a 95% interval of 98.7% to 99.4%, clearly better than the 96.5% a bell-curve plan delivers, and statistically indistinguishable from the 99% we promised. I scored the same quarter's shipments against the old brochure-plus-pad plan, so the season hits both plans equally. And I was explicit about the limit: on one lane a quarter is about thirteen shipments, so lane-level validation is the drift flag, not a quarterly test. Result. Primary metric: 99.1% on the held-out quarter against the 99% promise, then zero stockouts across the released lanes — record — with 580K units, about $1.5 million, released net.

20-SECOND VERSION NEW

I back-tested the p99 plan on a quarter it had never seen — 99.1%, interval 98.7 to 99.4 — scored the same shipments against the old plan so a quiet season could not flatter it, and kept a per-lane drift flag because a lane alone has too few shipments to validate.

NACF BRIDGE — ONE SENTENCE

For NACF placement and WOC targets I would want the same out-of-time test before any buffer comes out: train on the past, score a later quarter, and read the result as an interval, not a number.

SITUATION / TASK — THE PROBLEM SPACE

S — Elementum, six-site manufacturer. The measured p99 plans (S8a) and the formula (S8b) were ready; releasing 580K units of buffer was irreversible in practice, and the one thing a plan fitted to history will always do is look good on that history. The audience — Finance and Planning — would sign lane by lane only on evidence they could challenge.

T — Design the validation: an out-of-time test that could fail, an interval around the answer, a defense against "it was a quiet quarter," a live watch after release, and a drift flag that outlives me.

Baselines on the table: in-sample coverage 99% by construction (396 of 400 on the example lane) · no out-of-sample measurement existed · brochure-plus-pad plan late 7 in 100 on the example lane · no stockout metric tied to the release.

ACTION — FIVE GATED PHASES

- **Discovery** — named the failure mode — in-sample coverage is 99% by construction — and the second one, seasonality: a quarter that happens to be easy flatters any plan. Exit bar: Finance and Planning agree what result would make us stop.
- **Plan & Design** — train on twelve months so every season is inside the plan; hold out the next quarter; metric = arrival vs plan; score the same shipments against the old plan; pre-register the pass bar (99% promise, read as an interval) and the per-lane gate (99%, no pooling) before looking. Exit bar: bar and metric signed before the data is scored.
- **Build & Test** — scored the held-out quarter: 22 of 2,500 beyond plan, 99.1%; the paired score against the old plan on the same shipments; the power calculation. Exit bar: result reproduced by the data team; interval and power on the page.
- **Cutover & Go-Live** — release lane by lane under the signed rule; live watch — stockouts and expedites by lane — from day one; zero stockouts in the watch window. Exit bar: no lane released without its own accuracy ≥ 99%. VERIFY-WITH-RAJESH the watch window length.
- **Hypercare / Close** — drift flag per lane (p99 recomputed on refresh; decision re-opens if accuracy falls) and a standing report owned by planning; lesson recorded: shadow-mode the flags against history before they go live. Exit bar: planning re-runs the check on its own refresh.

Influence and ownership. I owned the validation design and the pass bars; the data team scored the lanes; Finance and Planning signed. The conflict was pace: releasing on the in-sample result would have been faster and indistinguishable on a slide — I held the release to the out-of-time test and wrote the per-lane gate so a good average could not hide a bad lane.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **%** — held-out quarter 99.1% against a 99% promise (loop packet; interval 98.7%–99.4% on an illustrative 2,500 shipments); lane accuracy 99% on 200 lanes (record).
- **#** — zero stockouts in the watch window (record); with 13 weeks × 150 releasing lanes = 1,950 lane-weeks that bounds the stockout rate below 0.15% per lane-week at 95% confidence (rule of three; window illustrative).
- **$** — the validation supports the 580K units / ~$1.5M release (record, one-time); it is not a separate saving. Cost of trusting the bell curve instead: ≈$79K/yr of expedites on the example inputs (modeled, a counterfactual, not a claim).
- **Quality** — a pre-registered pass bar, an interval, a paired replay, and a drift flag.
- **Customer** — no line stopped on a released lane.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	No definition of “the plan held”; I defined it as arrival vs plan, out of time, as an interval.
Scope & influence	Finance, planning and the data team; Finance’s lane-by-lane signature was the acceptance test.
Execution	Pre-registered the bar, ran the out-of-time score, paired replay, power check, live watch, drift flag.
Impact	The validation is what let 580K units / ~$1.5M come out with zero stockouts.
Process improvement	A standing out-of-time check and a drift flag per lane — the validation repeats on refresh.
Complexity	Small per-lane samples, seasonality, skew, a thin-lane tail — each named and bounded.
Root cause	The risk was not the plan; it was believing an in-sample number — validation addresses the belief.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: shipments per lane, 12 months training + the next quarter held out; arrival vs plan; same shipments scored against the old plan. Revealed: 22/2,500 beyond plan, 99.1%, interval 98.7–99.4%; the old plan late ≈ 7%.
2 · Solution design	Options: in-sample fit · hold-out quarter · rolling-origin back-test · live shadow. Chosen: out-of-time hold-out plus paired replay plus live watch plus per-lane drift flag; rolling-origin is the upgrade.
3 · Math & data logic	Math: Wilson interval · z vs 96.5% = 7.13, vs 99% = 0.60 · power 99% at a true 98.0% · expected out-of-sample coverage of the ranked p99 = r/(n+1) = 98.75% ± 0.55 · rule of three on zero stockouts.
4 · Implementation	Implementation: pre-registered bars → score → paired replay → lane-by-lane signature → watch → drift flag → standing report.
5 · Probe questions	Probe questions: interval and power · seasonality · per-lane n · what zero stockouts proves · what a thin lane does — see Trap Drills.

PROBING — CHALLENGE Q&A TUNED TO A STATISTICIAN

> **Analytics — A ranked p99 from 400 shipments, tested on new data. What coverage should you expect?**
>
> Not 99%. The r-th of n ranked observations covers, on average, r/(n+1) of the future distribution: 396/401 = 98.75% with a standard deviation of 0.55 points. So 99.1% is well inside what the method predicts — a 29% chance of a coverage that high or higher. On a thin lane the maximum of 50 shipments covers 50/51 = 98.04% ± 1.92 points and has a 7.7% chance of covering less than 95%.

> **Analytics — Is 99.1% different from 99%?**
>
> No. 22 late of 2,500; standard error under 99% is 0.20 points, z = 0.60. It is different from the bell-curve plan’s 96.5% (z = 7.13). The test is built to detect a shortfall, not a surplus: it rejects 99% only if coverage falls below 98.67%.

> **Analytics — How much power did the test have?**
>
> At 2,500 shipments: 99% power against a true coverage of 98.0%, 76% against 98.5%; the smallest shortfall it detects with 80% power is about 0.53 points (98.47% true coverage). So it is good at catching a plan that is clearly wrong and blind to a plan that is half a point short — which is why the drift flag exists.

> **Analytics — How do you know it was not seasonality?**
>
> Three defenses and one limit. The training window held every season; the metric is arrival against plan, not volume; and I scored the same held-out shipments against the old plan, so a quiet quarter flatters both plans equally — on the illustrative program numbers the old plan was late on 175 shipments and the new on 22. The limit: one held-out quarter is one draw; the upgrade is a rolling-origin back-test with a peak quarter in it. VERIFY-WITH-RAJESH which quarter was held out.

Analytics — Can you validate each lane the same way?

No. An average lane has about 50 shipments a year, roughly 13 a quarter; a lane whose true coverage is exactly 99% shows zero late shipments 87.8% of the time and one-or-more 12.2% of the time, so one late shipment in a quarter tells me nothing. Per-lane protection is the drift flag on rolling history; the quarterly test is pooled.

Shadow / closing — What would you do differently?

Shadow-mode the drift flags against history before they go live — replay a past year and count the false alarms — and start the rolling-origin back-test on day one so the peak quarter is in it.

BUSINESS IMPACT MODEL — WHAT VALIDATION PROTECTS

```
Release 580K units on a p99 plan fitted to history
+- Risk 1: in-sample 99% is by construction -> need out-of-time coverage
+- Risk 2: a quiet quarter flatters the plan -> same shipments scored vs old plan
+- Risk 3: a lane has ~13 shipments/quarter -> pooled test + per-lane drift flag
+- Risk 4: zero stockouts proves little in a short window -> rule-of-three bound
+- Payoff: Finance signs lane by lane; capital leaves; watch confirms
```

DRIVER	WITHOUT VALIDATION	WITH VALIDATION	$ / %	BASIS
Coverage evidence	in-sample 99% by construction	out-of-time 99.1% [98.7%, 99.4%]	—	loop packet (99.1%); n illustrative
Cost of the bell-curve plan	late 3.5% = 87.5 shipments / quarter	late 0.9% = 22 / quarter	≈$78,600/yr expedites avoided	modeled counterfactual (10% × $3,000)
Release supported	—	580K units / ~$1.5M	record	record (one-time)

If challenged. *"So validation saved $79K?"* — "No: that is what trusting the bell curve would have cost on illustrative inputs; the validation is the gate that made the $1.5M release signable."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: why in-sample evidence is not evidence.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	In-sample coverage	`rank ⌈0.99 × 400⌉ = 396 → 396/400 = 99.0%`	ranked p99 on the example lane (S8a)	99.0% by construction
2	Expected out-of-sample coverage	`r/(n+1) = 396/401 = 98.75%`	order-statistic (Beta(396, 5)) mean	98.75%
3	Spread of that coverage	`√[r(n+1−r)/((n+1)²(n+2))] = √[396 × 5/(401² × 402)] = 0.55 points`	Beta(396, 5) standard deviation	± 0.55 pts
4	Thin lane (n = 50)	`max covers 50/51 = 98.04% ± 1.92; P(coverage < 95%) = 0.95⁵⁰ = 7.7%`	maximum of 50 shipments as the "p99"	7.7% chance of a bad miss
5	Per-lane quarterly sample	`≈ 50/4 ≈ 13 shipments; P(0 late | 99%) = 0.99¹³ = 87.8%`	master's illustrative volume (≈50/lane/yr)	87.8% — one quarter says nothing per lane
6	The seasonality risk	`one held-out quarter = one draw; a quiet quarter flatters any plan`	design risk	needs a paired score and a peak quarter

Table 2 — the fix: the out-of-time test, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Held-out result	`2,478 of 2,500 within plan = 99.12% (22 late)`	loop packet 99.1%; n = 2,500 illustrative	99.12%

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	Interval	`Wilson 95% = [98.67%, 99.42%]`	binomial, n = 2,500	[98.67%, 99.42%]
3	Standard error under 99%	`√(0.99 × 0.01 ÷ 2,500) = 0.199 pts; reject if observed < 99% − 1.645 × 0.199 = 98.67%`	one-sided 5% test	cut-off 98.67%
4	vs the promise (99%)	`z = (99.12 − 99.00) ÷ 0.199 = 0.60`	H0: coverage = 99%	not distinguishable from 99%
5	vs the bell-curve plan (96.5%)	`z = (99.12 − 96.50) ÷ √(0.965 × 0.035 ÷ 2,500) = 7.13`	H0: coverage = 96.5%	clearly better
6	Power	`P(observed < 98.67% | true 98.0%) = 99%; true 98.5% → 76%; 80%-power shortfall ≈ 0.53 pts`	normal approximation, n = 2,500	blind below ≈ 0.5 pt shortfall
7	Paired replay (same shipments)	`old plan late 175 (7%); new late 22; discordant: old-only 157, new-only 4; McNemar χ² = (157 − 4)² ÷ (157 + 4) = 145.4`	old-plan share 7% = example lane applied program-wide; the 18 / 4 split illustrative	season hits both plans equally
8	Zero stockouts	`13 wks × 150 lanes = 1,950 lane-weeks; upper bound = 3 ÷ 1,950 = 0.154% per lane-week (95%)`	rule of three; window illustrative	< 0.15% / lane-week

Table 3 — after: the metrics, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Out-of-time coverage	none (in-sample 99% only)	99.1% [98.7%, 99.4%]	loop packet; interval illustrative
2	Late vs plan, same shipments	175 of 2,500 (7%) old plan	22 of 2,500 (0.9%)	illustrative program-level
3	Stockouts after release	unmeasured	0 in the watch window	record
4	Per-lane protection	none	drift flag, rolling history	design record
5	Pass bar	not defined	99% per lane, pre-signed; no pooling	record rule

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Late at the bell-curve plan	`3.5% × 2,500 = 87.5 shipments / quarter`	96.5% coverage (S8a)	87.5 / quarter
2	Late at the ranked plan	`22 / quarter`	row above	22 / quarter
3	Excess late	`87.5 − 22 = 65.5`	difference	65.5 / quarter
4	Expedite cost	`65.5 × 10% × $3,000 = $19,650 / quarter`	10% share and $3K premium illustrative (master)	$19,650
5	Annualized (counterfactual)	`$19,650 × 4 = $78,600`	modeled — a cost avoided by not using the bell curve, not a savings claim	≈$78,600/yr
6	The release it supports	`580,000 × $2.59 = $1,502,200`	record pair	≈$1.5M one-time (record)

SOLUTION FLOW — THE VALIDATION PIPELINE

```
shipments per lane (PO schedule vs goods receipt)
       | months 1-12: fit ranked p99          months 13-15: held out, never seen
       v                                                  v
  plan statistic per lane  ------------------->  arrival vs plan, scored out of time
                                                          |  same shipments scored vs old plan (paired)
                                                          v
      pooled coverage + Wilson interval + power  -> pass bar signed in advance?
                        | yes: release lane by lane (Finance signs)
                        v
      live watch: stockouts / expedites by lane  ->  rule-of-three bound
                        v
      drift flag per lane (rolling), standing report owned by planning
```

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Held-out shipments	1,000 → Wilson wider (about ±0.6 pt)	2,500 → [98.7%, 99.4%]	10,000 → about ±0.2 pt
Late shipments observed (of 2,500)	12 → 99.5%	22 → 99.1%	40 → 98.4%
Watch window × releasing lanes	8 wks × 150 → bound 0.25%	13 × 150 → bound 0.154%	26 × 150 → bound 0.077%
Expedite premium on the bell-curve counterfactual	$1,500 → $39,300/yr	$3,000 → $78,600/yr	$5,000 → $131,000/yr

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

“A plan fitted to a year of history looks perfect on that year, so I held out a quarter the plan had never seen: ninety-nine point one percent held, with an interval of about ninety-eight point seven to ninety-nine point four, against ninety-six and a half for a bell-curve plan. Same shipments scored against the old plan, so a quiet season could not flatter it. After release, zero stockouts across the released lanes — which on a thirteen-week window of about two thousand lane-weeks bounds the rate under about a sixth of a percent. What the test cannot show is any single lane — a lane has about thirteen shipments a quarter — so each lane carries a drift flag. The record is five hundred eighty thousand units, about one and a half million dollars, released with zero stockouts; the interval, power and paired replay are my design arithmetic on illustrative volumes.”

If challenged — the full answers

One quarter is one draw. “Agreed — I would have preferred a rolling origin with a peak quarter, and I say so; the paired score and the live watch are the other two legs.” Zero stockouts proves little. “It bounds the rate: zero in about two thousand lane-weeks is under 0.15% per lane-week at 95% — not zero risk, a number.” The numbers are illustrative. “The 99.1% and the zero stockouts are the record; the volumes behind the intervals are my illustrative inputs and I mark them.”

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 **Tell me about a time you tested your own decision and it could have failed.** ARE RIGHT, A LOT

Open (first two sentences): I was about to release 580K units on a plan fitted to history, and a plan fitted to history always looks perfect on that history. So I held a quarter out.

STAR (≈90 s): Situation: p99 plans per lane, buffer about to come out. Task: prove the plan held on shipments it had never seen. Action: trained on twelve months, scored the next quarter — 22 of 2,500 beyond plan, 99.1%, interval 98.7 to 99.4 — and scored the same shipments against the old plan so a quiet season could not flatter it. I wrote the pass bar before scoring. Result: Finance signed lane by lane; zero stockouts after release.

Land this number: 99.1% out of time vs 96.5% for a bell-curve plan. · **Pivot to NACF:** Before any NACF placement or WOC target change, the same out-of-time test on a held-back quarter. · **Do NOT say:** Do not say “proved”; say “held out of time, with an interval.”

Q2 **Tell me about a time you set a higher bar than you were asked to.** INSIST ON THE HIGHEST STANDARDS

Open (first two sentences): The bar was 99% per lane before any buffer released — not 99% on average. Below it, no release, written into the act before anyone asked.

STAR (≈90 s): Situation: a pooled 99% can hide a bad lane inside a good average. Task: a release rule that could not be gamed by pooling. Action: per-lane accuracy bar, signed in advance; lanes under it did not release; the drift flag re-opens the decision if accuracy falls. Result: 580K units out net, zero stockouts, and no lane released on an average.

Land this number: 99% per lane across 200 lanes; 580K units / ~$1.5M. · **Pivot to NACF:** A placement model can pass on region average and fail one FC-region pair — I gate per cell. · **Do NOT say:** Do not say the bar was Finance’s; it was written into the act as my design.

Q3 **Tell me about a time you had to earn trust from a skeptical partner.** EARN TRUST

Open (first two sentences): Finance signed lane by lane because I put what the test could not show on the same page as what it could.

STAR (≈90 s): Situation: Finance was being asked to sign a release on a statistical argument. Task: make the argument checkable. Action: the interval beside the number, the power beside the interval, and the limit beside the power — an average lane has about thirteen shipments a quarter, so per-lane validation is the drift flag, not the test. Result: Finance signed each lane and the watch confirmed zero stockouts.

Land this number: 99.1% [98.7, 99.4]; per-lane n ≈ 13. · **Pivot to NACF:** Amazon’s bar-raising partners trust the analyst who names the limit first. · **Do NOT say:** Do not oversell the 2,500; it is an illustrative program-level n.

Q4 **Tell me about something you learned that changed how you work.** LEARN AND BE CURIOUS

Open (first two sentences): What I learned is to shadow-mode the drift flags against history before they go live.

STAR (≈90 s): Situation: the drift flag — rolling p99 above plan by 2 days, σ above 1.25 times the base, or four late per hundred — looked right on a whiteboard. Task: know its false-alarm rate before planners had to answer it. Action (the lesson): replay a past year and count how often the flag would have fired. Result: the flag goes live with a known alert rate; the same replay is how I would tune any threshold now.

Land this number: Rule of three: 0 in 1,950 lane-weeks → < 0.154% per lane-week. · **Pivot to NACF:** In NACF, a new inbound alert would be replayed against a prior peak before it pages anybody. · **Do NOT say:** Do not say the replay was done unless it was — say “what I would now do first.” VERIFY-WITH-RAJESH

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Are Right, A Lot

Open: "A plan fitted to a year of history looks perfect on that year."

Headline metric: 99.1% out of time, [98.7, 99.4]; expected 98.75% ± 0.55

- Hold-out quarter, never seen.
- z vs 96.5% = 7.13; vs 99% = 0.60.
- Power 99% at a true 98.0%.
- Paired replay on the same shipments.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Insist on the Highest Standards

Open: "I would not release on a pooled average."

Headline metric: 99% per lane, no pooling

- The bar was signed before scoring.
- Lanes under the bar did not release.
- The drift flag re-opens the decision.
- VERIFY-WITH-RAJESH any pressure to release early.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Learn and Be Curious

Open: "My test could not tell me anything about a single lane."

Headline metric: per-lane quarterly n ≈ 13; P(0 late | 99%) = 87.8%

- One quarter is one draw — rolling origin is the upgrade.
- Zero stockouts bounds the rate; it does not prove zero.
- Shadow-mode the drift flags before go-live.
- VERIFY-WITH-RAJESH whether a peak quarter was ever held out.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 "99.1% — is that different from 99%? What was your power?"

The numbers: n = 2,500, 22 late; Wilson [98.67%, 99.42%]; z vs 99% = 0.60; power 99% at 98.0%, 76% at 98.5%; blind below ≈ 0.5 pt.

Say exactly: "It is not different from 99 — the interval is 98.7 to 99.4, z is 0.6 — and it is clearly different from the bell-curve plan at 96.5. The test catches a coverage of 98 percent about 99 times in a hundred and a half-point shortfall about three in four, so I do not read it as precise; that is why the drift flag exists."

T2 "How do you know it was not just an easy quarter?"

The numbers: Same 2,500 shipments: old plan late 175 (7%), new plan 22; McNemar $\chi^2 = 145.4$; training window 12 months, all seasons.

Say exactly: "I scored the same shipments against the old plan, so an easy quarter flatters both plans equally — the old plan was late on about a hundred and seventy-five, the new on twenty-two. One quarter is still one draw, and I would add a rolling origin with a peak quarter in it."

T3 "Zero stockouts — what does that prove?"

The numbers: 13 weeks × 150 lanes = 1,950 lane-weeks; upper bound 3 ÷ 1,950 = 0.154% per lane-week (95%); per-lane quarter n ≈ 13 → P(0 late | 99%) = 87.8%.

Say exactly: "It bounds the rate, it does not prove zero: no events in about two thousand lane-weeks puts the stockout rate under a sixth of a percent per lane-week at ninety-five percent confidence. And it says nothing about any one lane — at thirteen shipments a quarter a lane at exactly ninety-nine percent shows no late shipment 88 percent of the time."

S9 — Elementum — 1,000 routes asked, 597 built first: spend Pareto, a ≥95% valid-ASN gate held twice, completeness 60% → 95%, freight spend −10% ≈ $2M in six months NEW

LPs this story serves here: Have Backbone; Disagree and Commit (primary — re-scoped an executive's 1,000-in-two-months ask) · Insist on the Highest Standards (held a cohort twice) · Dive Deep (Pareto, lane scoring, EDI survey) · Invent and Simplify (the standard was the data, not the transport)
Process family: Source / Deliver seam — inbound visibility onboarding at scale (lane scoring → standard → gate → scorecard)

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Have Backbone; Disagree and Commit** — I took the 1,000-lanes-in-two-months ask two levels up with the spend curve and replaced it with 597 lanes = 80% of spend first — then committed fully.
- **Insist on the Highest Standards** — One cohort failed the ≥95% valid-ASN gate twice on a timestamp format; I held it both times — automating dirty data manufactures confident wrong answers.
- **Dive Deep** — I built the spend Pareto and lane scoring (spend × criticality × mode × volume) and the carrier EDI-capability survey before taking a scope to the executive.
- **Invent and Simplify** — The onboarding standard was the data set plus an XML/portal floor — "the standard was the data, not the transport" — and a parameterized harness replaced bespoke per-carrier maps.

Honesty label. I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor; engineers built the maps and harness to the standard; Transportation owned the carrier relationships.

60-SECOND PITCH

"**Situation.** The Global Head of Supply Chain wanted inbound visibility on 1,000 lanes in two months; shipment-data completeness was 60% and high-value lanes were blind. **Task.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor. **Action.** Discovery: spend Pareto — 597 lanes = 80% of freight spend — pre-validated with Transportation and Procurement. Plan & Design: X12 850/855/856/940/945/214/997 with an XML/portal floor; cohort gate ≥95% valid ASN. Build & Test: mapping-and-test harness; scorecards per lane. Cutover: 597 live in month one; one cohort failed twice on a timestamp format — I held it. Hypercare: completeness scorecards to 95%; ~200 next wave behind the same gate. Lesson: automating dirty data manufactures confident wrong answers — the gate is the product. **Result.** Primary metric: inbound shipment-data completeness — **60% → 95%**. $ impact: freight spend −10% ≈ $2M in six months (record); my model splits it into routing-guide compliance, mode discipline and expedites avoided, and adds ≈$0.4M/yr of modeled document-handling labor on top; 597 lanes = 80% of spend live in month one; adoption 8×."

20-SECOND VERSION NEW

A 1,000-lanes-in-two-months ask became 597 lanes carrying 80% of spend first behind a ≥95% valid-ASN gate I held twice: completeness 60% → 95%, freight spend −10% ≈ $2M in six months (record, not annualized), adoption 8×.

NACF BRIDGE — ONE SENTENCE

Inbound ASN completeness and validity are the first dependency of any NACF placement or WOC model — I would score vendors by volume Pareto and gate activation on valid-ASN rate, not on a date.

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — The customer's Global Head of Supply Chain wanted end-to-end inbound visibility on 1,000 global lanes in two months — scope, schedule and quality all fixed. Lanes differed by orders of magnitude in spend, volume and

criticality; carriers differed in EDI capability; completeness was 60% and high-value lanes were effectively blind.

T — Own the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor.

The goal: visibility that was actually trustworthy, at the spend that mattered first. The risk of taking the ask literally: automating dirty data manufactures confident wrong answers — 1,000 lanes at 60% completeness in two months would have shipped exactly that — and without a floor for carriers without EDI, the long tail gets priced out of the standard.

The metrics on the table, with their baselines: shipment-data completeness — 60%; high-value lane visibility — effectively blind, target 100% on the lanes carrying the spend; the activation standard — none existed, target a ≥95% valid-ASN cohort gate a lane had to clear before going live.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases:

- **Discovery** — built the spend Pareto and lane scoring model (spend × criticality × mode × volume) and the carrier EDI-capability survey; pre-validated the 597 / ~200 / remainder scope with Transportation and Procurement before taking it up. *Exit bar:* lane owners cannot contradict the cut; the executive accepts the phased plan in spend share, not lane count.
- **Plan & Design** — wrote the onboarding standard (document set + XML/portal floor — "the standard was the data, not the transport"), the ≥95% valid-ASN cohort gate, the per-lane scorecard spec (events received / expected; 997 rate; tracking-number joinability), the harness design. *Exit bar:* standard signed by Transportation; gate published to carriers before cohort 1; harness templates exist.
- **Build & Test** — held engineers to the harness (per-partner mapping templates, test scripts, DQ checklist), test files per carrier, a parallel run per lane, scorecards live. *Exit bar:* each cohort's lanes clear ≥95% valid ASN in test before activation.
- **Cutover & Go-Live (per cohort)** — activated cohorts only off the published gate; published the scorecards; recorded the hold decision when a cohort failed. *Exit bar:* cohort active only after the gate; a failing cohort held — twice, on a timestamp format.
- **Hypercare / Close** — weekly completeness scorecards; harness reused for later waves; method written up as the sourcing SOP (spend file in → phased plan out). *Exit bar:* completeness 60% → 95%; 100% visibility on high-value lanes; remaining waves behind the same gate.

The scope analysis and the standard were mine end to end — the spend Pareto and lane scoring model, the carrier EDI-capability survey, the onboarding standard, the gate, the scorecard spec and the hold decisions; engineers built the mapping templates, test scripts and harness, and Transportation held the carrier relationships and accepted the holds. The contribution: pre-validating the 597 cut with Transportation and Procurement so no lane owner could contradict it in the leadership meeting, and publishing the ≥95% gate to carriers before cohort one — a gate that only holds in private is a suggestion.

The obstacles: carriers unable to produce valid ASNs — answered with the XML/portal floor (“the standard was the data, not the transport”) and a hold rule; a cohort that failed the gate twice on a timestamp format — held both times, fixed, then activated; bespoke maps per partner — avoided by the parameterized harness. Priorities ran on spend share, not lane count — 597 lanes carrying 80% of spend first, ~200 next, the remainder after, each behind the same gate; buy-in came from the pre-validation, then the executive accepted the phased plan off the curve. The decision challenged was the 1,000-in-two-months ask itself — taken two levels up with the spend curve, deferring ~400 lanes carrying ~20% of spend, priced — and I refused activating any cohort that failed the gate.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: completeness **60%** → **95%**; 597 lanes = 80% of spend live in month one; freight −10% ≈ $2M in six months (record); adoption 8×.

- **$** — Freight spend −10% ≈ $2M in six months on the onboarded lanes (record); my three-way split (routing-guide compliance, mode discipline, expedites and detention) decomposes the record, never adds to it; ≈$0.4M/yr modeled document-handling labor on top.

- # — 597 of 1,000 lanes = 80% of freight spend live in month one; ~200 the next wave; the remainder behind the same gate (record).
- **%** — Completeness 60% → 95%; the ≥95% valid-ASN cohort gate — held twice on a cohort that failed on a timestamp format (record).
- **Time** — 597 lanes live in month one; the record $2M landed within six months (record).
- **Customer** — 100% visibility on high-value lanes; small carriers kept inside the standard by the XML/portal floor instead of being priced out (record).
- **Quality** — The gate is the product — automating dirty data manufactures confident wrong answers; adoption 8×; the method written up as the sourcing SOP (record).

Against the goal stated above: completeness 60 → 95%; 100% visibility on high-value lanes; 597 lanes = 80% of spend live in month one at ≥95% valid ASN, the remaining waves behind the same gate — with freight spend −10% ≈ $2M in six months funding the next wave. The trade was explicit and priced: ~400 lanes carrying ~20% of spend deferred so the 597 could go live at the highest data quality — and quality was never traded: a lane that slipped below its scorecard went back to “fix,” not “live.”

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	1,000 lanes in two months with scope, schedule and quality all fixed and no onboarding standard; I defined the standard and the gate.
Scope & influence	An executive two levels above my sponsor; Transportation, Procurement, carriers and engineers — none reporting to me.
Execution	597 lanes live in month one; one cohort held twice; scorecards per lane; harness parameterized for later waves.
Impact	Completeness 60% → 95%; freight spend −10% ≈ $2M in six months (record); adoption 8×.
Process improvement	The method is written up as the sourcing SOP: spend file in → phased plan out.
Complexity	Lanes differing by orders of magnitude in spend and criticality; carriers without EDI; per-partner formats.
Root cause	The root cause of blind high-value lanes was no onboarding standard and no gate — not a shortage of integrations.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: annual freight spend per lane, criticality, mode, volume; carrier EDI-capability survey; shipment-event completeness 60%. Revealed: 597 of 1,000 lanes carry 80% of spend; high-value lanes effectively blind.
2 · Solution design	Options: 1,000 lanes in two months · 597 first behind a gate · everything via EDI only. Chosen: Pareto-first, XML/portal floor, ≥95% valid-ASN cohort gate, scorecard per lane.
3 · Math & data logic	Math: S(597) = 80% (record); completeness 480/800 = 60%; implied spend base $2M ÷ 10% = $20M; document labor 90,000 × 6 min × $45/h ≈ $0.4M; gate lower bounds, grain test and control counterfactual below.
4 · Implementation	Implementation: Pareto pre-validated with Transportation and Procurement → standard and gate published to carriers → harness → cohorts activate only off the gate → weekly scorecards → SOP.
5 · Probe questions	Probe questions: is the $2M causal · what does a ≥95% gate mean with n lanes · event vs shipment grain · why 597 — see Trap Drills.

PROBING — CHALLENGE Q&A (MASTER RECORD) PLUS THE STATISTICAL QUESTIONS

***Challenge* — Why did you choose this story to illustrate Have Backbone; Disagree & Commit?**

Because I took the re-scope — 597 lanes first, not 1,000 — two levels up to the Global Head of Supply Chain with the spend curve and the lane owners' validation attached, then committed to delivering the remainder behind the same gate. And the gate held in public: one cohort failed twice on a timestamp format and I held it both times.

***Probing* — Why did you choose to focus on these results? What other results were important?**

Completeness 60 → 95% is the record the whole program was for, and freight −10% ≈ $2M in six months is what the executive funded the next wave on. The 597 = 80% of spend live in month one is the proof the re-scope was right.

***Probing* — You mentioned dollars — what percentage change is that year over year?**

Not YoY — the $2M is the record as measured: minus ten percent on the onboarded lanes' spend in six months, kept as stated rather than doubled into a run-rate. My three-way split of it decomposes the record, never adds to it.

***Probing* — I'm concerned about your split of the $2M — tell me more.**

The split is a judgment on how the ten points divide; to firm it I would tag every invoice line on the onboarded lanes by cause — off-guide carrier, mode change, expedite — for the six months and let the ledger set the shares.

***Challenge* — What were the lessons learned? What would you have done differently?**

Automating dirty data manufactures confident wrong answers — the gate is the product. Differently: the harness from cohort one, not cohort two.

***Challenge* — How would you implement this at Amazon?**

Vendor and carrier onboarding at scale is a Pareto plus a gate — spend share first, a published data-quality bar a partner must clear before activation, and a floor (portal/templates) so the long tail is kept inside the standard rather than priced out of it.

***Challenge* — How did these results compare to your actual goals?**

The ask was 1,000 lanes in two months; the delivery was 597 lanes = 80% of spend live in month one at ≥95% valid ASN, completeness 60 → 95%, the remaining waves behind the same gate — and the executive accepted the phased plan in spend share, not lane count.

PHASE DEEP-PROBE BANK — Q&A (13). *Answer the phase asked, then stop.*

Discovery — How did you know discovery was done?

When the spend Pareto was pre-validated by Transportation and Procurement and the scope proposal — 597 first, ~200 next, remainder after — could not be contradicted by any lane owner in the leadership meeting.

Who signed the scope?

The Global Head of Supply Chain accepted it after seeing the curve; the lane owners had already validated it. I priced the deferral in spend share, not lane count.

Plan & Design — What was in your risk register at design sign-off?

Carriers unable to produce valid ASNs, the scope being contradicted in the room, and bespoke maps per partner. Mitigations: XML/portal floor plus a hold rule, pre-validation, and a parameterized harness.

What was the entry bar for a cohort?

Standard and gate published to its carriers, mapping templates instantiated, test files in hand, a scorecard per lane. Then ≥95% valid ASN in test before activation.

How did you set the standard?

850/855/856/940/945/214/997 on X12, with XML or portal templates as the floor for carriers without EDI. The standard was the data, not the transport.

Build & Test — What was your definition of done for a lane?

Valid-ASN rate ≥95%, 997 acknowledgments flowing, tracking numbers joinable, completeness on its scorecard. Below that, the lane was fixed, not automated.

Describe your bug-bash equivalent.

The cohort test day: every carrier's file through the harness in one sitting; defects classified as format, timestamp, missing segment or join key, each with an owner. That is where the timestamp defect surfaced.

Cutover — Which cohort did you hold and how did the business take it?

One cohort failed twice on a carrier timestamp format; I held it both times. Transportation accepted because the 95% gate was published before cohort one — a gate that only holds in private is a suggestion.

What did each wave deliver?

Wave 1: 597 lanes, 80% of spend, live in month one. Wave 2: ~200 lanes on a harness that made the work mostly configuration. Wave 3: the tail, including non-EDI carriers on templates.

Hypercare — How did you know you were done? What did you watch?

Completeness 60% → 95%, 100% visibility on high-value lanes, the harness reused without me, the method adopted as the sourcing SOP. Watched weekly: per-lane completeness, the 997 rate, tracking-number joinability. A lane that slipped below its scorecard went back to "fix," not "live."

Cross-phase — Where did you trade scope for date? What did you escalate?

Deferred ~400 lanes carrying ~20% of spend so the 597 could go live at the highest data quality in month one — priced, and delivered later behind the same gate. Escalated the re-scope to the Global Head of Supply Chain — two levels up — with the spend curve and the lane owners' validation attached. Refused: activating a cohort that failed the gate.

Did you deliver the remaining 400? · Do differently?

Yes — later waves, same gate; the first 597 carried the spend and the decision. Differently: the harness from cohort one, not cohort two.

Analytics — What does “completeness 95%” count — events or shipments?

The record is shipment-data completeness 60% → 95%; my illustrative arithmetic counts four milestones per shipment (480 of 800 received = 60%). At the event grain 95% means 5% of milestones are missing; at the shipment grain, with independent misses, only 0.95^4 = 81.5% of shipments would be fully complete and the 60% start would be 0.6^4 = 13.0%. Misses cluster by carrier, so the truth sits between the union bound (80% at the 95% start) and 95%.
VERIFY-WITH-RAJESH which grain the 60% and 95% were measured at.

Analytics — A ≥95% valid-ASN gate — what does it mean with a finite test sample?

The record is the gate: the cohort must clear ≥95% valid ASN in test. The cohort sizes are not in the record and I do not quote them. As arithmetic: 38 valid of 40 is 95% observed but its one-sided 95% lower bound is 85.1%; 190 of 200 gives 91.7%; 194 of 200 gives 94.2%. To demonstrate ≥95% rather than observe it you need 196 of 200 (98%) at n = 200 (lower bound 95.5%); at n = 40 it cannot be demonstrated even at 40 of 40 (92.8%). I would add that bound to the scorecard. VERIFY-WITH-RAJESH test sizes and whether the bar was an observed or a bounded rate.

Analytics — How do you know the −10% freight spend came from visibility and not the market?

The record is −10% ≈ $2M on the onboarded lanes in six months; it carries no control. The deferred lanes are a free comparison group: about $5M of spend sits on the ~400 lanes that were not onboarded. If the deferred lanes' spend moved −2% on the market alone, the visibility effect is (−10% − (−2%)) × $20M = $1.6M; at −5% it is $1.0M. I would pull that before I defend a causal claim. VERIFY-WITH-RAJESH whether such a comparison was run.

STATISTICAL SPINE — PARETO, GRAIN, GATE AND CONTROL (EXTRA TABLES)

The record: 597 lanes = 80% of spend; completeness 60% → 95%; gate ≥95% valid ASN; freight −10% ≈ $2M in six months. The rows below are my modeled or illustrative arithmetic for a statistician's questions — labeled row by row. Cohort sizes and the number of cohorts are not in the record and are not quoted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Pareto fit through the record point	`S(k) = (k ÷ 1,000)^a with S(597) = 0.80 → a = ln 0.80 ÷ ln 0.597 = 0.4326`	one modeled curve through the one record point	a = 0.4326
2	Spend share at 300 lanes	`(0.300)^0.4326 = 59.4%`	modeled	59.4%
3	Wave 2 (~200 lanes, to 797)	`S(797) − 0.80 = 90.7% − 80.0% = 10.7%`	modeled; ~200 next wave is record	+10.7% of spend
4	Remainder (~200 lanes)	`1 − S(797) = 9.3%`	modeled; ~400 lanes ≈ 20% deferred is record	9.3% of spend
5	Marginal lane at the cut	`dS/dk = a × S ÷ k = 0.4326 × 0.80 ÷ 597 = 0.058% of spend per lane`	modeled	0.058% / lane
6	Completeness grain, independent misses	`0.60⁴ = 13.0% → 0.95⁴ = 81.5% of shipments fully complete`	four milestones per shipment (illustrative)	13% → 81.5% complete shipments
7	Completeness grain, bounds	`union bound at 95%: 1 − 4 × 0.05 = 80%; ceiling = 95%; at 60%: floor 0%, ceiling 60%`	no independence assumed	shipment-level at 95%: 80%–95%

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Gate: 38/40 valid ASN	`observed 95.0%; one-sided 95% lower bound (Clopper–Pearson) = 85.1%`	illustrative test sizes — cohort sizes are not in the record	85.1%
2	Gate: 40/40 valid ASN	`observed 100.0%; one-sided 95% lower bound (Clopper–Pearson) = 92.8%`	illustrative test sizes — cohort sizes are not in the record	92.8%
3	Gate: 95/100 valid ASN	`observed 95.0%; one-sided 95% lower bound (Clopper–Pearson) = 89.8%`	illustrative test sizes — cohort sizes are not in the record	89.8%
4	Gate: 190/200 valid ASN	`observed 95.0%; one-sided 95% lower bound (Clopper–Pearson) = 91.7%`	illustrative test sizes — cohort sizes are not in the record	91.7%
5	Gate: 194/200 valid ASN	`observed 97.0%; one-sided 95% lower bound (Clopper–Pearson) = 94.2%`	illustrative test sizes — cohort sizes are not in the record	94.2%
6	Gate: 196/200 valid ASN	`observed 98.0%; one-sided 95% lower bound (Clopper–Pearson) = 95.5%`	illustrative test sizes — cohort sizes are not in the record	95.5%
7	Gate: 490/500 valid ASN	`observed 98.0%; one-sided 95% lower bound (Clopper–Pearson) = 96.6%`	illustrative test sizes — cohort sizes are not in the record	96.6%
8	Implied spend base	`$2M ÷ 10% = $20M; total spend = $20M ÷ 0.80 = $25M; deferred = $5M`	derived from the record pair and S(597) = 80%	$5M deferred spend

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
9	Control: deferred lanes drift 0% / −2% / −5%	`(−10% − drift) × $20M = $2.0M / $1.6M / $1.0M`	illustrative difference-in-differences; the record has no control	$1.6M at −2%

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Inbound freight spend -10% ~= $2M in 6 months (record)
+- Visibility (stock): completeness 60% -> 95%, high-value lanes 100%
|    +- routing-guide compliance: off-guide carrier spend caught
|    +- mode discipline: air -> ocean/ground when ETA is trusted
|    +- expedites + detention: chase calls replaced by 214 events
+- Document handling (flow): 850/855/856/940/945/214 by X12, not by hand
+- DQ gate loop: >=95% valid ASN -> cohort activates -> scorecard per lane ->
     defects fixed as data -> harness parameterized -> cohort 3 mostly config -> adoption 8x
+- Scope loop: 597 = 80% spend first -> $ lands month one -> executive funds the next wave
```

DRIVER	BASELINE	AFTER	$	BASIS
Shipment-data completeness	60%	95%	—	record
Lanes live month one	0	597 = 80% of spend	—	record
Freight spend on onboarded lanes	~$20M over the six months (packet-derived)	−10%	≈$2M / 6 mo	record ($2M)
— routing-guide compliance	off-guide unseen	caught by 856/214	~$0.8M	modeled split of the record $2M
— mode discipline	premium modes by habit	ETA trusted	~$0.6M	modeled split
— expedite + detention avoided	blind lanes chased	events replace calls	~$0.6M	modeled split
Manual document handling	~90K docs/yr keyed	X12/XML validated	$0.4M/yr	modeled (90K × 6 min × $45/hr)

If challenged. *"Is the $2M annualized?"* — "No — it's the record as measured: minus ten percent on the onboarded lanes' spend in six months. I keep it as stated rather than doubling it into a run-rate." *"How do you know the split?"* — "I don't, to the dollar — the split is my model of where the ten percent came from, and the three slices sum to the record, never on top of it."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Completeness	`480 / 800 = 60%`	expected = 200 shipments × 4 milestones = 800; received 480 (illustrative volumes; 60% record)	60% — 2 of 5 events missing
2	What blindness costs	`off-guide spend unseen · premium modes by habit · chase calls per late shipment`	record problem	no instrument, no lever
3	Manual documents	`597 lanes × ~50 shipments × 3 docs ≈ 90,000 docs/yr keyed by hand`	illustrative volumes	~90K docs/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Spend Pareto	`S(k) = cumulative spend share; S(597) = 80%`	lanes sorted by annual freight spend (record)	597 of 1,000 lanes carry 80% of spend — do them first
2	Cohort gate	`validASN rate ≥ 95% in test before activation`	published to carriers before cohort one (record)	one cohort failed twice on a timestamp format — held both times
3	Long-tail floor	`XML / portal when EDI is absent`	“the standard was the data, not the transport” (record)	small carriers kept inside the standard
4	Completeness after	`760 / 800 = 95%`	illustrative volumes; 95% record	95%; high-value lanes 100%

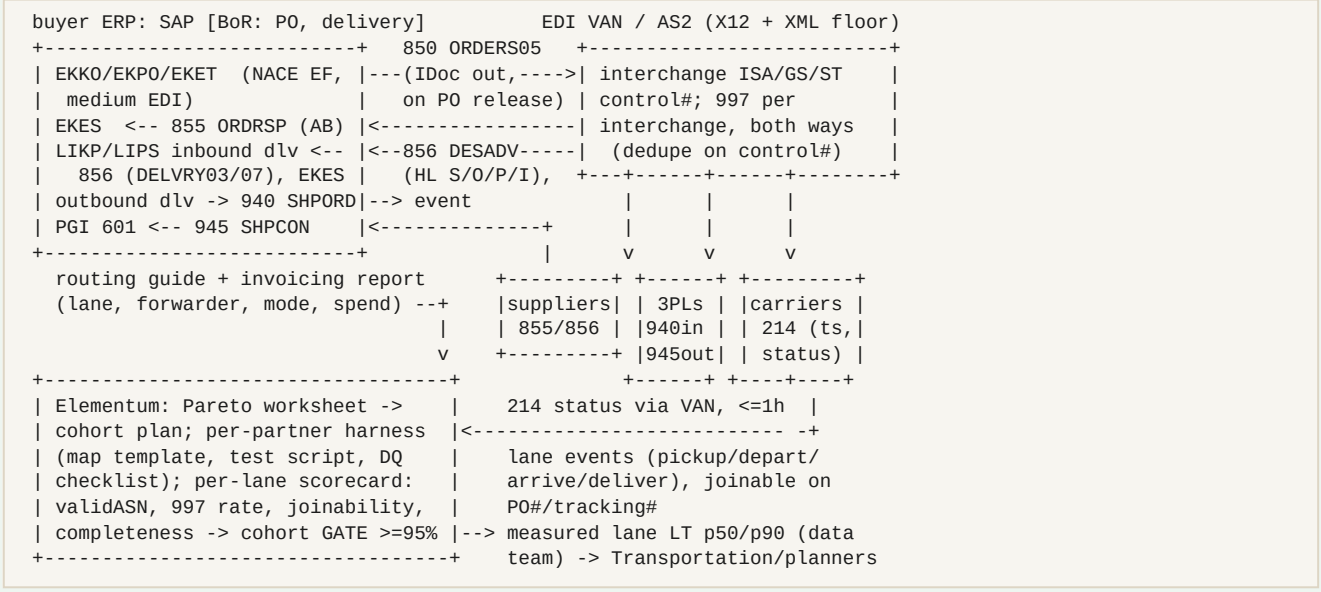
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Shipment-data completeness	60%	95%; high-value lanes 100%	record
2	Lanes live month one	0	597 = 80% of spend	record
3	Freight spend on onboarded lanes	base	−10% ≈ $2M in 6 months	record
4	Document handling	~90K docs/yr keyed	X12/XML validated	modeled

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Implied addressable base	`$2,000,000 / 10% = $20,000,000 of spend over the six-month window`	derived from the record pair, stated as such	≈$20M (six-month base)
2	The record result	`$20M × 10% = $2M in 6 months`	record	≈$2M
3	Slice — routing-guide compliance	`~$0.8M of the $2M`	off-guide spend caught by 856/214 (modeled split)	~$0.8M
4	Slice — mode discipline	`~$0.6M of the $2M`	premium modes → ocean/ground when ETA is trusted (modeled split)	~$0.6M
5	Slice — expedites + detention	`~$0.6M of the $2M`	events replace chase calls (modeled split)	~$0.6M
6	Split check	`0.8 + 0.6 + 0.6 = $2.0M`	the slices decompose the record — they never add to it	= the record $2M
7	Document labor	`90,000 × 6 min = 540,000 min; / 60 = 9,000 h; × $45 = $405,000`	illustrative volumes (modeled add-on)	≈$0.4M/yr

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Shipments/yr on 597 lanes (illustrative)	20,000	30,000	40,000
Minutes per manual document (benchmark)	4	6	8
Loaded coordinator rate (benchmark)	$40	$45	$55
Total modeled $/yr	≈$0.2M	≈$0.4M	≈$0.9M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The measured number is freight spend down ten percent, about two million dollars in six months on the five hundred ninety-seven lanes that carried eighty percent of spend, with completeness sixty to ninety-five percent and adoption eight x. Inside that two million I model roughly eight hundred thousand from catching off-guide carrier spend, six hundred thousand from mode discipline once the ETA was trusted, and six hundred thousand from expedites and detention that chase calls used to cause; those three are my split, the two million is the record. On top of it I model about four hundred thousand a year of document-handling labor that X12 replaced, and that piece is entirely mine."

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 **Tell me about a time you pushed back on a senior leader's request.**
HAVE BACKBONE; DISAGREE AND COMMIT

Open (first two sentences): The Global Head of Supply Chain wanted 1,000 lanes of inbound visibility in two months. I told him the schedule would deliver 1,000 lanes of confident wrong data — and showed him the spend curve instead.

STAR (≈90 s): Situation: scope, schedule and quality were all fixed, completeness was 60% and high-value lanes were blind. Task: own the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor. Action: I built the spend Pareto, pre-validated 597 / ~200 / remainder with Transportation and Procurement so no lane owner could contradict the cut, and took it up in spend share, not lane count. He accepted the phased plan off the curve; I committed completely. Result: 597 lanes carrying 80% of spend live in month one; ~400 lanes carrying ~20% deferred, priced, behind the same gate.

Land this number: 597 lanes = 80% of spend; ~400 deferred = ~20%. · **Pivot to NACF:** For NACF vendor onboarding I would lead with the volume Pareto and a gate, not a lane count. · **Do NOT say:** Do not name the executive's reaction beyond "accepted the phased plan"; do not quote cohort sizes.

Q2 **Tell me about a time you refused to compromise on quality under schedule pressure.**
INSIST ON THE HIGHEST STANDARDS

Open (first two sentences): A cohort failed our ≥95% valid-ASN gate twice on a carrier timestamp format. I held it both times.

STAR (≈90 s): Situation: month one, 597 lanes going live, the executive watching. Task: protect the standard without becoming the blocker. Action: the gate was published to carriers before cohort one — a gate that only holds in private is a suggestion; the failing cohort stayed in test, we fixed the format in the mapping template, retested, and activated off the gate. Result: completeness 60% → 95%; the gate is the product — automating dirty data manufactures confident wrong answers.

Land this number: Gate ≥95% valid ASN; held twice; completeness 60% → 95%. · **Pivot to NACF:** For NACF inbound the gate is "valid ASN rate before the vendor lane feeds placement." · **Do NOT say:** Do not say how many lanes were in the failed cohort.

Q3 **Tell me about a time you used data to decide where to focus.** DIVE DEEP

Open (first two sentences): Lanes differed by orders of magnitude in spend, volume and criticality, so I scored all 1,000 before I scoped anything.

STAR (≈90 s): Situation: a flat 1,000-lane plan would spend most of the effort on the long tail. Task: decide what to do first. Action: spend × criticality × mode × volume scoring, plus a carrier EDI-capability survey, so every lane had a score and a feasibility. Result: 597 lanes = 80% of spend first, ~200 next, the rest behind the same gate — and a method written up as the sourcing SOP.

Land this number: S(597) = 80%; scoring on four factors. · **Pivot to NACF:** Placement analytics starts the same way: rank by volume × cost × criticality, then gate. · **Do NOT say:** Do not call the modeled Pareto fit the record — only the 597 = 80% point is record.

Q4 **Tell me about a time you simplified a standard so more people could meet it.** INVENT AND SIMPLIFY

Open (first two sentences): Carriers without EDI would have been priced out of the standard, so I made the standard the data, not the transport.

STAR (≈90 s): Situation: some carriers could not produce valid ASNs over EDI. Task: keep the long tail inside one standard. Action: 850/855/856/940/945/214/997 on X12 with an XML or portal floor, parameterized per-partner mapping templates and a test harness instead of bespoke maps. Result: small carriers stayed in the standard; adoption 8×; the harness was reused for later waves.

Land this number: Adoption 8×; XML/portal floor. · **Pivot to NACF:** NACF vendors without EDI need the same floor: the standard is the data. · **Do NOT say:** Do not say I built the harness; engineers did to my standard.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: "597 of 1,000 lanes carried 80% of the spend."

Headline metric: S(597) = 80%; completeness 60% → 95%

- Pareto + four-factor lane scoring + EDI survey.
- Modeled wave 2 ≈ +10.7% of spend (~200 lanes).
- Shipment-grain completeness at 95% events: 80%–95%.
- Freight −10% ≈ $2M (record, not annualized).

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Have Backbone; Disagree and Commit

Open: "I did not take the 1,000-lane ask literally."

Headline metric: 597 live in month one; ~400 deferred, priced

- Took the curve two levels up; pre-validated with owners.
- Committed fully once accepted.
- Held a failing cohort twice.
- Deferral priced: ~20% of spend, same gate.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Insist on the Highest Standards; Learn and Be Curious

Open: "A cohort failed the gate twice on a timestamp format."

Headline metric: gate ≥95% valid ASN; failed twice, held twice

- What I would do differently: a timestamp-format check in the mapping template pre-test. VERIFY-WITH-RAJESH
- Gate as an observed rate, not a bound — I would add the lower bound.
- No control on the −10% — the deferred lanes are one ($1.6M at −2% drift).
- The record is the failure and the hold; the fix is mine to describe, not to inflate.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 "Is the $2M causal? What is your control group?"

The numbers: Record: −10% on ≈$20M ≈ $2M, no control. Deferred lanes ≈ $5M of spend; if they drift −2%, effect = (−10% + 2%) × $20M = $1.6M; at −5%, $1.0M.

Say exactly: "The two million is the record as measured on the onboarded lanes — it has no control. The four hundred deferred lanes, about five million dollars of spend, are the natural comparison; if they drifted two percent on the market alone, the visibility effect is about one point six million, and I would pull that before defending causality."

T2 "A ≥95% valid-ASN gate — what sample size makes that meaningful?"

The numbers: 38/40: observed 95%, lower bound 85.1%; 40/40: 92.8%; 190/200: 91.7%; 194/200 (97%): 94.2%; 196/200 (98%): 95.5%.

Say exactly: "The gate was an observed rate in test — ninety-five percent valid ASN. As statistics, ninety-five observed on forty only bounds the true rate above eighty-five, and on two hundred above ninety-two; to show at least ninety-five you need ninety-eight observed on two hundred, and on forty it cannot be shown at all. I would put the bound on the scorecard next to the rate — and I will not quote cohort sizes that are not in the record."

T3 "Completeness 60 → 95% — at which grain, and what does 95% of events mean for shipments?"

The numbers: Four milestones: 0.6^4 = 13.0% → 0.95^4 = 81.5% complete shipments if independent; bounds at 95%: 80%–95%.

Say exactly: "The record is shipment-data completeness, sixty to ninety-five. If that is events, then with four milestones a shipment is fully complete eighty-one percent of the time when misses are independent, and between eighty and ninety-five when they cluster by carrier — so I report both, and I would confirm the grain before quoting a shipment-level figure."

S19 — Elementum / Walmart Marketplace — the delivery promise: one governed buffer dial instead of a flat 1.10 days — fast-promise share 7.6% → 39.6% at the held 95% floor (engagement model) NEW

LPs this story serves here: Invent and Simplify (one dial) · Customer Obsession (a promise the customer can rely on) · Dive Deep (the Φ(5.0) vs Φ(0.69) audit) · Insist on the Highest Standards (shadow gate, peak re-solve)
Process family: Deliver seam — pre-purchase promise policy for seller-fulfilled delivery (Walmart Marketplace, an Elementum client; role R-2596147 Pre-Purchase Delivery Experience)

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Dive Deep** — I decomposed the 95.8% headline into two cohorts: the flat 1.10-day buffer bought Φ(5.0) — near-certainty — on the tightest cohort and Φ(0.69) ≈ 75% on the widest.
- **Invent and Simplify** — I replaced a hundred model micro-decisions with one governed dial, b = k·σ with k = 1.645, that a business owner can read every Monday.
- **Customer Obsession** — The customer blames the marketplace, not the seller, when Thursday becomes Saturday — the floor was a promise kept 95 times in 100 on every cohort.
- **Insist on the Highest Standards** — Four consecutive shadow weeks before any displayed promise, a published promotion bar and a warm rollback; and every figure said “the model” in the same breath.

Honesty label. Elementum engagement for a marketplace client — Walmart Marketplace's seller-fulfilled delivery program, where the client prints "Get it by Thursday" on inventory that sits in sellers' buildings. I led the promise-policy workstream: the decomposition, the buffer-policy alternatives, the validation gates and the governance design. The client's promise-engineering and data-science teams owned and built the promise engine and the transit models — I never configured or deployed any platform (the only platform I personally configured remains Manhattan DOM). The twelve-cohort book behind every number is the engagement's illustrative model, built so the argument could happen in numbers instead of adjectives.

60-SECOND PITCH

"**Situation.** A marketplace client promising delivery dates on inventory it never touches: promised date = order time + seller handling + predicted transit + a safety buffer — and the buffer was a flat 1.10 days on every lane regardless of how variable the lane was. In the engagement model — twelve lane cohorts, about a million orders a week, transit spread from 0.22 to 1.60 days — that flat buffer printed 95.8% accuracy but put only 7.6% of orders on a fast promise, because dense metro lanes were over-insured and fat-tailed rural lanes under-insured at the same time. **Task.** Engagement lead for promise policy: own the decomposition, the alternatives, the gates and the governance recommendation; the client's data science and engineering did not report to me. **Action.** Discovery: the buffer-waste audit — the flat buffer bought Φ(5.0), near-certainty, on the tightest cohort and only Φ(0.69) ≈ 75% on the widest. Plan & Design: four policies on one comparison table — uniform, σ-proportional with one governed dial, minimum-days at the floor, profit-optimal on priced breach economics. Build & Test: b = 1.645σ holds 95% on every cohort by construction; the equimarginal check proved the optimum; the Q4 drill showed a frozen summer dial breaking the floor. Cutover: four shadow weeks against logged live orders before any displayed promise. Hypercare: the dial published in the weekly review header, re-solved before peak. **Result.** Primary metric: fast-promise share — **7.6% → 39.6%, +32 points at the held 95% floor**; average promised day 3.09 → 2.75; +$255K a week, about $13.3M a year — all on the illustrative book, and I say 'the model' in the same breath."

20-SECOND VERSION NEW

A flat 1.10-day buffer printed 95.8% accuracy and put 7.6% of orders on a fast promise; one governed dial, b = 1.645σ, kept every cohort at 95% and lifted fast-promise share to 39.6% — +32 points on the engagement's illustrative model, never a production claim.

NACF BRIDGE — ONE SENTENCE

NACF promise and appointment buffers should be risk-proportional, not flat: buffer = k × the lane's own σ, one published dial, re-solved before peak — the same logic as INV-B's z·σ·√LT.

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — An Elementum engagement with Walmart Marketplace's seller-fulfilled delivery program, in the client years. The client underwrites a delivery promise on execution it does not own: the seller picks and packs, a carrier it only partially controls carries, and the customer blames the marketplace — not the seller — when Thursday becomes Saturday. The promise identity is a sum: promised date = order time + configured handling + predicted transit + buffer; the promise is kept when the handling error plus the transit error fit inside the buffer. The incumbent buffer policy was uniform — about 1.10 days on every lane — set once, owned by nobody between reviews.

T — Lead the promise-policy workstream: decompose where the incumbent buffer was wasted and where it was thin, put priced alternatives on one table, design the validation gates, and leave the client a governance instrument — not a tuning habit. Everything quantitative ran on the engagement's illustrative model: twelve lane cohorts spanning σ = 0.22 days (metro two-day) to 1.60 days (mountain rural), about a million orders a week.

Why it mattered: padding fails silently and cutting fails loudly, so every promise system drifts slow — the orders that never happened because the promise read five days file no tickets. The economics were stated up front and carried on every page: a breach costs about $12 (contact + refund risk + trust decay) against about $6.50 of contribution margin — both the model's stated assumptions, never a measured P&L. Left unfixed: the client keeps paying for insurance it doesn't need on tight lanes while underwriting known losses on loose ones, and the speed-versus-accuracy dial keeps being set implicitly by whoever tuned the last model.

The metrics on the table, with their baselines (all illustrative-model values): fast-promise share (orders promised in ≤2 days) — 7.6%; portfolio promise accuracy — 95.8% against a 95% floor; average promised day — 3.09; weekly modeled profit — $5.69M; and no governed owner of the speed-accuracy trade — the real baseline.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases, run as an engagement arc (the model, the table and the gates are the engagement record; no phase touched a production system — the client's teams built, I designed and governed):

- **Discovery** — wrote the promise identity and the accuracy factorization (accuracy ≈ P(seller ships in window) × P(carrier meets transit) × Φ(buffer/σ)); built the buffer-waste audit — days delivered early on kept promises, ranked by waste × volume — and the two-line indictment of the flat buffer: 1.10 days is 5.0 standard deviations on the tightest cohort and 0.69 on the widest. *Exit bar:* the client's own cohort table agreed in the room; "how much buffer" retired as a question, replaced by "where should it sit."
- **Plan & Design** — four policies written onto one comparison table on the same cohort set: A uniform (incumbent), B σ-proportional with one governed dial (b = k·σ), C minimum-promised-days at the floor (analytic optimum), D profit-optimal on the priced breach economics; the gates designed — shadow mode, a promotion bar, a rollback that keeps the incumbent computable. *Exit bar:* the table signed as the decision instrument; the dial (k) named as a governed, published policy instrument with an owner.
- **Build & Test** — the client's data-science team computed the table; I owned the proofs: Φ(1.645) = 0.95 holds the floor on every cohort by construction; the equimarginal condition for C (φ(b/σ)/σ equal across cohorts) verified to machine precision; the Q4 drill — σ inflated ~1.45× — showed the frozen summer dial printing 91.9%, below the floor. *Exit bar:* every policy's accuracy, speed and modeled profit reproduced independently by the client's team; the peak-regime rule written.

- **Cutover / Go-Live (design)** — shadow mode: the challenger policy computed and logged per order, never displayed, for four consecutive weeks; promotion only if would-be accuracy held the floor on logged live orders with no rise in contact signals. *Exit bar:* promotion bar published before the first shadow week; rollback = one flag back to the incumbent, kept warm.
- **Hypercare / Close** — the governance handoff: k printed in the weekly review header every Monday; tail metrics (p90 error, buffer-exceedance) on the page next to the average; the buffer-waste line standing; k re-solved per regime — published in October for peak, never frozen through it. *Exit bar:* the client's business owner — not data science — named as the dial's owner; the engagement closed on the instrument, not on a tuning.

The client's promise-engineering team owned the engine and its latency budget; their data-science team owned the transit models and σ estimates; their finance partner owned the breach price. Mine: the factorization, the waste audit, the four-policy table, the equimarginal and Q4 proofs, the shadow-mode gates and the governance design — the reframe from "a hundred ML micro-decisions" to one auditable dial a business owner reads on a Monday.

The obstacle was the room's instinct that accuracy at 95.8% meant the policy was healthy — the decomposition had to beat a green headline. The two-cohort arithmetic did it: near-certainty bought where σ was tiny, a coin-flip-plus where σ was fat. The decision I challenged: data science wanted per-lane learned buffers — more model, more value, invisible governance. I priced the difference and recommended the one-dial policy first: it captures most of the value, it is auditable, and σ improvements flow into speed automatically and visibly. The no went into the comparison table as a priced row, not a meeting minute.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: fast-promise share 7.6% → 39.6% (+32 points) at the identical 95% accuracy floor; average promised day 3.09 → 2.75; +$255K/week ≈ $13.3M/yr — every figure from the illustrative model, said with "the model" in the same breath.

- **$** — +$255K/wk vs the uniform policy ≈ $13.3M/yr (modeled on the illustrative ~1M-order book); the profit-optimal frontier (policy D: 97.8% accuracy, $6.07M/wk, ≈ +$20M/yr vs uniform) handed to the client's finance partner as a priced governance decision — shown, never claimed, never merged with the $13.3M.
- **#** — Twelve lane cohorts, ~1M orders/week, σ from 0.22 to 1.60 days; four policies on one table; one governed dial, k = 1.645 (illustrative model; the table and the dial are the engagement's design record).
- **%** — Fast-promise share +32.0 points at a held 95.0% floor; the incumbent's hidden split exposed: Φ(5.0) ≈ 99.99997% on the tightest cohort against Φ(0.69) ≈ 75.4% on the widest (model).
- **Time** — Average promised day 3.09 → 2.75 (−0.34); the tightest cohort's buffer 1.10 → 0.36 days; the widest cohort's buffer 1.10 → 2.63 days — longer on purpose, to stop underwriting known losses (model).
- **Customer** — The fast promise shown where it can be kept and withheld where it can't; the Q4 re-solve rule so peak variability never silently breaks the floor the customer experiences (engagement design record).
- **Quality** — The equimarginal check as the solver's own proof of optimality; p90 error and buffer-exceedance on the review page next to the mean, so a better average can never hide a worse tail (engagement design record).

Against the goal stated above: more fast promises without breaking the floor — +32 points at exactly 95.0%, a promise a third of a day shorter on average, and the speed-versus-accuracy trade converted from an implicit model parameter into one governed, published dial. The trade was explicit and priced: the widest cohorts got honestly slower promises so the whole book could get kept ones; nothing was bought with accuracy.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Nobody owned the speed-vs-accuracy trade; I defined it as one dial with a floor and an owner.
Scope & influence	Client promise engineering, data science, finance and seller programs — none reporting to me; the "no" to per-lane learned buffers went in as a priced row.
Execution	Four-policy table, two proofs (Φ(1.645) floor and equimarginal), Q4 drill, shadow gate and rollback designed.
Impact	+32 pts fast-promise share at a held 95% floor; +$255K/wk ≈ $13.3M/yr — all engagement model, labeled as such.
Process improvement	k published weekly and re-solved per regime — governance, not tuning.
Complexity	Twelve cohorts with σ from 0.22 to 1.60 days; heavy tails; a peak regime that breaks a frozen dial.
Root cause	A flat buffer ignores variance: it over-insures dense lanes and under-insures fat-tailed ones at once.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: twelve lane cohorts, σ 0.22–1.60 days, ~1M orders/week (the engagement's illustrative model); promise identity = order + handling + transit + buffer. Revealed: flat 1.10 d = 5.0σ on the tight cohort, 0.69σ on the wide one.
2 · Solution design	Options: A uniform · B σ-proportional with one dial · C minimum-days at the floor · D profit-optimal. Chosen: B first (auditable, one dial); D shown and never claimed.
3 · Math & data logic	Math: P(kept) = Φ(b/σ); b = kσ with Φ(k) = 0.95 → k = 1.645; tight 0.36 d, wide 2.63 d; Q4 σ × 1.45 → z = 1.13 → 87.2% per cohort, 91.9% portfolio (record-from-docs); price-of-floor, estimation-error and heavy-tail arithmetic below.
4 · Implementation	Implementation: waste audit → four-policy table → proofs → 4-week shadow → published promotion bar → k in the weekly review header; re-solved in October.
5 · Probe questions	Probe questions: σ estimation error · heavy tails · why not policy D · how the floor behaves at peak — see Trap Drills.

PROBING — CHALLENGE Q&A (MASTER RECORD) PLUS THE STATISTICAL QUESTIONS

> ***Challenge* — Why did you choose this story to illustrate Dive Deep?**
>
> Because the finding only exists below the headline: portfolio accuracy printed a healthy 95.8%, and the depth — per-cohort z-scores — showed the same policy was simultaneously over-insuring metros at five standard deviations and under-insuring rural lanes at 0.69. A team that stops at the portfolio number never finds thirty-two points of free speed.

> ***Probing* — Why did you choose to focus on these results? What other results were important?**
>
> The +32 points at a held floor leads because it survives any parameter change — it comes from re-allocating risk, not spending money. Also important: the honest cost (the widest cohorts' promises got longer), the Q4 regime rule, and the governance artifact — a dial the business reads weekly — which outlives any one solve.

> ***Probing* — You mentioned dollars — what percentage change is that year over year?**
>
> I don't give this one as year-over-year — every dollar here is the model's: +$255K/wk is about 4.5% of the modeled $5.69M weekly contribution, ≈ $13.3M annualized on the illustrative book. There is no production dollar in this engagement and I never promote one.

> ***Probing* — I'm concerned the whole story rests on an illustrative cohort set — tell me more.**
>
> It does, deliberately — the model exists so the argument happens in numbers instead of adjectives. If the real σ looks nothing like the model's, k changes and the lane ranking changes; what survives any parameter set is the structure: buffers should be risk-proportional, the last buffer-day should buy the same accuracy everywhere, and a floor that disagrees with the breach economics is a governance finding.

***Challenge* — What were the lessons learned? What would you have done differently?**

The lesson: padding is not safety — a flat buffer is slow and unsafe at the same time, and only the decomposition shows it. Differently: price the breach bundle with the finance partner in discovery rather than design — the $12-versus-$6.50 conversation is what turned the floor from theology into economics, and it came one phase late.

***Challenge* — How would you implement this at Amazon?**

The same shape at the promise seam: decompose the promise identity, audit where buffer is wasted on kept promises, one governed dial per regime with the tail metrics on the page, shadow before display, re-solve before peak. At Amazon's scale a tenth of a point of buffer waste is real money — which is why the governance matters more than any single solve.

***Challenge* — How did these results compare to your actual goals?**

The goal was more fast promises without breaking the floor. The model delivered +32 points at exactly the floor and a promise a third of a day shorter — and the frontier beyond it (policy D at 97.8%) was handed to Finance as a priced decision rather than smuggled in as a default.

PHASE DEEP-PROBE BANK — Q&A (12). *Answer the phase asked, then stop.*

Discovery — How did you know discovery was done?

When the room accepted one sentence of arithmetic: the flat 1.10-day buffer is 5.0 standard deviations on the tightest cohort and 0.69 on the widest. Once that was agreed on the client's own cohort table, "how much buffer" was dead as a question and "where should it sit" replaced it.

Why a buffer-waste audit instead of starting from accuracy?

Accuracy printed 95.8% — a healthy-looking headline hiding the split. Waste — days delivered early on kept promises — is where the recoverable speed lives, and it is unarguable because it only counts promises that were kept. Ranked by waste × volume, the audit is the trim roadmap with no new models needed.

Plan & Design — Why four policies instead of recommending one?

Because the client's real decision was governance, not mathematics. A is what you run today; B is one auditable dial; C is the fastest promise the floor allows; D is what the economics want. All four on one cohort set made the trade-offs priced choices instead of expert opinion — and made B's simplicity a visible decision, not a default.

Why recommend B over the profit-optimal D?

D's optimum moves every time the breach price or margin assumption moves, and it is harder to audit. B is one dial a business owner reads on a Monday and it captures most of the value. I recommended running B and letting Finance decide whether to chase the D frontier with its own signature on the $12.

Build & Test — What exactly is the equimarginal check?

The optimality condition for policy C: the marginal accuracy bought by the last unit of buffer — $\varphi(b/\sigma)/\sigma$ — must be equal on every cohort; if a buffer-day buys more accuracy on one lane than another, moving it is free improvement. The solver's output showed the condition holding to machine precision — the model proving its own optimum.

What was the Q4 drill and what did it change?

Inflate every cohort's σ by ~1.45× — the model's peak regime — and hold the summer dial: portfolio accuracy printed 91.9%, under the floor (on a single lane taking the full inflation, the whiteboard version is $\Phi(1.645/1.45) = \Phi(1.13) \approx 87\%$ — the portfolio mix lands at 91.9%). The drill wrote the rule: k is re-solved and published before peak, never frozen through it.

Why normal errors — freight tails are skewed?

Said before anyone asked: the normal is the whiteboard version; the production recommendation was the empirical quantile per lane. The normal makes the argument legible; the quantile makes it true. Nothing in the recommendation depends on normality — only the clean Φ numbers do.

Cutover — What was the promotion bar out of shadow mode?

Four consecutive shadow weeks with would-be-promise accuracy at or above the floor on logged live orders and no rise in contact signals. A policy that can't clear its own floor in shadow has no business touching a displayed date — the same rehearse-before-commit discipline as my mock cutovers (S18).

What was the rollback?

The incumbent uniform buffer stayed computable the whole time — one flag back. A promise policy is the easiest class of change to reverse if you keep the old policy warm, and the easiest to make irreversible if you don't.

Hypercare — Why publish the dial in the review header?

Because k *is* the speed-versus-accuracy trade, and a trade the business can't see is a trade it didn't agree to. One number in the header every Monday makes the floor a standing decision instead of a buried parameter.

What keeps the policy honest after handoff?

Three standing lines on the review page: the dial, the tail metrics (p90 error and buffer-exceedance) next to the average, and the buffer-waste line. A policy whose tail and waste are public decays far more slowly than one graded on its mean.

Cross-phase — Where did you trade scope for the date? What did you refuse?

Traded: per-lane learned buffers deferred in favor of the one dial — sophistication for governance, knowingly leaving model value on the table for the first cycles. Refused: promoting on the model's average error (the tail breaks promises, so the tail went on the page), and any record dollar — the book is illustrative and I said so in every read-out.

Analytics — What does the floor cost? Price each step.

On the widest cohort (σ = 1.60 d) the buffer is 2.63 d at a 95% floor, 3.14 d at 97.5% (+0.50 d) and 3.72 d at 99% (+1.09 d); on the tightest cohort (σ = 0.22) the same steps cost 0.07 d and 0.15 d. The dial is linear in σ — so a point of floor is nearly free on dense lanes and expensive on rural ones, which is the whole argument for a risk-proportional rule. Engagement model.

Analytics — Your σ is estimated. What does that do to the floor?

With n orders in a cohort, σ̂ has relative standard error 1 ÷ √(2(n − 1)): 7.1% at n = 100, 2.2% at n = 1,000. If σ̂ is one SE low, a dial of 1.645 realises z = 1.645 ÷ (1 + SE): 1.54 at n = 100 → 93.8%; 94.6% at n = 1,000. Holding the floor within half a point at a one-SE miss needs about 584 orders per cohort per window. At about a million orders a week that is trivially met; the real exposure is regime change, which is why Q4 is the drill.

Analytics — Normal errors — what if the errors are heavy-tailed?

Production uses the empirical quantile; the whiteboard uses the normal. As a sensitivity, take Student-t errors with 5 degrees of freedom scaled to the same σ: the 95% quantile is 1.561σ, below 1.645 — the normal dial is conservative there and delivers 95.6% — but the 99% quantile is 2.607σ, above 2.326, so a normal 99% dial would deliver only 98.5%. The heavier the tail the more the empirical quantile matters at the high floors.

Analytics — With a million orders a week, is the weekly accuracy noise a problem?

No: the standard error of a 95% weekly accuracy on 1,000,000 orders is √(0.95 × 0.05 ÷ 1,000,000) = 0.022 points; on a 100,000-order cohort 0.069. A reading of 91.9% in Q4 is not noise — it is a σ regime change. At this scale the question is never “is it significant,” it is “which factor moved.”

STATISTICAL SPINE — PRICE OF THE FLOOR, Σ-ESTIMATION ERROR, HEAVY TAILS (EXTRA TABLES)

The engagement model is the record-from-docs; these tables are my sensitivity arithmetic on the same cohorts and dial — every figure engagement-model or textbook-normal, none a production measurement.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Floor 90% → k	`Φ(k) = 0.90 → k = 1.2816`	standard normal table	tight 0.28 d · wide 2.05 d

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	Floor 95% → k	`Φ(k) = 0.95 → k = 1.645`	record-from-docs dial	tight 0.36 d · wide 2.63 d
3	Floor 97.5% → k	`Φ(k) = 0.975 → k = 1.96`	standard normal table	tight 0.43 d · wide 3.14 d
4	Floor 99% → k	`Φ(k) = 0.99 → k = 2.326`	standard normal table	tight 0.51 d · wide 3.72 d
5	Price of 95% → 97.5% (wide cohort)	`(1.96 − 1.645) × 1.60 = 0.50 d`	σ = 1.60 d (illustrative model)	+0.50 d of promise
6	Price of 95% → 99% (wide cohort)	`(2.326 − 1.645) × 1.60 = 1.09 d`	σ = 1.60 d (illustrative model)	+1.09 d of promise
7	Incumbent accuracy, tight / wide	`Φ(1.10 ÷ 0.22) = Φ(5.0) ≈ 100%; Φ(1.10 ÷ 1.60) = Φ(0.69) = 75.4%`	flat buffer b = 1.10 d	75.4% on the wide cohort

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	σ̂ standard error, n = 100	`1 ÷ √(2 × 99) = 7.1%`	large-sample approximation	7.1%
2	Realised accuracy at a one-SE miss, n = 100	`z = 1.645 ÷ (1 + 0.0711) = 1.536; Φ = 93.8%`	σ̂ one SE below the truth	93.8% (floor 95%)
3	Same, n = 1,000	`z = 1.645 ÷ (1 + 0.0224) = 1.609; Φ = 94.6%`	same method	94.6%
4	Orders needed for ≥ 94.5% at a one-SE miss	`z* = Φ⁻¹(0.945) = 1.5982; 1 + SE = 1.645 ÷ 1.5982; n = 1 + 1 ÷ (2 × SE²) = 584`	design arithmetic	≈ 584 orders per cohort-window
5	Heavy tails, t_5 scaled to unit variance	`95%: 2.015 ÷ √(5/3) = 1.561; 99%: 3.365 ÷ √(5/3) = 2.607`	Student-t table; sensitivity only	normal dial delivers 95.6% at 95%, 98.5% at 99%
6	Q4 regime, frozen dial	`z = 1.645 ÷ 1.45 = 1.134; Φ = 87.2% per fully-inflated cohort`	σ × 1.45 (illustrative model)	87.2% per cohort; 91.9% portfolio (record-from-docs)
7	Accuracy noise at scale	`√(0.95 × 0.05 ÷ 1,000,000) = 0.022 pts; at 100,000: 0.069 pts`	binomial	noise ≪ any real signal

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Marketplace contribution (modeled, illustrative book)
+- Conversion on fast promises (flow) -> fast share 7.6% -> 39.6%
|    fast share = SUM_cohorts volume x [handling + transit_pred + buffer <= 2d]
|    (loop) faster promise -> higher CVR -> more orders -> more seller GMV
+- Breach cost (flow, $/order) -> floor held at 95.0%
|    breach EV = (1 - Phi(b/sigma)) x $12   [contact + refund risk + trust decay]
+- Buffer waste (stock: days early on kept promises) -> harvested on tight lanes
+- Governance -> one dial k, published weekly; re-solved per regime (Q4)
+- Frontier (not claimed) -> policy D: 97.8% accuracy, +$0.38M/wk vs A
```

DRIVER	BASELINE	AFTER	$/WK	BASIS
Weekly modeled profit, policy A (uniform 1.10d)	$5.69M	—	$5.69M	illustrative model output
Weekly modeled profit, policy B (b = 1.645σ)	—	$5.94M	$5.94M	illustrative model output

DRIVER	BASELINE	AFTER	$/WK	BASIS
Fast-promise share	7.6%	39.6% (+32 pts)	inside the profit lines	illustrative model
Average promised day	3.09	2.75	inside the profit lines	illustrative model
Frontier: policy D (profit optimum)	—	97.8% accuracy, $6.07M/wk	+$0.38M vs A	illustrative; handed to Finance, never claimed

If challenged. *"Your whole result is a model."* — "Yes, and I say so in the same breath — the model exists so the client's decision could be quantitative. The structure survives any parameter set: risk-proportional buffers, equal marginal accuracy per buffer-day, a floor reconciled with the breach economics. The parameters are the client's to correct." *"Why is the $12 breach cost credible?"* — "It's the model's stated assumption — a contact, refund and return risk, and trust decay bundled — put on the table precisely so Finance could attack it. At $12 the profit optimum holds 97.8% accuracy; if Finance believes $6, the optimum falls toward the floor. Fight about the number, not the theology." *"Did any of this ship?"* — "The engagement delivered the policy table, the gates and the governance design; the client's teams owned the build. I claim the instrument and the modeled deltas, labeled — never a production P&L."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The promise identity	`EDD = t_order + handling + transit_pred + b; kept ⇔ (handling_err + transit_err) ≤ b`	per order; b = buffer in days (record-from-docs identity)	the buffer is the only policy term
2	Kept-promise probability per lane	`P(kept) = Φ(b / σ)`	ε ~ N(0, σ²) per cohort — whiteboard version; production uses the empirical quantile (record-from-docs)	accuracy is a function of b and σ only
3	Incumbent buffer	`b = 1.10 days on every lane`	uniform policy A (illustrative model)	1.10 d everywhere
4	Tightest cohort, z-score	`z = b / σ = 1.10 / 0.22 = 5.0; Φ(5.0) ≈ 0.9999997`	metro cohort σ = 0.22 d (illustrative model)	near-certainty bought — promise needlessly slow
5	Widest cohort, z-score	`z = 1.10 / 1.60 = 0.6875; Φ(0.69) ≈ 0.754`	mountain-rural cohort σ = 1.60 d (illustrative model)	≈75.4% kept — a known-loss book
6	Portfolio print	`accuracy 95.8% · fast share 7.6% · avg promised day 3.09 · profit $5.69M/wk`	order-weighted across 12 cohorts, ~1M orders/wk (illustrative model)	a green headline hiding both failures
7	The asymmetry	`padding fails silently (lost conversion files no ticket); cutting fails loudly (breaches do)`	record-from-docs — why the system drifted slow	bias toward slow, unowned

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	One governed dial	`b_lane = k × σ_lane; Φ(k) = floor ⇒ k = 1.645 for 95%`	Φ(1.645) = 0.95 under normal errors (record-from-docs design)	k = 1.645, one number for the book

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	Tightest cohort re-buffered	`b = 1.645 × 0.22 = 0.36 days`	σ = 0.22 d (illustrative)	buffer 1.10 → 0.36 d; 0.74 d of speed harvested
3	Widest cohort re-buffered	`b = 1.645 × 1.60 = 2.63 days`	σ = 1.60 d (illustrative)	buffer 1.10 → 2.63 d — slower on purpose, kept honest
4	Every cohort at the floor	`P(kept) = Φ(1.645 × σ / σ) = Φ(1.645) = 0.95 on each cohort`	by construction (design)	95.0% everywhere; no cohort subsidizes another
5	Portfolio after	`accuracy 95.0% · fast share 39.6% · avg day 2.75 · profit $5.94M/wk`	illustrative model output, policy B	+32 pts fast share at the floor
6	Equimarginal proof (policy C)	`φ(b/σ)/σ equal across all 12 cohorts (spread < 1e-15)`	solver optimality condition (record-from-docs)	the last buffer-day buys the same accuracy everywhere
7	Peak regime rule	`σ_Q4 ≈ 1.45 × σ; frozen k ⇒ z = 1.645/1.45 = 1.13; Φ(1.13) ≈ 0.87 per fully-inflated lane; portfolio prints 91.9% < 95%`	Q4 drill (illustrative model)	k re-solved and published in October, never frozen
8	Shadow gate	`4 consecutive weeks logged ≥ floor, contacts flat → promote; else hold`	promotion bar (design)	rehearse against reality before any displayed date

Table 3 — after: the metric, before → after.

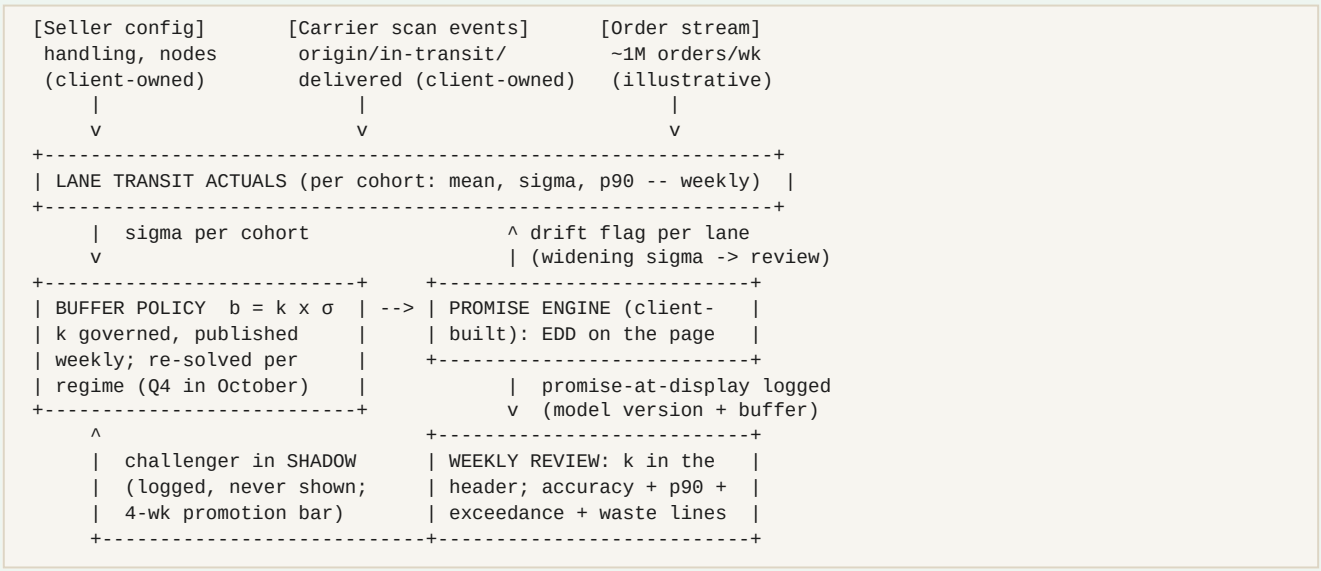
STEP	METRIC	BEFORE	AFTER	BASIS
1	Fast-promise share (≤2-day)	7.6%	39.6% (+32 pts)	illustrative model
2	Portfolio promise accuracy	95.8% (split: ~100% / 75.4%)	95.0% — the floor, held on every cohort	illustrative model
3	Average promised day	3.09	2.75 (−0.34)	illustrative model
4	Weekly modeled profit	$5.69M	$5.94M (+$255K)	illustrative model
5	Owner of the speed-accuracy trade	implicit, in the last model tune	one governed dial, published weekly, re-solved per regime	engagement design record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Weekly profit, policy A	`$5,690,000 / wk, taken from the model`	illustrative model output	$5.69M/wk
2	Weekly profit, policy B	`$5,940,000 / wk, taken from the model`	illustrative model output	$5.94M/wk
3	Weekly delta	`$5,940,000 − $5,690,000 = $250,000 (the model's precise print: +$255K)`	rows 1, 2 (illustrative)	≈ +$255K/wk
4	Annualized	`$255,000 × 52 = $13,260,000`	row 3 × 52 weeks (modeled)	≈ $13.3M/yr — "the model," said in the same breath
5	Frontier: policy D delta	`$6,070,000 − $5,690,000 = $380,000/wk; $380,000 × 52 = $19,760,000`	illustrative model output; ×52 modeled	≈ $20M/yr — handed to Finance, never claimed

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
6	Why D holds 97.8%	`breach EV $12 > margin-per-promise-day trade at $6.50 ⇒ optimum above the 95% floor`	$12 and $6.50 are the model's stated assumptions	the floor and the economics disagree — a governance finding
7	Headline	`≈ $13.3M/yr modeled; record dollars: none`	rows 3–4	≈ $13.3M/yr, always labeled

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Breach cost bundle (model assumption)	$6	$12	$18
Profit-optimal accuracy (policy D)	falls toward the 95% floor	97.8%	above 97.8% — accuracy gets cheaper to justify
σ dispersion across cohorts (drives the free lunch)	narrow — gain shrinks	0.22–1.60 d	wider — gain grows
Fast-share gain from B vs A	≈ +10 pts	+32 pts	≈ +40 pts
Modeled $/yr vs uniform	≈ $4–5M	≈ $13.3M	≈ $20M (the D frontier)

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"On the engagement's illustrative book — twelve lane cohorts, about a million orders a week — moving from a flat 1.10-day buffer to risk-proportional buffers with one governed dial bought thirty-two points of fast-promise share at the identical ninety-five percent accuracy floor, took a third of a day off the average promise, and was worth about two hundred fifty-five thousand dollars a week, call it thirteen million a year. Every one of those numbers is the model's, and I say that in the same breath — there is no production dollar here. What I claim as mine is the instrument: the decomposition, the four-policy table, the shadow gates, and a dial the business reads every Monday."

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 Tell me about a time a headline metric hid the real problem. DIVE DEEP

Open (first two sentences): Promise accuracy printed 95.8% — green — and the room believed the policy was healthy. I split it into two cohorts.

STAR (≈90 s): Situation: a marketplace promised delivery dates on inventory it never touched, with a flat 1.10-day buffer on every lane. Task: decompose where the buffer was wasted and where it was thin. Action: z = buffer ÷ σ — 1.10 ÷ 0.22 = 5.0 on the tightest cohort (near-certainty, a needlessly slow promise) and 1.10 ÷ 1.60 = 0.69 on the widest (75.4% kept, a known-loss book). The average was hiding both failures. Result: the question changed from “how much buffer” to “where should it sit,” and the one-dial policy followed. All of this is the engagement’s illustrative model.

Land this number: Φ(5.0) vs Φ(0.69) = 75.4%; headline 95.8%. · **Pivot to NACF:** NACF appointment and ASN promise lanes: check the tightest and widest vendor lanes before trusting the average. · **Do NOT say:** Do not call any figure a production result.

Q2 Tell me about a time you simplified a complex decision. INVENT AND SIMPLIFY

Open (first two sentences): Data science wanted per-lane learned buffers. I recommended one dial — b = k·σ — and priced the difference.

STAR (≈90 s): Situation: a hundred ML micro-decisions, invisible governance. Task: leave the client an instrument, not a tuning habit. Action: four policies on one table — uniform, σ-proportional with one dial, minimum-days at the floor, profit-optimal — and the one-dial policy first because it captured most of the value and was auditable; σ improvements flow into speed automatically. The “no” to per-lane models went into the table as a priced row. Result: fast-promise share 7.6% → 39.6% at the held 95% floor on the model.

Land this number: +32 pts at a 95% floor; k = 1.645. · **Pivot to NACF:** A NACF dial a manager can read on Monday beats a model nobody can audit. · **Do NOT say:** Do not claim I built the promise engine; the client’s teams did.

Q3 Tell me about a time you protected the customer’s experience when the numbers could have been juiced. CUSTOMER OBSESSION

Open (first two sentences): Fast-promise share can be juiced by bleeding accuracy, so I pinned accuracy at 95% on every cohort and let speed float.

STAR (≈90 s): Situation: the customer blames the marketplace when Thursday becomes Saturday. Task: more fast promises without breaking the floor. Action: floor first — Φ(1.645) = 0.95 on each cohort by construction — then harvest speed: the tight cohort’s buffer 1.10 → 0.36 d, the wide cohort’s 1.10 → 2.63 d, slower on purpose. Result: +32 points of fast-promise share at exactly 95.0% on the model; no cohort subsidizes another.

Land this number: 95.0% on every cohort; wide-cohort buffer 2.63 d. · **Pivot to NACF:** The NACF promise is OTIF: protect the floor per lane, not per average. · **Do NOT say:** Do not say $13.3M a year as fact — say the model, in the same breath.

Q4 Tell me about a time you set a high bar on how a change rolled out. INSIST ON THE HIGHEST STANDARDS

Open (first two sentences): Nothing I designed went live without four consecutive shadow weeks, a promotion bar published before week one and a rollback kept warm.

STAR (≈90 s): Situation: padding fails silently and cutting fails loudly, so every promise system drifts slow. Task: roll out a tighter promise without a loud failure. Action: shadow mode — the challenger computed and logged per order, never displayed — promoted only if would-be accuracy held the floor on logged live orders with no rise in contact signals; and the Q4 drill showed a frozen summer dial printing 91.9%, so k is re-solved and published in October. Result: a governance instrument the client’s business owner — not data science — owns.

Land this number: 4 shadow weeks; Q4 σ × 1.45 → 91.9%. · **Pivot to NACF:** Any NACF promise change shadows a peak before it displays. · **Do NOT say:** Do not say the shadow ran — it is the design of the engagement.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: “The headline was 95.8% and the policy was broken twice.”

Headline metric: Φ(5.0) vs Φ(0.69) = 75.4%

- Flat 1.10 d = 5.0σ on one cohort, 0.69σ on another.
- Price of the floor on the wide cohort: +0.50 d for 97.5%.
- σ-estimation: ≈ 584 orders per cohort-window to hold within half a point.
- All model figures.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Have Backbone; Invent and Simplify

Open: “Data science wanted more model; I recommended one dial first.”

Headline metric: +32 pts at 95.0%; D at 97.8% shown, never claimed

- The “no” was a priced row in the table.
- D ≈ +$20M/yr rides on a $12 breach cost I cannot observe.
- B holds a floor I can defend.
- One dial, owner named.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Learn and Be Curious; Are Right, A Lot

Open: “A frozen dial broke the floor in the Q4 drill — 91.9%.”

Headline metric: Q4 σ × 1.45 → 91.9%

- Re-solve k per regime, published in October.
- Normal errors understate the 99% dial; the empirical quantile fixes it.
- No production dollar — design engagement.
- VERIFY-WITH-RAJESH any failure beyond the drill is not in the record.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 “Your σ is estimated. How many orders per cohort before the 95% floor is safe?”

The numbers: SE(σ̂) = 1 ÷ √(2(n−1)): n = 100 → 7.1% → realised 93.8%; n = 1,000 → 94.6%; ≥ 94.5% at a one-SE miss needs n ≈ 584.

Say exactly: “With a hundred orders a cohort, a σ one standard error low delivers about 94 percent against the 95 floor; to stay within half a point at a one-SE miss you need roughly 584 orders per cohort-window — trivial at a million orders a week, so the real risk is regime change, not estimation.”

T2 “Real transit errors are heavy-tailed. Does b = 1.645σ still hold 95%?”

The numbers: Student-t_5 scaled to unit variance: 95% quantile 1.561σ (normal dial delivers 95.6%); 99% quantile 2.607σ (normal dial delivers 98.5%).

Say exactly: “At the 95 percent floor the normal dial is slightly conservative on heavy tails — it delivers about 95.6 — but at 99 it would fall to about 98.5, so production uses the empirical quantile and I treat the normal rule as the whiteboard version.”

T3 “Policy D is worth $20M a year. Why didn’t you ship it?”

The numbers: B: +$255K/wk × 52 = $13.3M/yr at a 95% floor. D: ($6.07M − $5.69M) × 52 = $19.8M/yr at 97.8% — on a $12 breach cost vs $6.50 margin, both model assumptions; at $6 the optimum falls toward the 95% floor.

Say exactly: “Policy D is the priced governance decision I handed to the client’s finance partner — it rests on a twelve-dollar breach cost nobody can observe; B holds a floor I can defend and is auditable, so I recommended B first and showed D without ever claiming it.”

S20 — Elementum / Walmart Marketplace — the weekly business review for pre-purchase delivery experience: 7 outputs over 10 inputs, factor math that names the owner in minutes, and an Andon that cuts detection-to-action from ~30 days to under 24 hours (engagement design) NEW

LPs this story serves here: Dive Deep (primary — factor math, the aggregate trap) · Ownership (one owner per output) · Invent and Simplify (7 outputs, 10 inputs, one joint headline) · Bias for Action (the Andon)
Process family: Plan / Deliver seam — metric architecture and control loop for the pre-purchase promise (Walmart Marketplace; role R-2596147)

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Dive Deep** — I decomposed accuracy into O1 ≈ I3 × I4 × Φ(b/σ) = 0.97 × 0.985 × 0.996 ≈ 95.2% so a red week names its factor — a regional carrier miss (97% national, 82% in the Southeast) was attributed in minutes.
- **Ownership** — Promise accuracy was owned by a committee, so by nobody between Tuesdays; every output now has one named owner and every input a roll-up and an owner.
- **Invent and Simplify** — Seven outputs over ten inputs and one joint headline, Promise Yield = fast share × accuracy; “two dashboards is zero dashboards” is a written rule.
- **Bias for Action** — The Andon — a 7-day control chart per seller-lane cell, act within 24 hours — replaces a 30-day rolling window that let a degrading cell burn customers for weeks.

Honesty label. Elementum engagement for Walmart Marketplace's pre-purchase delivery-experience organization. I designed the weekly business review — the metric architecture, the decomposition math, the thresholds and the control loop — and led the program that stood it up; the client's engineering built the pipelines and dashboards and their teams own the lines. I never configured or deployed any platform, and I never claim a production output metric moved — the engagement delivered the instrument, its sample-data proof, and the Andon design. The production-record version of this muscle is S11 (Flexport), and I keep the two separate.

60-SECOND PITCH

"**Situation.** A marketplace client whose delivery promise was owned by a committee: data science owned the model, engineering the engine, seller ops the handling, carrier programs the lanes — and the one number the customer feels, promise accuracy, was owned by nobody between Tuesdays. The scorecards ran on 30-day rolling windows, so a degrading seller or carrier did weeks of customer damage before any floor bit, and a red week was a blame negotiation. **Task.** Design and stand up the weekly business review for the pre-purchase delivery experience — outputs a customer feels, input levers a named owner can turn, and the math that connects them — plus a control loop faster than the 30-day clock. **Action.** Discovery: one glossary, one event source, audited denominators. Plan & Design: seven output metrics over ten input levers, each input with a roll-up to the output it drives and a named owner; the factorization — accuracy ≈ seller-ships-in-window × carrier-meets-transit × Φ(buffer/σ) — as the decomposition; red/green at ±5% relative week-over-week with hard reds on policy floors. Build & Test: the sample-week proof — 0.97 × 0.985 × 0.996 ≈ 95.2% — and the drill: a regional carrier miss (97% nationally, 82% in the Southeast) decomposed to its factor in minutes. Cutover: Monday data freeze, Tuesday review, reds pre-wired with cause, owner, date. Hypercare: mix-adjusted twins next to every raw output, and the Andon — a 7-day control chart per seller-lane cell, act within 24 hours. **Result.** Primary metric: red-week attribution — **a blame negotiation → minutes, by factor math**; detection-to-action 30 days → under 24 hours by design; one joint headline, Promise Yield = fast share × accuracy, baseline 41.5% on the docs' sample. The only dollar is modeled: about $580K per caught degradation episode, and I say 'modeled' in the same breath."

20-SECOND VERSION NEW

The delivery promise was owned by a committee and scored on a 30-day window; I designed the weekly review — seven outputs over ten owned inputs, the factor identity that names the red, and an Andon that acts in under 24 hours — proven on sample data; the only dollar is modeled, about $0.58M per caught episode.

NACF BRIDGE — ONE SENTENCE

The NACF version is INV-E's OTIF chain: ship-in-window × carrier-meets-transit × buffer-absorbs-residual — a weekly review where every red names a factor and an owner.

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — The same Elementum client engagement — Walmart Marketplace's seller-fulfilled delivery program, pre-purchase side. The promise the customer sees decomposes across three parties: the seller ships in a window, a carrier meets a transit time, a model plus a buffer absorbs the residual. The operating problem was measurement and ownership, not effort: promise accuracy, on-time delivery and late shipment lived on 30-day rolling windows that smear a step change into a slow drift; aggregate metrics hid the variance (a carrier can print 97% nationally while running 82% in the Southeast after a hub consolidation — the engagement's own worked example); three teams kept three clocks (order date, ship date, promise date); and when a week went red, the review discovered it live and negotiated blame instead of reading a decomposition.

T — Design the weekly business review for the pre-purchase delivery experience and lead the program that stood it up: the output metrics a customer feels, the input levers a named owner can turn this week, the math that connects them, the thresholds, and a control loop faster than the 30-day scorecard — then hand it to owners and leave.

Why it mattered: an output metric owned by a committee is owned by nobody between Tuesdays, and a 30-day window means a degrading cell burns customers for weeks before policy floors bite — enforcement lag was a named strategic issue in the engagement record. Left unfixed: reds without owners, action items that age into a diary, mix-shift illusions celebrated as wins (portfolio accuracy improving while every cohort degrades, because volume shifted to easy metro lanes), and the speed-versus-accuracy trade argued as theology because nobody priced a breach.

The metrics on the table, with their baselines: red-week attribution time — days of cross-team negotiation (baseline condition); detection-to-action on a degrading cell — up to ~30 days on the rolling scorecard; output ownership — committee; the joint headline — none (fast share and accuracy lived on separate lines with separate owners, tradeable against each other invisibly); the docs' sample baseline for that joint number: fast share 43.6% × accuracy 95.2% ≈ 41.5% Promise Yield.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (the metric definitions, formulas, thresholds and the Andon parameters are the engagement's design record; the sample numbers are the docs' own illustrative set — six weeks of sample data proved the mechanics before any live line):

- **Discovery** — one glossary: every metric got one definition, one owner, one grain, one date basis (order date for operating cohorts; the 30-day windows kept only for seller-facing consequences); audited the denominators (cancelled orders quietly leaving the OTD base, suppressed listings leaving penetration). *Exit bar:* three teams' three clocks reconciled to one event source; no metric without a named definition owner.
- **Plan & Design** — the architecture: seven outputs (promise accuracy ≥95%, on-time delivery ≥90% policy floor, fast-promise penetration +2 pts/qtr, late shipment ≤5%, cancellation ≤2%, WISMO contacts −10% QoQ, A/B-measured conversion lift) over ten input levers (SSS share of GMV, ML-recommended handling coverage, seller handling adherence, carrier lane SLA adherence, transit-model error and freshness, buffer calibration, inventory-feed accuracy, eligibility coverage, engine latency/cache, onboarding throughput + 30-day retention) — every input carrying a roll-up to the outputs it drives and a named owner; red/green at ±5% relative week-over-week, hard reds on the policy floors regardless of trend; one joint headline, Promise Yield = fast share × accuracy. *Exit bar:* every output decomposable into named inputs with owners; definitions frozen in writing.

- **Build & Test** — the client's engineering built the datasets and dashboard; I owned the proof on sample data: the factor identity (accuracy ≈ I3 × I4 × Φ(b/σ)) computed live — 0.97 × 0.985 × 0.996 ≈ 95.2% — then the drill: drop the Southeast carrier cell to 82% on a tenth of volume and watch the decomposition name the factor and the owner in minutes. *Exit bar:* the six-step loop (measure → variance → decompose → root-cause → act → verify) executed end-to-end on sample weeks; a red that arrives without cause, owner and date counted as a process miss.
- **Cutover / Go-Live (design)** — the operating rhythm: Monday data freeze on the Mon–Sun week, Tuesday review, pod leads pre-wire reds before the room; the action log with owner, date and visible ageing; "the meeting decides; it never discovers." *Exit bar:* two consecutive sample-cycle reviews run entirely pre-wired; the output block and the action log assigned to one named owner.
- **Hypercare / Close** — the honesty layer and the control loop: mix-adjusted twins printed next to every raw output (fixed base-quarter cohort weights; the delta *is* the mix effect) so Simpson's paradox can't celebrate volume migrating to easy lanes; and the Andon — a 7-day EWMA/CUSUM control chart per seller-lane cell, tripwire at a sustained 2× baseline breach rate with n ≥ 30 orders, action within 24 hours (widen that cell's buffer or pull its fast badge), auto-restore on recovery, designed false-trip rate under 5%. *Exit bar:* the verify step closing the loop — if an input moved and its output didn't, the roll-up hypothesis is declared wrong and re-examined, in writing.

The client's engineering built every pipeline and surface; data science owned the models behind the transit and buffer lines; seller ops and carrier programs owned their levers. Mine: the glossary and denominator audit, the output/input architecture and the roll-up map, the factorization as the decomposition instrument, the thresholds and floors, the six-step loop, the mix-adjusted twins, the Andon design and its parameters, and the handoff that left the output block owned by a person, not a committee.

The obstacle was that every team already had a dashboard and nobody wanted a new clock — overcome by making the review cheaper than the status quo: the decomposition turned the worst meeting of the week (the blame negotiation) into a five-minute read, and the pre-wire rule moved the argument out of the room. The decision I challenged: the program wanted more metrics; I held the board to seven outputs and ten inputs — a metric nobody owns or acts on came off the board, and "two dashboards is zero dashboards" went into the design note as a rule, not a preference.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: red-week attribution — from a cross-team blame negotiation to minutes, by factor math (proven on the sample-data drill); detection-to-action on a degrading cell — ~30 days → under 24 hours by design (the Andon). The only dollar is modeled: ≈$0.58M per caught episode.

- **$** — ≈$0.58M per caught degradation episode, modeled: a Southeast-type carrier miss costs ≈$192K/week in excess breaches; caught and corrected in ≤1 week instead of ~4, three weeks × $192K ≈ $576K avoided (inputs below; the $12 breach cost is the model's stated assumption). No annual figure — episode frequency is the client's data, and I won't invent it.
- **#** — 7 output metrics over 10 input levers, every input with a roll-up and a named owner; 1 joint headline (Promise Yield); 1 action log; 6 sample weeks proving the mechanics (engagement design record on the docs' illustrative set).
- **%** — Promise Yield baseline 43.6% × 95.2% ≈ 41.5% on the docs' sample, four-quarter design target 50%; red/green at ±5% relative WoW; hard reds on the floors (OTD < 90%, late shipment > 5%, cancellation > 2%); Andon false-trip design bar < 5%.
- **Time** — Red-week attribution: days → minutes (the factor math); detection-to-action: ~30 days of rolling-window lag → < 24 hours from a sustained signal (designed); Monday freeze → Tuesday review, reds pre-wired before the room.
- **Customer** — The Andon protects customers at the grain where variance lives (the seller-lane cell) instead of letting a degrading cell burn them for a month; WISMO contacts sit on the output board so promise failures are read in customer contacts, not just percentages.
- **Quality** — Mix-adjusted twins next to every raw output (the mix effect printed as its own line); denominators audited; the verify step that falsifies the lever map itself when an input moves and its output doesn't — the

mechanism that keeps the instrument honest after I leave.

Against the goal stated above: one owner of the output block instead of a committee, a decomposition that names the factor and the owner in minutes, a control loop 30× faster than the scorecard clock, and a joint headline that makes the speed-accuracy trade un-gameable at the KPI layer. The trade was explicit: seller-facing consequences stayed on the fair 30-day window — the Andon acts on cells and badges, never on scorecards — stability for sellers, speed for customers.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	No agreed glossary or owner for the promise output; I defined one definition, one owner, one grain, one date basis per metric.
Scope & influence	Data science, engineering, seller ops and carrier programs — none reporting to me; every team already had a dashboard and nobody wanted a new clock.
Execution	Metric architecture, decomposition proof, red-week drill, Monday freeze / Tuesday review, Andon design and hypercare rule.
Impact	Red-week attribution days → minutes (drill); detection-to-action ~30 days → under 24 hours by design; no production metric claimed.
Process improvement	A standing review with pre-wired reds, mix-adjusted twins and an action log with owners and dates.
Complexity	Aggregates hiding cells; mix-shift illusions; rolling-window lag; a gameable pair of metrics.
Root cause	The root cause was ownership and clocks, not data: three teams on three clocks and no joint metric.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: the engagement's sample set — six weeks, seven outputs, ten inputs; fast share 43.6% × accuracy 95.2%; carrier 97% national vs 82% Southeast. Revealed: the aggregate hid the cell; the rolling window hid the lag.
2 · Solution design	Options: more metrics · fewer, owned metrics · 30-day windows · a 7-day cell chart. Chosen: 7 outputs / 10 inputs / one joint headline / Andon per seller-lane cell.
3 · Math & data logic	Math: O1 ≈ I3 × I4 × Φ(b/σ) = 0.9516; episode: I4′ = 0.9685 → O1′ = 93.6%; $192K/wk × 3 wks = $576K; Andon binomial false-trip arithmetic; mix-shift example; Promise Yield gaming test.
4 · Implementation	Implementation: glossary → architecture → sample-data proof → Monday freeze, Tuesday review → pre-wired reds → Andon with auto-restore → mix-adjusted twins.
5 · Probe questions	Probe questions: Andon false-trip rate · mix-shift · whether the joint headline can be gamed · any production claim — see Trap Drills.

PROBING — CHALLENGE Q&A (MASTER RECORD) PLUS THE STATISTICAL QUESTIONS

> ***Challenge* — Why did you choose this story to illustrate Insist on the Highest Standards?**
>
> Because the standards here are numbers with teeth, published in advance: hard reds on policy floors regardless of trend, a pre-wire rule that makes an unexplained red a process miss, and an Andon whose tripwire, sample floor and false-trip bar were written before the first live week. A standard that can't stop a program is decoration.

> ***Probing* — Why did you choose to focus on these results? What other results were important?**
>
> Attribution speed and detection speed lead because they are what a review is for — a WBR that can't say which lever moved is a viewing, not a review. Also important: the joint headline (Promise Yield) that stops the two factor teams trading against each other invisibly, and the mix-adjusted twins that keep a growth quarter from flattering the numbers.

***Probing* — You mentioned dollars — what percentage change is that year over year?**

None claimed — the ≈$0.58M is per caught episode, modeled from the sample-data drill, and I don't multiply it by an invented episode frequency. The year-over-year story here is mechanism speed: 30 days of enforcement lag to under 24 hours.

***Probing* — I'm concerned your baseline Promise Yield is from sample data — tell me more.**

Correct, and I say so: 43.6% × 95.2% ≈ 41.5% is the engagement docs' own illustrative six-week set, used to prove the mechanics before live lines existed. The design target — 50% in four quarters — is a target on the real book, and the first live job is re-baselining it.

***Challenge* — What were the lessons learned? What would you have done differently?**

Lesson: the decomposition is the peace treaty — attribution math is what turns a blame negotiation into a five-minute read. Differently: I'd have run the denominator audit in week one rather than mid-design; two of the "metrics" we inherited changed meaning once cancelled orders stopped quietly leaving the base.

***Challenge* — How would you implement this at Amazon?**

This is Amazon's own grammar — input levers under output metrics, reds pre-wired with an ask, COE-style mechanism fixes — so the transplant is the control loop: factor math on the promise seam, mix-adjusted twins beside raw outputs, and cell-grain Andon where a 30-day window is too slow for the customer.

***Challenge* — How did these results compare to your actual goals?**

The signed goals were one owner of the output block, decomposition in minutes, and a sub-24-hour control loop — all three delivered as the standing instrument, proven on six sample weeks, with the verify step written so the lever map itself can be falsified and corrected.

PHASE DEEP-PROBE BANK — Q&A (12). ***Answer the phase asked, then stop.***

Discovery — How did you know discovery was done?

When three teams' three clocks — order date, ship date, promise date — reconciled to one event source, and every metric on the inherited boards had exactly one definition owner or was struck. The exit memo listed the denominator findings with owners, not hopes.

What did the denominator audit find?

The classic games, running unmalevolently: cancelled orders quietly leaving the on-time base, suppressed listings leaving the penetration base. Neither was fraud — both were defaults nobody owned. The rule that fixed it: metric governance audits denominators, not just numerators.

Plan & Design — Why seven outputs and ten inputs, not more?

Because a board nobody can hold in their head is a board nobody reads. The cut rule was ruthless: an input had to be provably upstream of an output, turnable this week, and owned by a name — a metric nobody owns or acts on came off. Everything else lives in drill-down, not on the board.

Why is Promise Yield the headline instead of accuracy?

Because either factor alone is gameable: chase accuracy and you pad buffers until every promise is slow and safe; chase penetration and you shave buffers until promises are fast and broken. The product punishes both games — and the board still decomposes it one line below, because a composite that can't be decomposed is a hiding place.

Why keep the 30-day window at all?

Fairness. A seller consequence — a scorecard, a suspension — deserves a stable clock that one bad day can't trip. The design splits the clocks: 30-day for consequences, 7-day panels for operations, and the Andon acting on cells and badges in hours. Each clock does the job it's fair at.

Build & Test — Walk me through the sample-week factor proof.

Seller-ships-in-window 0.97, carrier-meets-transit 0.985, buffer absorption Φ(b/σ) 0.996 — product 95.2%, matching the sample week's printed accuracy. Then the drill: Southeast carrier adherence to 82% on 10% of volume → the carrier factor falls to 0.9685, accuracy to 93.6%, and the week-over-week factor deltas name the carrier PM in one read.

How did you test the Andon's false-trip rate?

On the sample set: replayed stable cells through the EWMA/CUSUM chart and counted trips. The n ≥ 30 floor and the sustained-signal requirement (CUSUM, not single-week thresholds) are what hold designed false trips under 5% — a noise-tripped Andon punishes small sellers and kills the mechanism's legitimacy.

Cutover — What does "pre-wired" mean mechanically?

Before Tuesday, every red line arrives with its factor decomposition, the named input that moved, the owner, and a dated action. The meeting's job is to decide on the ask — never to discover the cause. A red discovered live is logged as a process miss on the review itself.

Who could stop what?

Hard reds on policy floors stop expansion into the affected cohort automatically — eligibility growth doesn't board on a floor breach. The Andon stops a cell's fast badge without waiting for a meeting. And the output-block owner can hold any launch that would change a metric's meaning mid-flight.

Hypercare — What are mix-adjusted twins and why print both?

Every output twice: raw, and re-weighted on fixed base-quarter cohort weights. The delta between them is the mix effect, printed as its own line — so the room stops celebrating "improvements" that are volume migrating to easy metro lanes, and stops flogging teams for "degradation" that is growth into hard geographies. Re-based quarterly, by written policy.

What does the verify step do when an input moved and the output didn't?

It declares the roll-up hypothesis wrong, in writing, and re-examines the lever map — that is how the map stays honest. A WBR that never falsifies its own causal claims degrades into numerology with owners.

Cross-phase — What did you refuse?

A bigger board (metrics without owners), a faster clock for seller consequences (unfair), an annual dollar claim for the Andon (no episode-frequency basis), and running the first live weeks myself — the instrument had to be owned by the client's named owner from day one, or it would have stayed mine and died when I left.

Analytics — What is the false-trip rate of an Andon that fires on “7-day breach rate ≥ 2× baseline, n ≥ 30”?

The record carries EWMA/CUSUM parameters and a false-trip design bar under 5%; the binomial analog shows why n ≥ 30 alone cannot meet it. Baseline breach rate 5% (the 95% floor): at n = 30 a “2×” trigger is 3 breaches, which a healthy cell produces 18.8% of weeks and catches a true 2× cell only 59% of weeks; the smallest trigger under 5% at n = 30 is 5 breaches (3.3× baseline, false trips 1.6%). At n = 100, “2×” is 10 breaches: false trips 2.8%, power 55%. A 2× rule is only clean from about n = 103; below that, “sustained” — several consecutive days — has to do the work.
VERIFY-WITH-RAJESH the exact EWMA/CUSUM parameters.

Analytics — Show me a mix-shift illusion.

Two cohorts, metro and rural. Before: 97% and 88% at a 50/50 mix = 92.5%. After: each cohort degrades a point — 96% and 87% — but volume shifts to 70% metro: 93.30%. The raw number rises 0.80 points while every cohort falls; at the old mix the same week is 91.5%, down 1.0. That is the mix-adjusted twin, printed beside every raw output. Illustrative numbers.

Analytics — Can Promise Yield be gamed?

Yes, if it is the only constraint. Yield = fast share × accuracy = 0.436 × 0.952 = 41.5%. Push fast share up 2 points and let accuracy fall to 94.2% and yield rises to 43.0% — a trade the product rewards and the customer pays for. That is why the floors are hard reds regardless of trend (accuracy ≥ 95%), and why yield is a headline, not a target.

STATISTICAL SPINE — ANDON FALSE-TRIP ARITHMETIC, MIX-SHIFT, HEADLINE GAMING (EXTRA TABLE)

The record: 7 outputs over 10 inputs; the factor identity; a 7-day Andon with false-trip design bar under 5% and EWMA/CUSUM parameters; mix-adjusted twins; hard floors. The table below is my binomial and mix arithmetic on illustrative cells — it explains the design, it is not a production measurement.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Healthy-cell breach rate	$p_0 = 1 - 0.95 = 5\%$	accuracy floor 95% (record-from-docs)	$p_0 = 5\%$
2	"2× baseline" at n = 30	trigger c = 2 × 0.05 × 30 = 3 breaches; false trips P(X ≥ 3 | 30, 0.05) = 18.8%; power at p = 10%: 59%	binomial analog of the 7-day cell chart	false trips 18.8% — over the 5% bar
3	Smallest trigger under 5% at n = 30	c = 5 breaches = 3.3× baseline; false trips 1.6%; power at 10%: 18%	same method	5/30
4	"2×" at n = 100	c = 10; false trips 2.8%; power at 10%: 55%	same method	false trips 2.8% — under the bar
5	Smallest trigger under 5% at n = 300	c = 22 = 1.47× baseline; false trips 4.86%	same method	1.47× suffices
6	Orders for a clean "2×" rule	$n = 0.0475 \times (2.33 \div 0.05)^2 = 103$	$p_0(1 - p_0) = 0.0475$; 2.33σ margin to 10%	≈ 103 orders per cell-week
7	Mix-shift: before	0.5 × 97% + 0.5 × 88% = 92.5%	illustrative cohorts	92.5%
8	Mix-shift: after, raw	0.7 × 96% + 0.3 × 87% = 93.30%	both cohorts −1 pt; mix shifts to metro	93.30% (up 0.80)
9	Mix-shift: after, old mix	0.5 × 96% + 0.5 × 87% = 91.5%	mix-adjusted twin	91.5% (down 1.0)
10	Promise Yield gaming	0.436 × 0.952 = 41.5%; 0.456 × 0.942 = 43.0%	fast share +2 pts, accuracy −1 pt (illustrative)	yield up 1.4 pts while accuracy falls

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Customer promise kept + operating speed (primary outcome)
+- Attribution speed (time) -> days of negotiation -> minutes
|    O1 ≈ I3 x I4 x Phi(b/sigma): the biggest factor WoW delta names the owner
+- Detection-to-action (time) -> ~30d rolling lag -> <24h (Andon per cell)
|    (loop) late detection -> weeks of excess breaches -> WISMO + trust decay
+- Excess-breach cost per episode ($) -> modeled below
+- Gaming resistance -> Promise Yield = fast share x accuracy (one headline)
+- Honesty -> mix-adjusted twins; audited denominators; verify step
     falsifies the lever map itself
```

DRIVER	BASELINE	AFTER	$	BASIS
Red-week attribution	cross-team negotiation, days	factor math, minutes	—	engagement design record (sample-data drill)
Detection-to-action on a degrading cell	up to ~30 days	< 24 h from sustained signal	—	designed (Andon: EWMA/CUSUM, 2× baseline, n ≥ 30)
Excess breaches during one Southeast-type episode	≈16,000/wk for ~4 wks	≤1 wk	≈$0.58M avoided per episode	modeled ($12 breach assumption; illustrative book)

DRIVER	BASELINE	AFTER	$	BASIS
Joint headline	none (two gameable lines)	Promise Yield 41.5% (sample) → 50% 4-qtr target	—	record-from-docs (sample set); target is design
Annual Andon value	not claimed — episode frequency is the client's data		—	deliberately uninvented

If challenged. *"You never ran this live."* — "Correct, and I say so — the engagement delivered the instrument proven on six sample weeks; the client's owners run it. The production record for running this cadence weekly for a year is S11, with record numbers." *"The $580K rests on a $12 breach cost."* — "The $12 is the model's stated bundle — contact, refund risk, trust decay — put on the table so Finance could attack it. At $6 the episode is still ≈$290K; the mechanism argument — act in hours, not weeks — survives any price." *"Isn't a sub-24-hour Andon rash?"* — "It acts on a cell's buffer or badge — reversible in one flag, auto-restored on recovery — never on a seller's scorecard. The n ≥ 30 floor and the sustained-signal test are the guards; the 30-day window stays for consequences."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The accuracy identity	`O1 ≈ I3 × I4 × Φ(b/σ) — seller ships in window × carrier meets transit × buffer absorbs the residual`	record-from-docs factorization; three factors, three owners	every breach belongs to exactly one factor
2	Healthy sample week	`O1 = 0.97 × 0.985 × 0.996 = 0.9516 ≈ 95.2%`	I3 = 0.97, I4 = 0.985, Φ = 0.996 (docs' illustrative sample)	95.2% — matches the sample print
3	The aggregate trap	`carrier prints 97% nationally while one region runs 82%`	Southeast after a hub consolidation (record-from-docs example)	the average hides the cell
4	The rolling-window lag	`a step change entering a 30-day window moves the average ≈1/30 per day`	30-day scorecard mechanics (record-from-docs issue)	weeks before a floor bites — enforcement lag
5	The gameable pair	`fast share 43.6% and accuracy 95.2% on separate lines, separate owners`	docs' sample; either can be juiced by bleeding the other	no joint headline
6	Baseline joint number	`Promise Yield = 0.436 × 0.952 = 0.415`	rows 2, 5 (docs' sample)	≈ 41.5%

Table 2 — the fix: step by step, values substituted (the red-week drill).

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The episode	`Southeast cell: carrier adherence 0.985 → 0.82 on w = 10% of volume`	82% is the docs' example; the 10% weight is illustrative	one regional carrier miss
2	New carrier factor	`I4' = 0.90 × 0.985 + 0.10 × 0.82 = 0.8865 + 0.082 = 0.9685`	volume-weighted blend (illustrative)	I4: 0.985 → 0.9685 (−1.7%)
3	New accuracy	`O1' = 0.97 × 0.9685 × 0.996 = 0.9357 ≈ 93.6%`	rows 1–2; other factors unchanged	95.2% → 93.6% — a red week
4	The attribution read	`WoW deltas: I3 0.0% · I4 −1.7% · Φ 0.0%`	computed per factor (the decomposition)	the red names I4 and the carrier PM — in minutes
5	Segmentation	`factor red isolated to one region ⇒ C-family cause, not model or seller`	lane/cohort cut of I4 (design)	cause class identified before the meeting

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
6	The Andon trigger	`cell 7-day breach rate sustained ≥ 2 × baseline, n ≥ 30 ⇒ act < 24 h`	EWMA/CUSUM parameters (record-from-docs design)	buffer widened or badge pulled on the cell, same day
7	Auto-restore	`control chart recovers ⇒ badge/buffer restored, no meeting needed`	design	surgical, reversible, seller-fair

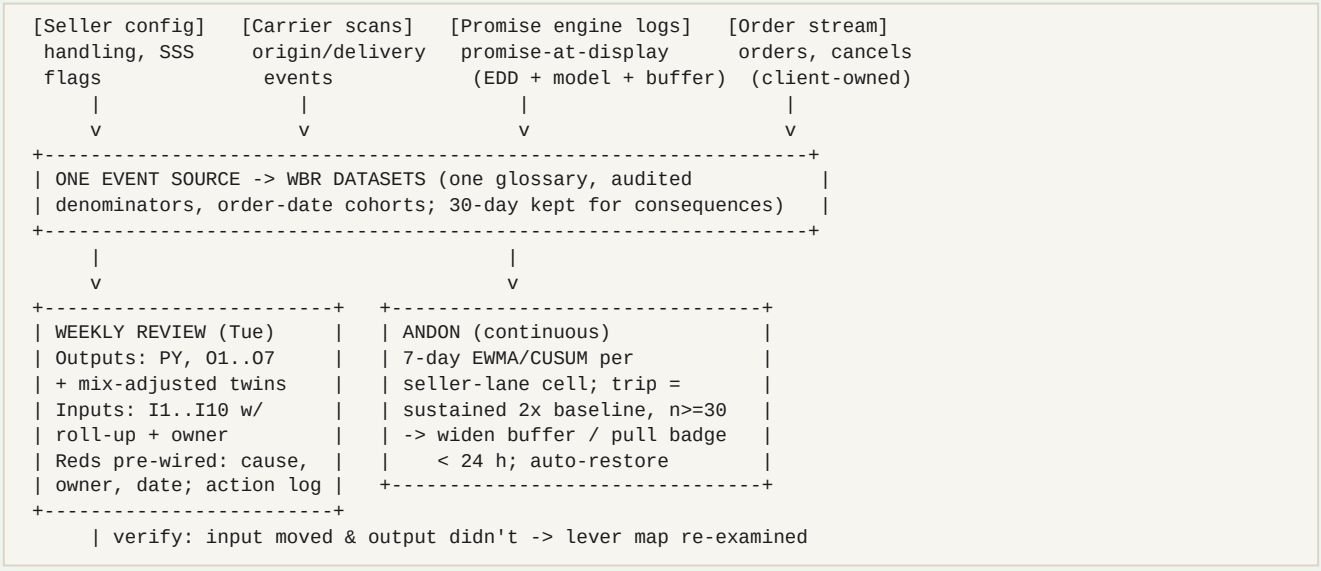
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Red-week attribution time	days (blame negotiation)	minutes (factor math)	engagement design record, proven on the drill
2	Detection-to-action, degrading cell	up to ~30 days	< 24 h from sustained signal	designed (Andon)
3	Output ownership	committee	one named owner of the output block + log	engagement design record
4	Joint headline	none	Promise Yield, 41.5% sample baseline → 50% 4-qtr target	docs' sample; design target
5	Mix honesty	raw outputs only	mix-adjusted twin + mix line per output	engagement design record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Episode accuracy loss	`95.2% − 93.6% = 1.6 points`	Table 2 rows 2–3 (illustrative episode)	1.6 pts of excess breaches
2	Excess breaches per week	`1,000,000 × 1.6% = 16,000`	~1M orders/wk illustrative book	≈16,000 breaches/wk
3	Cost per breach	`$12, taken as stated`	the model's assumption: contact + refund risk + trust decay (record-from-docs)	$12 each
4	Excess cost per week	`16,000 × $12 = $192,000`	rows 2 × 3 (modeled)	≈$192K/wk
5	Detection advantage	`~4 weeks (rolling-window era) − 1 week (Andon era) = 3 weeks`	row 4 of Table 1; Andon design	3 weeks earlier action
6	Avoided per episode	`$192,000 × 3 = $576,000`	rows 4 × 5 (modeled)	≈$0.58M per caught episode
7	Annual total	`not computed — $576,000 × (episodes/yr) requires a frequency I don't have`	deliberately uninvented	per-episode only, labeled modeled

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Breach cost (model assumption)	$6	$12	$18
Episode size (regional weight × adherence drop)	5% × −10 pts	10% × −16.5 pts	15% × −20 pts
Excess cost per week	≈$50K	≈$192K	≈$430K
Weeks of detection advantage	2	3	4
Avoided per caught episode	≈$0.1M	≈$0.58M	≈$1.7M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"What the engagement delivered is an instrument: seven output metrics a customer feels over ten input levers with named owners, one joint headline — fast share times accuracy — and the factor math that attributes a red week to seller, carrier or model in minutes instead of a week of negotiation. The control loop went from a thirty-day rolling window to under twenty-four hours at the seller-lane cell. The one dollar I'll quote is modeled and I'll label it as I say it: a regional carrier degradation of the kind in the engagement record costs about a hundred ninety thousand a week in excess breaches at the model's twelve-dollar breach cost, and catching it three weeks earlier is worth roughly five hundred eighty thousand per episode. I don't annualize it — the episode frequency is the client's data, not mine."

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 Tell me about a time you found the root cause of a problem everyone was arguing about. DIVE DEEP

Open (first two sentences): Carrier on-time printed 97% nationally and the Southeast was at 82%. The review used to spend a week arguing about whose fault that was; the factor math named the cell in minutes.

STAR (≈90 s): Situation: promise accuracy was owned by a committee and a red week was a blame negotiation. Task: design a review where a red names its cause. Action: the identity accuracy ≈ seller ships in window × carrier meets transit × Φ(buffer ÷ σ) — 0.97 × 0.985 × 0.996 ≈ 95.2% — and the drill: drop the Southeast carrier cell to 82% on a tenth of volume and watch the decomposition: carrier factor 0.985 → 0.9685, accuracy 95.2% → 93.6%, the carrier factor named. Result: attribution in minutes on the sample data, with a named owner.

Land this number: 0.985 → 0.9685; 95.2% → 93.6%; attribution days → minutes. · **Pivot to NACF:** INV-E's OTIF chain is the same decomposition for NACF inbound. · **Do NOT say:** Do not say any production metric moved — the engagement delivered the instrument and its sample-data proof.

Q2 Tell me about a time you took ownership of something with no clear owner. OWNERSHIP

Open (first two sentences): The one number the customer feels — promise accuracy — was owned by nobody between Tuesdays.

STAR (≈90 s): Situation: data science owned the model, engineering the engine, seller ops the handling, carrier programs the lanes. Task: a weekly review with one owner per output and a named owner per input. Action: one glossary — one definition, owner, grain and date basis per metric — seven outputs, ten inputs with roll-ups, red/green at ±5% relative week-over-week and hard reds on policy floors, reds pre-wired with cause, owner and date before the room. Result: the review became cheaper than the status quo — a five-minute read instead of the worst meeting of the week.

Land this number: 7 outputs · 10 inputs · every one with an owner. · **Pivot to NACF:** Every NACF review metric needs a definition owner and a grain. · **Do NOT say:** Do not say the client adopted it in production.

Q3 Tell me about a time you pared something down to make it work. INVENT AND SIMPLIFY

Open (first two sentences): The program wanted more metrics. I held the board to seven outputs and ten inputs and wrote "two dashboards is zero dashboards" into the design note.

STAR (≈90 s): Situation: every team already had a dashboard and nobody wanted a new clock. Task: make the review cheaper than what it replaced. Action: a metric nobody owns or acts on comes off the board; one joint headline, Promise Yield = fast share × accuracy, 43.6% × 95.2% ≈ 41.5% on the docs' sample; mix-adjusted twins beside each raw output. Result: a board small enough to run in a meeting; four-quarter design target 50% for Promise Yield.

Land this number: 41.5% sample baseline → 50% design target. · **Pivot to NACF:** Same discipline for any placement scorecard: fewer owned metrics. · **Do NOT say:** Do not present the 50% as a result — it is a design target.

Q4 Tell me about a time you built something to act faster than the process allowed. BIAS FOR ACTION

Open (first two sentences): A 30-day rolling scorecard lets a degrading seller or carrier burn customers for weeks. I designed a control that acts in 24 hours.

STAR (≈90 s): Situation: a step change entering a 30-day window moves the average about 1/30 per day, so floors bit weeks late. Task: close the loop faster without a meeting. Action: the Andon — a 7-day control chart per seller-lane cell; sustained breach at ≥ 2× baseline with n ≥ 30 triggers a same-day action — widen the buffer or pull the badge — and auto-restores when the chart recovers. Result: detection-to-action designed from ~30 days to under 24 hours; ≈$0.58M per caught episode modeled.

Land this number: ~30 days → < 24 hours (design); ≈$0.58M per episode (modeled). · **Pivot to NACF:** NACF inbound: a per-vendor-lane control chart with an auto-restore beats a monthly scorecard. · **Do NOT say:** Do not annualize the $0.58M — episode frequency is the client's data.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: "The carrier was 97% nationally and 82% in one region — the average hid the cell."

Headline metric: accuracy 95.2% → 93.6% via one factor

- O1 ≈ I3 × I4 × Φ(b/σ).
- WoW deltas by factor: I4 −1.7%.
- Mix-adjusted twin: raw +0.8 pt, adjusted −1.0 pt.
- Andon false-trip arithmetic.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Have Backbone; Ownership

Open: "The program wanted more metrics; I held it to seven and ten."

Headline metric: 7 outputs / 10 inputs / 1 joint headline

- A metric nobody owns comes off the board.
- Hard floors regardless of trend.
- Reds pre-wired before the room.
- No production claim.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Are Right, A Lot; Learn and Be Curious

Open: "A multiplicative headline can still be gamed."

Headline metric: yield 41.5% → 43.0% while accuracy falls to 94.2%

- The joint headline needs hard floors beside it.
- n ≥ 30 alone gives 19% false trips at 2×.
- This is a design engagement — nothing in production moved.
- VERIFY-WITH-RAJESH any live Andon results are not in the record.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 "Your Andon fires at 2× baseline with n ≥ 30. What is the false-trip rate?"

The numbers: p_0 = 5%: n = 30, c = 3 → false trips 18.8%, power 59%; need c = 5 for < 5%; n = 100, c = 10 → 2.8%; clean 2× rule from n ≈ 103.

Say exactly: "On a single week at thirty orders a healthy cell trips about 19 percent of the time at two-times baseline and a true doubling is caught only about 59 percent — so the record's EWMA and "sustained" logic carries the false-trip bar under five percent; a clean two-times rule needs about 103 orders a cell-week."

T2 "Raw accuracy improved. Is it improving?"

The numbers: Before 0.5 × 97 + 0.5 × 88 = 92.5%; after 0.7 × 96 + 0.3 × 87 = 93.30%; at the old mix 91.5%.

Say exactly: "Not necessarily — in my example the raw number rises about 0.8 of a point while every cohort falls a point, because volume moved to the easy lanes; at the old mix it is down a full point. That is why a mix-adjusted twin prints beside every raw output."

T3 "Promise Yield is fast share × accuracy. Can a team game it?"

The numbers: 0.436 × 0.952 = 41.5%; fast share +2 pts, accuracy −1 pt → 0.456 × 0.942 = 43.0%.

Say exactly: "Yes, if it is the only constraint — two points of speed for one point of accuracy raises it from forty-one and a half to forty-three — so the accuracy floor is a hard red regardless of trend, and the yield is a headline, not a target."

S21 — Elementum / Walmart Marketplace — the stalled managed-shipping program: an incentive diagnosis, an all-or-nothing insurance contract and a shadow gate — about $3.7M a year per ten points of enrolled GMV, modeled NEW

LPs this story serves here: Dive Deep (primary — adverse selection in declared handling) · Invent and Simplify (an all-or-nothing contract) · Earn Trust (I claim the contract, never the curve) · Insist on the Highest Standards (shadow gate and graduation criteria)
Process family: Deliver seam — seller-fulfilled delivery: transit settings, handling and the promise (Walmart Marketplace; role R-2596147)

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Dive Deep** — I read the stall as adverse selection, not marketing: sellers declared one-day handling to win placement and hit it about 85% of the time — the miss became the marketplace's broken promise.
- **Invent and Simplify** — I rebuilt the offer as one insurance contract — transit automation + observed handling + a participating label — with the benefit priced in the seller's own dollars.
- **Earn Trust** — I claim the contract design and wave-one mechanics only — the adoption curve after wave one belongs to the client — and I say "modeled" in the same breath as $3.7M.
- **Insist on the Highest Standards** — Shadow-run a seller's orders under managed settings before flipping; a wave that fails a published criterion does not board the next cohort.

Honesty label. Elementum engagement for Walmart Marketplace's seller-fulfilled program. The client's "Simplified Shipping Settings" (SSS) lets the marketplace manage a seller's transit settings and absorb carrier-delay accountability; adoption had stalled at roughly 38–42% of seller-fulfilled GMV. I led the adoption-program design: the incentive diagnosis, the insurance-contract offer, the wave mechanics and the measurement pair. The client's product and engineering teams built Seller Center, the enrollment flow and the observed-handling computation — I never configured or deployed any platform, and I claim the contract design and the wave-one mechanics, never the adoption curve that followed.

60-SECOND PITCH

"**Situation.** A marketplace client owning the delivery promise on inventory in sellers' buildings. Its managed-shipping program — the marketplace picks the transit settings and absorbs carrier-delay accountability — had stalled at roughly 38 to 42 percent of seller-fulfilled GMV, so the majority of promise risk still rode on manual seller configuration. And that configuration was optimistic by construction: sellers declared one-day handling to win placement and hit it about 85 percent of the time — adverse selection, not bad luck; the miss became the marketplace's broken promise. **Task.** Lead the adoption-program design: diagnose why sellers refused, rebuild the offer, design the rollout and its gates, and leave the client a measurement pair that can't be gamed. **Action.** Discovery: enrollment-funnel and refusal interviews — sellers were trading away control for a benefit that was real but unpriced and weakly communicated; an incentive-design problem, not a feature-marketing problem. Plan & Design: the offer rebuilt as an all-or-nothing insurance contract — transit automation plus observed handling plus a participating label, full bundle earns scorecard protection — with a personalized GMV-delta line in Seller Center making the benefit concrete in the seller's own dollars. Build & Test: shadow-run a seller's orders under managed settings before flipping, so day one is verified, not hoped. Cutover: waves behind published graduation criteria — a wave that fails one doesn't board the next cohort. Hypercare: the honest pair — activations per week *and* 30-day retention; enrollments without retention is churn wearing a growth costume. **Result.** Primary deliverable: the signed contract design and wave-one mechanics; the modeled economics — **about $3.7M a year per ten points of enrolled GMV** from handling-miss reduction alone — and I say 'modeled' in the same breath; the adoption curve after wave one belongs to the client, not to my record."

20-SECOND VERSION NEW

The managed-shipping program stalled at ~38–42% of GMV because sellers traded control for an unpriced benefit; I rebuilt it as an all-or-nothing insurance contract with a shadow gate and published wave criteria — about $3.7M a year per ten points of enrolled GMV, modeled; the adoption curve is the client's.

NACF BRIDGE — ONE SENTENCE

The NACF analog is vendor-managed inbound: where a vendor's declared lead time or handling is optimistic by incentive, I would move to observed performance and price the trade for the vendor.

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — The same Elementum client engagement — Walmart Marketplace's seller-fulfilled program, adoption side. The structural fact: the marketplace underwrites the delivery promise, but the inputs that keep it — handling time, ship nodes, carrier choices — are configuration data that thousands of independent sellers set themselves, optimistically, and update rarely. The managed-settings program existed precisely to fix this, and it had stalled at ~38–42% of seller-fulfilled GMV (record-from-docs). The refusal logic was rational: adoption asked sellers to give up control (their carrier rates, their handling declarations) for a benefit — accountability transfer — that was real but weakly priced and weakly communicated.

T — Lead the adoption-program design: diagnose the stall, rebuild the offer so adoption becomes the seller's rational choice, design the wave rollout with gates that protect both customer and seller, and define the measurement that keeps the program honest after handoff.

Why it mattered: with ~60% of GMV on manual configuration, the single largest driver of broken promises was seller-declared handling — sellers declared one day and hit it ~85% of the time (record-from-docs), and the 15% became the marketplace's broken promise at the marketplace's brand cost. Left unfixed: the promise team keeps buffering around seller optimism (taxing every customer on the lane to absorb one seller's miss), the scorecard keeps punishing sellers for carrier variance they don't control (eroding the program's legitimacy), and adoption stays stalled because the offer prices neither the benefit nor the trade.

The metrics on the table, with their baselines: managed-settings share of seller-fulfilled GMV — ~38–42%, stalled (record-from-docs); declared-handling adherence — ~85% on 1-day declarations (record-from-docs); the offer's value statement — unpriced (no seller could see what enrollment was worth in their own dollars); rollout mechanics — none designed; the measurement — enrollments counted, retention not.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (the baselines, the contract structure and the gate design are the engagement record; wave one ran as the pilot; the adoption curve beyond it is the client's, never claimed):

- **Discovery** — the refusal diagnosis: enrollment-funnel walk plus structured interviews with enrolled, refused and churned sellers (the refusals teach the most); the finding — fear of cost, loss of carrier control, distrust of the scorecard — an incentive-design problem, not a feature-marketing problem; the handling-adherence baseline written: declared 1-day, hit ~85%. *Exit bar:* the stall explained in the sellers' own words with the funnel's biggest drop-off named; the client's product owner co-signed the diagnosis.
- **Plan & Design** — the offer rebuilt as a priced insurance contract: (a) transit automation (the marketplace picks the service), (b) observed handling time (the marketplace computes the seller's real handling speed from scan history — the seller stops declaring, killing self-reported optimism at the source), (c) a participating label purchase — full bundle earns carrier-delay excusal from the seller's scorecard and the fast badge where the math clears; partial adoption earns nothing, so the hedge is all-or-nothing and self-selecting. Plus the pricing surface: a personalized line in Seller Center — "enrollment would raise your projected GMV by $X/month," computed by the client's team from the fast badge's measured conversion lift on that seller's assortment. *Exit bar:* the contract and its protection rules signed by product, seller ops and the scorecard owner; the design reference set (the dominant marketplace's equivalent bundle) documented, with the explicit decision *not* to copy its enforcement aggression while the client's own fulfillment alternative was subscale.

- **Build & Test** — the client's engineering built the enrollment flow, the observed-handling computation and the Seller Center line; I owned the verification design: shadow-run each enrolling seller's recent orders under managed settings before flipping — projected promise accuracy and badge eligibility computed on their own history, so day one is verified, not hoped. *Exit bar:* shadow results within tolerance of projection on the pilot cohort; the GMV-delta line reproducible by the client's analytics from logged lift data.
- **Cutover / Go-Live (wave one)** — wave mechanics: cohorts boarded behind published graduation criteria (shadow-verified settings, observed-handling coverage, inventory-feed accuracy at the seller's nodes, scorecard-protection rules armed); a wave that fails a criterion does not board the next cohort — publicly. *Exit bar:* wave one boarded and stable; no enrolled seller's scorecard damaged by a carrier miss the contract had excused; the refusal-reason codes flowing back into the offer design.
- **Hypercare / Close** — the measurement handoff: the honest pair on the program board — activations per week *and* 30-day retention — plus share of enrolled GMV under full-bundle protection; the kill condition written (a protection promise the scorecard fails to honor stops enrollment until fixed). *Exit bar:* the client's adoption PM owning the funnel, the pair and the gates; the engagement closed on the contract and the mechanics, with the curve explicitly left to the client's record.

The client's product team owned Seller Center and the enrollment surfaces; engineering built the observed-handling computation and the protection rules in the scorecard pipeline; seller ops ran the cohorts; their analytics computed each seller's GMV-delta line. Mine: the refusal diagnosis, the insurance-contract structure and its all-or-nothing logic, the pricing surface concept, the shadow-verification gate, the wave mechanics and graduation criteria, the measurement pair, and the recommendation on enforcement posture — carrots over sticks until the client's own fulfillment network was a credible alternative at scale.

The obstacle was the program's instinct to market the feature harder — more banners, more emails — when the stall was structural: an unpriced trade. The reframe that won the room: a seller who trusts their own ops should rationally keep control; a seller who doesn't should rationally convert — the offer's job is to make that self-selection visible in dollars, not to persuade everyone. The decision I challenged: requiring the label purchase as a hard cliff — sellers with negotiated carrier rates would read it as a tax. The recommendation: tier the protection with bundle depth rather than cliff it, and revisit when the fulfillment alternative scales. The trade went into the contract document as a priced row, not a hallway compromise.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary deliverable: the signed insurance-contract design, the shadow-verification gate, the wave-one mechanics and the measurement pair — with the modeled economics labeled: ≈$3.7M/yr per +10 points of enrolled GMV from handling-miss reduction alone. The adoption curve after wave one is the client's record, not mine.

- **$** — ≈$3.7M/yr per +10 pts of seller-fulfilled GMV enrolled, modeled: 100K orders/wk × (15% − 3%) handling-miss reduction × 50% breach-through × $12 × 52 (full chain in Table 4; the 3% and 50% are illustrative, the $12 is the model's stated assumption). No adoption "after" dollar — designed, not banked.
- **#** — One contract (three components, all-or-nothing), one pricing line per seller, one wave boarded; refusal interviews across enrolled, refused and churned cohorts; graduation criteria published before wave one (engagement design record).
- **%** — Baseline ~38–42% of seller-fulfilled GMV enrolled, stalled (record-from-docs); declared 1-day handling hit ~85% (record-from-docs); the designed measurement: % of enrolled GMV under full-bundle protection, alongside the pair.
- **Time** — Shadow verification before any seller's flip (day one verified, not hoped); wave cadence gated on criteria, not calendar; 30-day retention as the clock that keeps enrollment honest.
- **Customer** — The customer's promise stops depending on a seller's optimistic declaration: observed handling replaces declared at the source; the fast badge appears only where the math clears — the promise shown is the promise kept.
- **Quality** — The protection promise armed and audited (no enrolled seller's scorecard damaged by an excused carrier miss in wave one — the wave's exit bar); refusal reason codes flowing back into the offer; the kill condition written before launch.

Against the goal stated above: the stall diagnosed as incentive design, the offer rebuilt so the trade is explicit and priced in the seller's own dollars, wave one boarded behind published gates, and the program handed over measuring the pair that can't be gamed. The trade behind the design was itself priced: enforcement tuned down from the reference mechanism's aggression — a churned seller is a strategic loss to a marketplace still in assortment-growth mode — carrots first, the stick deferred until the fulfillment alternative is credible.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Why adoption stalled was unknown; I framed it as an incentive-design problem and tested it with funnel and refusal interviews.
Scope & influence	Client product and engineering, seller programs and sellers — none reporting to me; the contract needed all of them.
Execution	Offer redesign, shadow gate, wave criteria and the measurement pair; wave one boarded.
Impact	≈$3.7M/yr per ten points of enrolled GMV, modeled; no claim on the adoption curve.
Process improvement	Graduation gates and an honest measurement pair (activations and 30-day retention) with a kill condition.
Complexity	Adverse selection, all-or-nothing bundling, shadow replay against declared settings.
Root cause	Declared handling is a bid for placement; the stall was incentives, not awareness.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data: enrollment funnel, refusal interviews, declared vs observed handling (85% hit rate), the GMV share on managed settings (~38–42%). Revealed: optimistic declaration; an unpriced benefit.
2 · Solution design	Options: more marketing · a priced all-or-nothing contract · mandatory enrollment. Chosen: insurance contract with shadow gate and published graduation criteria.
3 · Math & data logic	Math: 1M × 60% × 15% × 50% × $12 = $540,000/wk on the manual book; +10 pts → 100K × 12% × 50% × $12 × 52 = $3,744,000/yr; flip rule c = 11 misses in 200 shadow orders.
4 · Implementation	Implementation: diagnose → contract → shadow gate → waves behind criteria → activations × 30-day retention.
5 · Probe questions	Probe questions: interval on 85% · sensitivity of $3.7M · false-pass of the shadow gate · churn vs growth — see Trap Drills.

PROBING — CHALLENGE Q&A (MASTER RECORD) PLUS THE STATISTICAL QUESTIONS

> ***Challenge* — Why did you choose this story to illustrate Think Big?**
>
> Because the move is structural, not incremental: the program stopped marketing a settings toggle and started selling an insurance contract — accountability transfer, priced per seller, self-selecting at marketplace scale. Thinking big here meant redesigning the incentive so thousands of independent sellers each make the right call on their own.

> ***Probing* — Why did you choose to focus on these results? What other results were important?**
>
> The contract design leads because it is the thing I can honestly claim — the diagnosis, the structure, the gates, the measurement. Also important: the enforcement-posture recommendation (carrots until the alternative scales) and the kill condition — a protection promise the scorecard fails to honor stops enrollment, because one broken protection kills the program's legitimacy faster than slow adoption does.

> ***Probing* — You mentioned dollars — what percentage change is that year over year?**
>
> None claimed — the ≈$3.7M/yr is per ten points of enrolled GMV, modeled from the handling baseline, and the curve itself is the client's record. I give the unit economics so the investment argument is quantitative; I don't book the outcome.

***Probing* — I'm concerned you're taking credit for a program you didn't run to completion — tell me more.**

That's exactly why the claim is scoped: mine is the diagnosis, the contract, the gates, wave one and the measurement pair; the client's is the curve. I'd rather under-claim a design than over-claim an outcome — the same discipline as S17, where I refuse the award dollar on the carrier bid.

***Challenge* — What were the lessons learned? What would you have done differently?**

Lesson: adoption stalls are usually incentive-design problems wearing feature-marketing costumes — price the trade and the funnel moves. Differently: start the refusal interviews before touching the funnel data; the sellers' own words located the drop-off faster than the analytics did, and I ran them second.

***Challenge* — How would you implement this at Amazon?**

Amazon runs the reference version of this contract — observed handling, settings automation, protected labels, all-or-nothing accountability transfer. What I'd bring is the posture judgment: knowing when the contract's teeth fit the business (a mature fulfillment alternative, churn tolerance) and when they don't — and the measurement pair, because enrollments without retention is churn wearing a growth costume at any company.

***Challenge* — How did these results compare to your actual goals?**

The engagement's signed goals were a diagnosed stall, a rebuilt offer, a gated wave one and an honest measurement handoff — all four delivered. The goal was never "move the share N points under my name"; that outcome belongs to the owners who run the waves.

PHASE DEEP-PROBE BANK — Q&A (12). *Answer the phase asked, then stop.*

Discovery — How did you know discovery was done?

When the stall had one explanation that survived the sellers' own words, the funnel data and the product team's review: a real benefit, unpriced and weakly communicated, traded against a visible loss of control. The exit memo named the funnel's biggest drop-off and the three refusal reasons in rank order.

Why interview churned sellers, not just refusers?

Because churn is the expensive refusal — someone who tried the program and left knows exactly which promise broke. Wave-one's protection-audit exit bar came directly from a churned seller's story about a scorecard ding the program should have excused.

Plan & Design — Why all-or-nothing instead of à la carte?

Because partial adoption keeps the adverse selection: a seller who keeps declared handling keeps the optimism that breaks promises. Cede all three controls and the marketplace can underwrite you; keep any one and you keep the risk. The bundle is what makes the contract insurable — and self-selecting.

Why does observed handling matter so much?

It kills self-reported optimism at the source. Declared handling is a bid for placement (declare one day, hit it 85%); observed handling is a measurement. Moving the promise onto what sellers actually do — rather than what they say — is the single largest accuracy lever in the seller-execution family.

What was the reference mechanism and what did you refuse to copy?

The dominant marketplace's equivalent: the same three-part bundle with full accountability transfer, enforced aggressively. We copied the contract structure and the grain. We did not copy the enforcement aggression or the churn tolerance — this client was still in assortment-growth mode, its fulfillment alternative subscale, so a churned seller was a strategic loss. Tiered protection instead of a cliff, revisited when the alternative scales.

Build & Test — What exactly did the shadow run verify?

Each enrolling seller's recent orders replayed under managed settings: projected promise accuracy, badge eligibility, and the scorecard-protection rules firing where they should. Day one verified on the seller's own history, not hoped — the same calibration-before-onboarding toll as my Flexport waves (S11).

How was the GMV-delta line computed and policed?

The client's analytics computed it per seller from the fast badge's measured conversion lift applied to that seller's assortment — their model, reproducible from logged lift data. My rule on it: the line is a projection and labeled as one in Seller Center; an unlabeled promise there would be the program lying to its own sellers.

Cutover — What were the wave-one graduation criteria?

Shadow results within tolerance of projection; observed-handling coverage on the cohort; inventory-feed accuracy at the seller's ship-from nodes (a stale feed causes cancellations — the worse defect); and the protection rules armed and audited. A wave that failed a criterion did not board the next cohort — publicly.

What was the kill condition?

A protection promise the scorecard fails to honor — an enrolled seller dinged for an excused carrier miss — stops enrollment until fixed. The contract's credibility is the product; one broken protection spends it faster than any growth banks it.

Hypercare — Why activations *and* 30-day retention?

Because enrollments without retention is churn wearing a growth costume. The pair is the honest read: throughput says the funnel works; retention says the contract kept its promise. A program graded on activations alone will board sellers it should have fixed first.

What flowed back into the offer after wave one?

Refusal and churn reason codes, as data: which component sellers feared (the label requirement led), which protection they doubted, where the GMV-delta line read as marketing instead of math. Overrides with reasons are data; silent refusals are a funnel you can't fix.

Cross-phase — What did you refuse?

An adoption target under my name (the curve is the client's), a hard label-purchase cliff (a tax on sellers with negotiated rates — tiered instead), enrollment pushes ahead of the shadow gate (day one must be verified), and counting activations without retention. Every refusal is in the contract document as a priced row.

Analytics — How sure are you that handling adherence is 85%?

The 85% is a record-from-docs baseline. As statistics: a seller with 100 shipments at 85% on time has a 95% Wilson interval of 76.7% to 90.7%; with 400 shipments, 81.2% to 88.2%. That is why “observed handling” needs enough scan history per seller before it replaces a declaration.

Analytics — What does the shadow gate actually test?

Replay a seller’s recent orders under managed settings and flip only within tolerance of the projection. At a projected 3% miss rate on 200 orders, flipping when misses ≤ 10 has a 4.0% false-reject rate on a seller who truly is at 3%, and detects a seller truly at 6% with 66% probability. Illustrative gate; the record is shadow-run-before-flip.

STATISTICAL SPINE — INTERVALS, SENSITIVITY, SHADOW GATE, RETENTION (EXTRA TABLES)

The record is the engagement design and the modeled economics; the tables below add interval and sensitivity arithmetic. The adoption curve after wave one is the client’s and is not claimed.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Orders on manual configuration	`1,000,000 × (1 − 40%) = 600,000/wk`	enrolled ≈ 38–42%, 40% midpoint (record-from-docs)	600,000/wk
2	Seller-made handling misses	`600,000 × 15% = 90,000/wk`	85% hit rate (record-from-docs)	90,000/wk
3	Breach-through	`90,000 × 50% = 45,000/wk`	50% illustrative	45,000/wk
4	Cost of the manual book	`45,000 × $12 = $540,000/wk`	$12 breach bundle (model assumption)	≈$540,000/wk

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	+10 points of enrolled GMV	`1,000,000 × 10% = 100,000 orders/wk`	illustrative book	100,000/wk
2	Handling misses avoided	`100,000 × (15% − 3%) = 12,000/wk`	3% post-fix illustrative	12,000/wk
3	Breaches avoided	`12,000 × 50% = 6,000/wk`	50% illustrative	6,000/wk
4	Per week / per year	`6,000 × $12 = $72,000/wk; × 52 = $3,744,000`	modeled	≈$3.7M/yr per 10 pts
5	Sensitivity — miss reduction 6 / 12 / 15 pts	`$1.87M / $3.74M / $4.68M`	one-at-a-time	$1.87M–$4.68M
6	Sensitivity — breach-through 25% / 50% / 75%	`$1.87M / $3.74M / $5.62M`	one-at-a-time	$1.87M–$5.62M
7	Sensitivity — breach cost $6 / $12 / $18	`$1.87M / $3.74M / $5.62M`	one-at-a-time	$1.87M–$5.62M
8	Shadow gate	`n = 200, p₀ = 3%: reject if misses ≥ 11; false-reject 4.0%; power at 6% = 66%`	binomial, illustrative	c = 11
9	Churn in a growth costume	`200 enrollments × 50% retained = 100; 100 × 90% = 90`	illustrative weekly cohorts	raw enrollments say 2×; retained say 1.1×
10	Retention estimate, one wave	`40 of 50 retained = 80%; Wilson [67%, 89%]`	illustrative wave size	80% ± ≈12 pts

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Promise kept at the source + adoption that sticks (primary outcome)
+- Handling-miss rate on manual-config GMV (flow) -> declared ~85% kept
|    miss -> breach-through -> broken promise -> WISMO + trust decay
+- Enrolled GMV share (stock) -> ~38-42% stalled -> waves behind gates
|    (loop) priced offer -> rational enrollment -> observed handling
|           -> fewer misses -> badge earned -> conversion lift -> more GMV
+- Protection credibility (stock) -> kill condition; audit per wave
+- Measurement honesty -> activations/wk x 30-day retention (the pair)
+- Unit economics -> ~$3.7M/yr per +10 pts enrolled GMV (modeled)
```

DRIVER	BASELINE	AFTER (DESIGNED)	$/YR	BASIS
Enrolled share of seller-fulfilled GMV	~38–42%, stalled	waves behind gates; curve unclaimed	—	record-from-docs baseline
Handling adherence on declared 1-day	~85%	observed handling replaces declared; ~97% modeled	inside the unit economics	85%: record-from-docs; 97%: illustrative
Unit economics per +10 pts enrolled GMV	—	≈6,000 breaches/wk avoided	≈$3.7M	modeled (Table 4; $12 assumption)
Offer pricing surface	benefit abstract	personalized GMV-delta line per seller	—	engagement design record (client-computed)
Program honesty	enrollments counted	activations/wk × 30-day retention + full-bundle share	—	engagement design record

If challenged. *"Your result is a design, not an outcome."* — "Yes — and I say exactly that. Mine is the diagnosis, the contract, the gates, wave one and the measurement pair; the adoption curve is the client's record. The production-record version of me running an adoption machine end-to-end is S11: 7 to 68 behind published gates." *"The $3.7M rests on two illustrative rates."* — "It does — the 3% post-fix miss rate and the 50% breach-through are labeled

illustrative; the 85% baseline and the $12 are the engagement record's own figures. Halve both assumptions and the unit is still about a million a year per ten points — the argument survives." *"Why not just mandate enrollment?"* — "Because a mandate spends seller trust the marketplace hasn't earned yet — and a churned seller is a strategic loss in assortment-growth mode. The contract makes enrollment the rational hedge; mandates are for after the fulfillment alternative is credible."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	GMV on manual configuration	`1 − enrolled share = 1 − 40% = 60%`	enrolled ~38–42% (record-from-docs); 40% midpoint used	≈60% of seller-fulfilled GMV on declared settings
2	Orders on manual config	`1,000,000 × 60% = 600,000 / wk`	~1M orders/wk illustrative book	≈600K orders/wk
3	Handling misses	`600,000 × (1 − 85%) = 600,000 × 15% = 90,000 / wk`	declared 1-day handling hit ~85% (record-from-docs)	≈90K seller-made misses/wk
4	Breach-through	`90,000 × 50% = 45,000 / wk`	a handling miss of a day typically exhausts the buffer; 50% breach-through (illustrative)	≈45K broken promises/wk from handling alone
5	Cost of the manual book	`45,000 × $12 = $540,000 / wk`	$12 breach bundle (the model's stated assumption)	≈$540K/wk of handling-driven breach cost
6	Why it persists	`declared handling = a bid for placement; adoption = control traded for an unpriced benefit`	adverse selection + unpriced trade (record-from-docs diagnosis)	a stall no banner fixes

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The contract	`protection ⇔ (transit automation) AND (observed handling) AND (participating label)`	all-or-nothing bundle (engagement design)	full bundle → carrier-delay excusal + badge where the math clears
2	Observed handling effect	`miss rate 15% → ~3% on enrolled orders`	handling set from scan history, not declaration; 3% illustrative	−12 points of handling misses on enrolled GMV
3	Pricing the benefit	`seller line = conversion lift × seller's eligible assortment GMV = "$X/month"`	client-computed from measured badge lift (design)	the trade visible in the seller's own dollars
4	Self-selection logic	`trusts own ops ⇒ keeps control and risk; doesn't ⇒ converts`	incentive design, not persuasion	adoption as the rational hedge
5	Shadow gate	`replay seller's recent orders under managed settings; flip only within tolerance of projection`	verification design	day one verified, not hoped
6	Wave mechanics	`cohort boards ⇔ all published criteria pass; fail one ⇒ hold, publicly`	graduation gates (design; the S11 motion)	adoption earns its expansion
7	The honest pair	`program health = activations/wk AND 30-day retention AND full-bundle share`	measurement design	growth that can't be faked by churn

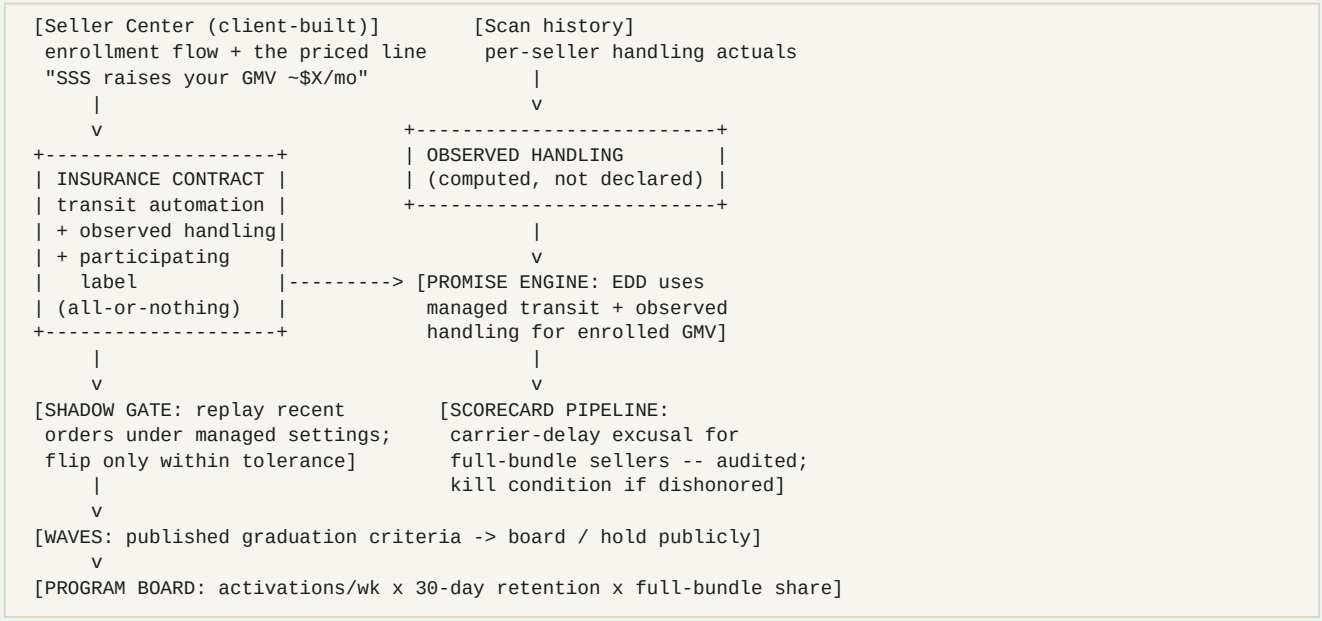
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	The offer	settings toggle, benefit abstract	priced insurance contract, benefit in seller dollars	engagement design record
2	Handling input on enrolled GMV	declared (15% miss)	observed (~3% miss, modeled)	15%: record-from-docs; 3%: illustrative
3	Rollout mechanics	none (push marketing)	shadow gate + waves behind published criteria; wave one boarded	engagement design record
4	Program measurement	enrollments	activations/wk × 30-day retention × full-bundle share; kill condition armed	engagement design record
5	Enrolled GMV share	~38–42% stalled	curve unclaimed — the client's record	record-from-docs baseline; honesty rule

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Orders in +10 pts of enrolled GMV	`1,000,000 × 10% = 100,000 / wk`	~1M orders/wk illustrative book	100K orders/wk move onto the contract
2	Handling-miss reduction	`15% − 3% = 12 points`	15% record-from-docs; 3% illustrative post-fix rate	12 pts fewer misses
3	Misses avoided	`100,000 × 12% = 12,000 / wk`	rows 1 × 2	≈12K misses/wk avoided
4	Breaches avoided	`12,000 × 50% = 6,000 / wk`	50% breach-through (illustrative)	≈6K broken promises/wk avoided
5	Weekly value	`6,000 × $12 = $72,000 / wk`	$12 breach bundle (the model's stated assumption)	≈$72K/wk
6	Annualized	`$72,000 × 52 = $3,744,000`	row 5 × 52 (modeled)	≈$3.7M/yr per +10 pts enrolled
7	Not claimed	`adoption curve × unit economics = the client's outcome`	honesty rule	unit economics only; conversion-lift upside also excluded

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Post-fix handling-miss rate (illustrative)	6%	3%	2%
Breach-through on a handling miss (illustrative)	30%	50%	70%
Breach cost (model assumption)	$6	$12	$18
Breaches avoided per +10 pts (/wk)	≈2,700	≈6,000	≈9,100
Modeled $/yr per +10 pts enrolled	≈$0.8M	≈$3.7M	≈$8.5M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The honest version first: what I delivered is the contract, not the curve. The program was stalled around forty percent of seller-fulfilled GMV because it asked sellers to trade control for an unpriced benefit — so we rebuilt it as insurance: give up transit settings, declared handling and the label, and the marketplace absorbs carrier-delay accountability and shows you what the fast badge is worth on your own assortment, in dollars. The unit economics are modeled and I label them: on the engagement record's own baseline — sellers declaring one-day handling and hitting it eighty-five percent of the time — every ten points of GMV that enrolls is worth about three point seven million dollars a year in avoided broken promises at the model's twelve-dollar breach cost. Wave one boarded behind published gates; the adoption curve after that belongs to the client's owners, and I say so."

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 **Tell me about a time you found the real reason a program stalled.** DIVE DEEP

Open (first two sentences): Adoption was stuck near forty percent and everyone wanted a better banner. The data said sellers were making a rational trade.

STAR (≈90 s): Situation: a managed-shipping program stalled at ~38–42% of seller-fulfilled GMV. Task: diagnose why. Action: enrollment-funnel and refusal interviews; the pattern was control traded for an unpriced, weakly communicated benefit, and declared one-day handling that sellers hit about 85% of the time — adverse selection. Result: the diagnosis moved the work from marketing to incentive design; the contract followed.

Land this number: 85% handling adherence; ~38–42% enrolled. · **Pivot to NACF:** NACF vendor ASN accuracy has the same incentive: declared dates are bids. · **Do NOT say:** Do not claim the adoption curve after wave one.

Q2 **Tell me about a time you redesigned an offer instead of pushing the existing one.** INVENT AND SIMPLIFY

Open (first two sentences): I made it one contract: give up transit settings, declared handling and take the label — and the marketplace absorbs carrier delays.

STAR (≈90 s): Situation: sellers refused a toggle. Task: a design they would choose. Action: an all-or-nothing insurance bundle — transit automation + observed handling + participating label — with a personalized GMV-delta line in Seller Center showing the benefit in the seller's own dollars. Result: signed contract design and wave-one mechanics; modeled ≈$3.7M a year per ten points of enrolled GMV.

Land this number: ≈$3.7M/yr per 10 pts (modeled). · **Pivot to NACF:** A priced, bundled commitment beats a toggle at NACF too. · **Do NOT say:** Do not call $3.7M a result.

Q3 **Tell me about a time you were precise about what was yours.** EARN TRUST

Open (first two sentences): I claim the contract design and the wave-one mechanics. The adoption curve after that belongs to the client, and I say so.

STAR (≈90 s): Situation: a program whose headline number — enrolled GMV — was moved by many hands. Task: keep my claim to what I did. Action: the honesty pair — activations per week and 30-day retention — so growth cannot be faked by churn; and in every retelling, "modeled" is in the same breath as the dollar. Result: a claim that survives a reference check.

Land this number: Wave one boarded; $3.7M modeled. · **Pivot to NACF:** In an Amazon interview, the line between "I designed" and "we shipped" is the line. · **Do NOT say:** Do not annualize or extrapolate the curve.

Q4 **Tell me about a time you held a high bar on rollout quality.** INSIST ON THE HIGHEST STANDARDS

Open (first two sentences): Day one is verified, not hoped: we replayed a seller's orders under managed settings before flipping.

STAR (≈90 s): Situation: flipping a seller wrongly breaks a promise. Task: a gate. Action: shadow-run before flip within tolerance of the projection; waves behind published graduation criteria — fail one, hold, publicly. As a design check: at n = 200 orders and a 3% projection the gate rejects a good seller 4.0% of the time and catches a 6% seller 66% of the time. Result: wave one boarded behind the criteria.

Land this number: c = 11 misses in 200 orders (illustrative). · **Pivot to NACF:** A NACF vendor-lane activation gate on valid-ASN rate, as in S9. · **Do NOT say:** Do not quote gate numbers as the record.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep

Open: “Declared handling was a bid for placement.”

Headline metric: 85% hit rate; manual book ≈$540K/wk breach cost (model)

- 1M × 60% × 15% × 50% × $12.
- 85% interval at n = 100: [76.7%, 90.7%].
- Contract: observed handling replaces declaration.
- Sensitivity $1.9M–$5.6M on miss reduction.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Have Backbone; Earn Trust

Open: “I would not put the adoption curve on my résumé.”

Headline metric: claim = contract + wave-one mechanics

- The curve is the client’s.
- “Modeled” in the same breath.
- Kill condition armed on the measurement pair.
- Honest pair: activations × retention.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Learn and Be Curious

Open: “The incentive diagnosis was not my first hypothesis.”

Headline metric: refusal interviews changed the frame from marketing to incentives

- The funnel alone suggested awareness.
- Interviews showed an unpriced trade.
- I would instrument seller-level retention earlier.
- VERIFY-WITH-RAJESH that this was the order of discovery.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 “Your $3.7M — which assumption dominates?”

The numbers: +10 pts: 100,000 × 12% × 50% × $12 × 52 = $3,744,000; miss reduction 6/15 pts → $1.87M/$4.68M; breach-through 25/75% → $1.87M/$5.62M; breach cost $6/$18 → $1.87M/$5.62M.

Say exactly: “It is a product of four inputs, so a halving or doubling of any one moves it proportionally — from about $1.9 million to $5.6 million on breach-through, $1.9 to $5.6 on the breach cost — and I call it modeled in the same breath; the client’s own data replaces each input.”

T2 “Eighty-five percent — what is the interval, and is 15% → 3% plausible?”

The numbers: n = 100: [76.7%, 90.7%]; n = 400: [81.2%, 88.2%]; 3% is illustrative.

Say exactly: “On a hundred shipments eighty-five percent is anywhere from about 77 to 91; the fifteen-to-three-point effect is an illustrative target — observed handling is the mechanism, and I would not quote three as measured.”

T3 “Enrollments are up — is the program working?”

The numbers: 200 enrollments × 50% retained = 100; 100 × 90% = 90; Wilson at 40/50 retained: [67%, 89%].

Say exactly: “Not necessarily — two hundred enrollments at fifty percent retention retain a hundred and a hundred enrollments at ninety retain ninety; raw enrollments say twice as good, retained activations say about the same, so the pair is activations per week and thirty-day retention, with a kill condition.”

E7 — Elementum design engagement (new-market broadband build) — equipment forecast-to-buy: a newsvendor first buy, risk-pooled central stock and a ±20% two-week re-forecast trigger, with the statistics behind each (design, not run) NEW

LPs this story serves here: Learn and Be Curious (primary — the toolkit applied, labeled) · Think Big (a repeatable launch template) · Are Right, A Lot (a decision under maximum uncertainty) · Customer Obsession (install delays caused by stockouts → 0)
Process family: Plan / Source seam — forecast-to-buy for a market launch (master E7; Google Fiber-style field-install build; finished-kit grain, no component explosion)

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH NEW

- **Learn and Be Curious** — I applied the newsvendor critical ratio and risk pooling from the MIT MicroMasters toolkit to a cold-start launch — labeled as the toolkit, not as a program I ran.
- **Think Big** — The design is a template for market N+1: build plan → adoption curve → requirement → measured lead times → first buy held central → forward-deploy as sell-through proves the curve.
- **Are Right, A Lot** — Under maximum uncertainty I priced the decision — Cu ≈ $300 vs Co ≈ $3/month → CR 0.99 — and wrote the trigger that proves the curve wrong in two weeks, not two quarters.
- **Customer Obsession** — The metric is install delays caused by stockouts → 0 in the first 90 days — a missed install is a customer waiting, priced at ≈$300 in the model.

Honesty label. Design engagement, not a run record. I have not owned a market-launch forecast end to end; the forecast number belongs to Planning — I own the equation, the measurement and the triggers (master E7). The newsvendor and pooling formulas are the MIT MicroMasters toolkit, labeled. All volumes, CVs and correlations here are illustrative. The master's E-series does not name the client; the brief ties them to Google Fiber. VERIFY-WITH-RAJESH that you are comfortable saying "Google Fiber" aloud — otherwise say "a broadband build-out client." Finished-kit grain only — no component explosion.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. A broadband client launches a new market: build sites ramp by month, demand follows an adoption curve with no local history, and long-lead equipment takes weeks to months to arrive. A missed install means a re-roll and deferred revenue; extra stock means holding cost and stranding risk. Task. Design the forecast-to-buy chain so installs are never delayed by stockouts and stranded stock stays inside a band. Action. Four decisions, each with its arithmetic. First buy: a newsvendor — a missed install costs about $300, a unit held a month about $3, so the critical ratio is 0.990 and I order the 99th percentile of the first quarter's demand for the core unit; accessories with a lower miss cost sit at the 70th. Held central: one pool covers several markets' variance, √k instead of k — if the markets are uncorrelated, which I would measure. Trigger: actual take-rate versus the curve beyond ±20% for two consecutive weeks forces a new forecast version. Demand enters planning once, through the engine — no spreadsheets beside it. Result. Primary metric: install delays caused by stockouts → 0 in the first 90 days of a launch. Illustrative dollar: 2,000 installs a month × 3% reschedules × $300 ≈ $18K a month, about $216K a year per market, plus about $100K of stranding avoided — design arithmetic, not a program result.

20-SECOND VERSION NEW

For a new-market build I would buy the core unit at the 99th percentile of first-quarter demand because a missed install costs 100× a unit-month of holding, hold it central so uncertainty pools, and re-forecast when the take-rate leaves ±20% of the curve two weeks running.

NACF BRIDGE — ONE SENTENCE

NACF placement for a new FC or a new product launch has the same shape: a critical-ratio first buy, central-first, and a variance trigger — INV-C's placement model plus INV-B's service-level logic.

SITUATION / TASK — THE PROBLEM SPACE

S — Market N launches: build sites ramp by month from the project system's build plan; demand follows an adoption curve with no local history; kits and long-lead equipment have vendor lead times of weeks to months. The books: the project system for the build plan, Planning for the adoption curve, SAP MRP for supply.

T — Design the forecast-to-buy chain so installs are never delayed by stockouts and stranded stock stays inside a band — the equation, the measurement and the triggers; Planning owns the forecast number.

Baselines on the table (illustrative): 2,000 installs a month per market · 3% of installs rescheduled for stockouts · missed install ≈ $300 · holding ≈ $3/month · first buy ≈ $500K.

ACTION — FIVE GATED PHASES (DESIGN)

- **Discovery** — restated the problem as a stock-and-flow map — build plan → sites passed → adoption curve × marketing → install demand → requirement by SKU × week → MRP on measured lead times → POs → central pool → market transfers. Exit bar: the three books of record named; the primary metric agreed.
- **Plan & Design** — newsvendor first buy; central-first with risk pooling; re-forecast trigger ±20% for two weeks; invariants: demand enters planning once, lead times are measured distributions (the S8 rule), floors with hold-until dates so one market cannot steal another's launch. Exit bar: each rule has its arithmetic on one page.
- **Build & Test** — the arithmetic below — newsvendor cost curve, pooling under correlation, trigger false-alarm and detection — run on illustrative inputs; failure modes listed (a borrowed adoption curve, a brochure lead time, manual loads). Exit bar: every rule survives its own sensitivity.
- **Cutover & Go-Live (design)** — launch plan: first buy held central; forward-deploy via transfers as sell-through proves the curve; bias sign published weekly from month two. Exit bar: install delays caused by stockouts = 0 in the first 90 days.
- **Hypercare / Close (design)** — MAPE and bias per SKU × market × week; days-of-cover inside the band; the template re-run for market N+1. Exit bar: a floor breach is a governed screen with a reason code, not an email.

Influence and ownership. This is a design I owned the equation of; Planning owns the forecast, Finance the unit costs, Marketing the adoption curve. The backbone content is the invariant: demand enters planning exactly once, through the engine — manually loaded requirements beside it are blocked — and need floors stop one market starving another.

RESULTS — DESIGN OUTCOMES, ALL ILLUSTRATIVE

- **Primary metric (design target)** — install delays caused by stockouts → 0 in the first 90 days of a launch.
- **$** — 2,000 installs/month × 3% × $300 = $18,000/month ≈ $216,000/yr per market; 20% × $500K = $100,000 of stranding avoided by holding central (master, illustrative) — never a program result.
- **Statistics** — critical ratio 0.990 → z = 2.33 → first-quarter buy 9,495 units on a 6,000-unit mean; pooling saves 50% of buffer at ρ = 0 and 31% at ρ = 0.3 (k = 4).
- **Quality** — MAPE and bias per SKU × market × week from month two; bias sign before magnitude.
- **Honest limit** — not run end to end; the forecast is Planning's.

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	A launch with no history; I restated it as a stock-and-flow problem with a priced objective.
Scope & influence	Planning, Finance, Marketing, vendors and field operations; the design is the influence.
Execution	Newsvendor, pooling, trigger and invariants each with arithmetic and a failure-mode list.
Impact	Design arithmetic: ≈$216K/yr per market of avoided reschedules plus ≈$100K stranding avoided (illustrative).
Process improvement	A launch template for market N+1; demand enters once; floors with hold-until dates.
Complexity	Cold-start demand, correlated markets, long-lead supply, a borrowed adoption curve risk.
Root cause	Launch stockouts come from an unpriced first buy and an untested curve — not from vendor behavior.

THE FIVE FACETS NEW — data analysis · solution design · math & data logic · implementation · probe questions; each labeled so the interviewer can see all five are present

1 · Data analysis	Data (design): build plan by month, adoption curve by archetype market, install-reschedule reasons joined to the supply view, vendor lead-time distributions (S8). Revealed (illustrative): the 3% baseline reschedule rate corresponds to ordering near the 79th percentile.
2 · Solution design	Options: forecast mean · p79 · p99 first buy; forward-deploy vs central-first; trigger at ±10% / ±20% / ±30%. Chosen: p99 on the core unit, p70 accessories, central-first, ±20% two weeks.
3 · Math & data logic	Math: CR = 300 ÷ 303 = 0.9901; z = 2.330; Q = 6,000 + 2.330 × 1,500 = 9,495; expected short σ·L(z) = 5.0; pooling σ√(k + k(k−1)ρ); trigger DP on consecutive exceedances.
4 · Implementation	Implementation: stock-and-flow map → rules with arithmetic → invariants → launch plan → weekly MAPE/bias → template for N+1.
5 · Probe questions	Probe questions: why p99 · what if the CV is wrong · does pooling survive correlation · trigger false alarms — see Trap Drills.

PROBING — CHALLENGE Q&A TUNED TO A STATISTICIAN

Analytics — Show the newsvendor logic with numbers.

Critical ratio = Cu ÷ (Cu + Co) = 300 ÷ 303 = 0.9901, so order the 99.0th percentile of first-quarter demand: z = 2.330. With mean 6,000 and σ = 1,500 (CV 25%, illustrative) that is 9,495 units; expected shortfall σ·L(z) = 5.0 units, expected leftover 3,500; expected cost $12,009, versus $58,234 at the 79th percentile.

Analytics — Why does the master's 3% reschedule rate matter?

Because it pins the baseline: a 3% shortfall on a 25% CV is L(z) = 0.03 ÷ 0.25 = 0.12 → z ≈ 0.8, the 79th percentile — a first buy near the 79th percentile reproduces the 3% and the $18K a month. The design moves it to the 99th percentile: expected shortfall 0.08% of installs. Illustrative CV; the master's 3% is the anchor.

Analytics — Your CV is a guess. What if it is 35%?

Then a buy sized for 25% at the 99th percentile — 9,495 — sits at z = 1.66 on the true spread: a 4.8% chance of running short and an expected shortfall of 42 units, versus 5.0. The policy is robust in direction and fragile in tail: so the buy is re-sized as soon as the first weeks' CV is observed, and the trigger exists for the same reason.

Analytics — Why is the central pool better — and when does it stop being better?

Pooling k markets cuts the buffer from k·zσ to z·σ·√(k + k(k−1)ρ): at k = 4 and independence, 50%; at ρ = 0.3, 31.1%; at ρ = 0.6, 16.3%; at ρ = 1, nothing. New markets share a product, a campaign and an adoption archetype, so ρ is probably not zero — I would measure it on analog markets before promising the saving. VERIFY-WITH-RAJESH

Shadow / closing — What would you do differently?

Measure the cross-market correlation on analog markets before quoting a pooling saving, and archetype-review the adoption curve with Marketing first — a curve borrowed from a non-analog market is the failure mode that voids everything else.

BUSINESS IMPACT MODEL — WHAT THE FORECAST-TO-BUY DESIGN IS FOR

```
Install delays caused by stockouts -> 0 (first 90 days)
+- First buy at the critical ratio: p99 core unit, p70 accessories
+- Central pool: one buffer covers several markets (sqrt k, if uncorrelated)
+- Re-forecast trigger: take-rate vs curve beyond +/-20% for 2 weeks
+- Invariants: demand enters once; measured lead times; floors with hold-until
+- Dollars (illustrative): reschedules avoided + stranding avoided
```

DRIVER	BASELINE	AFTER	$	BASIS
Reschedules from stockouts	3% of 2,000 installs/month	→ ≈0	$18,000/month ≈ $216,000/yr per market	master illustrative
Stranding avoided	20% of a $500K first buy in-market	held central	≈$100,000	master illustrative
Expected first-quarter cost, core unit	p79: $58,234	p99: $12,009	$46,225 lower	newsvendor arithmetic (illustrative)

If challenged. *"Did you save $216K a year?"* — "No — it is design arithmetic on illustrative inputs; I did not run this launch."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the baseline in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Baseline (master)	`2,000 × 3% × $300 = $18,000/month; × 12 = $216,000/yr`	master E7 illustrative	$216,000/yr per market
2	What the 3% implies	`shortfall fraction = CV × L(z) → 0.03 = 0.25 × L(z) → L(z) = 0.12 → z ≈ 0.8 (p79)`	standard normal loss function	baseline ≈ p79 first buy
3	The 3-month demand	`μ = 2,000 × 3 = 6,000; σ = 25% × 6,000 = 1,500`	CV 25% illustrative	σ = 1,500
4	Stranding at risk	`20% × $500,000 = $100,000`	master illustrative	$100,000

Table 2 — the fix: the newsvendor first buy, every row substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Critical ratio	`Cu ÷ (Cu + Co) = 300 ÷ 303 = 0.9901 → z = Φ⁻¹ = 2.330`	Cu ≈ $300; Co ≈ $3/month (master)	order the 99.0th percentile
2	Buy at p50	`z = 0.0000: Q = 6,000 + 0.0000 × 1,500 = 6,000; short σL = 598.413; left σ(z+L) = 598.413; cost = 598.413 × $300 + 598.413 × $3 = $181,319`	normal demand, one-month holding (illustrative)	$181,319
3	Buy at p79	`z = 0.8000: Q = 6,000 + 0.8000 × 1,500 = 7,200; short σL = 180.311; left σ(z+L) = 1,380.311; cost = 180.311 × $300 + 1,380.311 × $3 = $58,234`	normal demand, one-month holding (illustrative)	$58,234
4	Buy at p90	`z = 1.2816: Q = 6,000 + 1.2816 × 1,500 = 7,922; short σL = 71.007; left σ(z+L) = 1,993.407; cost = 71.007 × $300 + 1,993.407 × $3 = $27,282`	normal demand, one-month holding (illustrative)	$27,282

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
5	Buy at p95	`z = 1.6450: Q = 6,000 + 1.6450 × 1,500 = 8,468; short σL = 31.328; left σ(z+L) = 2,498.828; cost = 31.328 × $300 + 2,498.828 × $3 = $16,895`	normal demand, one-month holding (illustrative)	$16,895
6	Buy at p99 (CR)	`z = 2.3301: Q = 6,000 + 2.3301 × 1,500 = 9,495; short σL = 5.027; left σ(z+L) = 3,500.146; cost = 5.027 × $300 + 3,500.146 × $3 = $12,009`	normal demand, one-month holding (illustrative)	$12,009
7	Buy at p99.9	`z = 3.0900: Q = 6,000 + 3.0900 × 1,500 = 10,635; short σL = 0.416; left σ(z+L) = 4,635.416; cost = 0.416 × $300 + 4,635.416 × $3 = $14,031`	normal demand, one-month holding (illustrative)	$14,031
8	Accessory at the 70th	`Cu = 0.70 ÷ 0.30 × $3 = $7.00 per missed unit`	implied by the master's p70	Cu ≈ $7

Table 3a — pooling and the trigger (substituted).

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Pooling, k = 4, ρ = 0	`σ_pool = 500 × √(4 + 4·3·0) = 1,000.0; buffer = 2.33 × 1,000.0 = 2,330 vs 4 × 2.33 × 500 = 4,660`	month-1 σ = 500 per market (illustrative)	saves 50.0%
2	Pooling, k = 4, ρ = 0.3	`σ_pool = 500 × √(4 + 4·3·0.3) = 1,378.4; buffer = 2.33 × 1,378.4 = 3,212 vs 4 × 2.33 × 500 = 4,660`	month-1 σ = 500 per market (illustrative)	saves 31.1%
3	Pooling, k = 4, ρ = 0.6	`σ_pool = 500 × √(4 + 4·3·0.6) = 1,673.3; buffer = 2.33 × 1,673.3 = 3,899 vs 4 × 2.33 × 500 = 4,660`	month-1 σ = 500 per market (illustrative)	saves 16.3%
4	Pooling, k = 4, ρ = 1	`σ_pool = 500 × √(4 + 4·3·1) = 2,000.0; buffer = 2.33 × 2,000.0 = 4,660 vs 4 × 2.33 × 500 = 4,660`	month-1 σ = 500 per market (illustrative)	saves 0.0%
5	Trigger, no real shift (false alarms)	`p = 1 − Φ(2.0) = 2.28% per week per side; P(trigger within 52 weeks) = 2.55% per side → 0.051 false triggers per series-year (both sides)`	weekly noise 10% of curve (illustrative); consecutive-run recursion	0.051 / series-year
6	Trigger, true take-rate +10% vs curve	`p = P(week beyond ±20%) = 1 − Φ((0.20 − 0.10) ÷ 0.10) = 15.87% per week (one side); P(2 in a row within 4 / 12 weeks) = 6.8% / 22.0%`	weekly noise 10% of curve (illustrative); consecutive-run recursion	7% by week 4
7	Trigger, true take-rate +25% vs curve	`p = P(week beyond ±20%) = 1 − Φ((0.20 − 0.25) ÷ 0.10) = 69.15% per week (one side); P(2 in a row within 4 / 12 weeks) = 77.3% / 99.3%`	weekly noise 10% of curve (illustrative); consecutive-run recursion	77% by week 4
8	Trigger, true take-rate +40% vs curve	`p = P(week beyond ±20%) = 1 − Φ((0.20 − 0.40) ÷ 0.10) = 97.72% per week (one side); P(2 in a row within 4 / 12 weeks) = 99.8% / 100.0%`	weekly noise 10% of curve (illustrative); consecutive-run recursion	100% by week 4

Table 3 — after: the metrics, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Install delays from stockouts	3% of installs (baseline)	≈ 0.08% at the p99 buy	illustrative
2	First-quarter expected cost	$58,234 (p79)	$12,009 (p99)	newsvendor, illustrative
3	Buffer for 4 markets (independent)	4,660 units unpooled	2,330 pooled (−50%)	toolkit, illustrative
4	Re-forecast detection of a +25% shift	quarterly review	77% within 4 weeks	design arithmetic

STEP	METRIC	BEFORE	AFTER	BASIS
5	Owner of the forecast number	Planning	Planning (unchanged); I own equation, measurement, triggers	master honest line

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Reschedule cost avoided	`2,000 × (3% − 0.0838%) × $300 = $17,497/month`	p79 → p99 first buy	≈ $209,967/yr per market
2	Holding cost added	`p99 leftover 3,500 × $3 = $10,500 (one month)`	newsvendor row	$10,500
3	First-quarter cost difference	`$58,234 − $12,009 = $46,225`	expected cost, p79 vs p99	$46,225 per market per quarter
4	Stranding avoided (master)	`20% × $500K = $100,000`	held central	≈ $100,000
5	Master's annual spine	`$216,000/yr + $100,000 one-time`	illustrative; never a claim	design arithmetic

SOLUTION FLOW — THE FORECAST-TO-BUY CHAIN (PROPOSED)

```
[Build plan] --milestones--> [Sites passed] --adoption curve x marketing--> [Install demand]
      --kit requirement at finished-kit grain--> [Need by SKU x week, as PIRs]
      --MRP (measured lead times, confirmed dates)--> [Planned orders -> POs]
      --ASN--> [Central 3PL pool] --STOs as sell-through proves the curve--> [Markets]

First buy: Q = mu + z*sigma, z = PhiInv(Cu/(Cu+Co)) ; central, pooled
Trigger: |take-rate / curve - 1| > 20% for 2 consecutive weeks -> new forecast version
Weekly: MAPE and bias per SKU x market x week; bias sign before magnitude
```

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Cost of a missed install (Cu)	$150 → CR 0.9804	$300 → CR 0.9901	$600 → CR 0.9950
Demand CV (first buy sized for 25%)	CV 15% → surplus	CV 25% → 5.0 short	CV 35% → 42 short, 4.8% risk
Cross-market correlation ρ (k = 4)	0 → 50% saved	0.3 → 31.1%	0.6 → 16.3%
Trigger: weekly noise (true shift +25%)	5% → fires fast	10% → 77% by week 4	15% → slower

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"This is a design, not a run record: I have not owned a launch forecast end to end, and the forecast number belongs to Planning. What I own is the equation, the measurement and the triggers. A missed install costs about three hundred dollars and a unit held a month about three, so the critical ratio is point nine nine and the first buy sits at the ninety-ninth percentile of the first quarter's demand; held central it pools — half the buffer for four uncorrelated markets, about thirty percent if they share a correlation of point three, which I would measure. A take-rate outside plus or minus twenty percent for two weeks forces a new forecast. The dollars — about two hundred sixteen thousand a year per market and a hundred thousand of stranding avoided — are the master's illustrative arithmetic, never a result."

If challenged — the full answers

Did you run it? “No — a design engagement; Planning owns the forecast, I own the equation, the measurement and the triggers.” Is the CV real? “Illustrative; the buy is re-sized on the first weeks’ observed spread.” Is that Google Fiber? “The master’s design engagements are Elementum client work; I would name the client only if confirmed.”

VERIFY-WITH-RAJESH

BEHAVIORAL PACK NEW — FOUR QUESTIONS THIS STORY ANSWERS, FOUR SEPARATE 90-SECOND ANSWERS (EACH FROM A DIFFERENT MOMENT OF THE STORY)

Q1 Tell me about a time you learned a new method to solve a problem. LEARN AND BE CURIOUS

Open (first two sentences): I took the newsvendor model from my MIT MicroMasters coursework and applied it to a launch problem — and I label it as the toolkit.

STAR (≈90 s): Situation: a new-market build with no demand history. Task: design the first buy. Action: Cu ≈ $300 for a missed install, Co ≈ $3 a month for a unit held, critical ratio 0.990, so the 99th percentile of first-quarter demand for the core unit and the 70th for lower-cost accessories. I then stress-tested it: if the true CV is 35% rather than 25% the shortfall chance goes from about 1% to 4.8%, which is why the trigger exists. Result: a design with arithmetic, not a run record.

Land this number: CR 0.990; z 2.33; Q = 9,495 on a 6,000 mean. · **Pivot to NACF:** A new FC or product launch at NACF is the same critical-ratio problem. · **Do NOT say:** Do not say I ran the launch; do not present the CV as data.

Q2 Tell me about a time you designed for scale beyond the immediate problem. THINK BIG

Open (first two sentences): I designed the launch chain as a template for market N+1, not a one-off for market N.

STAR (≈90 s): Situation: every market launch repeats the same chain — build plan, adoption curve, requirement, supply. Task: a design that a second and third market can reuse. Action: a stock-and-flow map, invariants — demand enters planning once, lead times are measured distributions, need floors with hold-until dates — and a weekly MAPE/bias loop. Result: a template; the forecast stays with Planning.

Land this number: One chain, three books of record, five invariants. · **Pivot to NACF:** A reusable placement template for each new NACF node. · **Do NOT say:** Do not claim a deployed template.

Q3 Tell me about a decision you made with very little data. ARE RIGHT, A LOT

Open (first two sentences): With no local history, the right question was not “what is demand” but “what does being wrong cost on each side.”

STAR (≈90 s): Situation: maximum uncertainty at launch. Task: first buy. Action: price both errors, buy to the critical ratio, hold it central so uncertainty pools, and write the trigger that proves the curve wrong fast: ±20% for two weeks. The trigger catches a true 25% shift within four weeks 77% of the time and false-fires about once every 20 series-years. Result: a decision with its own error rates.

Land this number: 77% detection by week 4 at +25%; 0.051 false triggers per series-year. · **Pivot to NACF:** Placement under uncertainty: price both errors, then set the trigger. · **Do NOT say:** Do not quote the weekly noise (10%) as measured.

Q4 Tell me about a time you put the end customer first in a supply decision. CUSTOMER OBSESSION

Open (first two sentences): The metric was install delays caused by stockouts — a customer waiting for a technician.

STAR (≈90 s): Situation: a missed install means a re-roll and deferred revenue. Task: make the stockout the thing the whole chain is measured on. Action: the primary metric joined reschedule reasons to the supply view; need floors so one market could not starve another’s launch; a governed screen with a reason code instead of an email. Result: design target zero delays caused by stockouts in the first 90 days.

Land this number: Target 0 delays in 90 days; ≈$300 per missed install (model). · **Pivot to NACF:** The NACF customer is the promise date; same joined-metric discipline. · **Do NOT say:** Do not claim installs were delivered.

SAME STORY, THREE ANGLES NEW — THREE DIFFERENT OPENERS, HEADLINE METRICS AND LPS FROM THE SAME FACTS

(I) DATA / ANALYTICAL

LP: Dive Deep; Are Right, A Lot

Open: "A 3% reschedule rate tells you the baseline was a 79th-percentile buy."

Headline metric: CR 0.990; p99 buy 9,495; cost $12,009 vs $58,234

- L(z) = 0.12 → z ≈ 0.8.
- Expected cost curve is flat near the optimum, steep to the left.
- CV 35% turns a 1% risk into 4.8%.
- All illustrative.

(II) CONFLICT / INFLUENCE / BACKBONE

LP: Have Backbone

Open: "Demand enters planning exactly once — manual loads beside the engine are blocked."

Headline metric: one engine, one demand path; floors with hold-until dates

- A floor breach is a governed screen with a reason code.
- One market cannot starve another's launch.
- A borrowed adoption curve is a stop-the-line item.
- Design stance, not a run record.

(III) FAILURE — LEARNING — WHAT I WOULD DO DIFFERENTLY

LP: Learn and Be Curious

Open: "I have not owned a launch forecast end to end."

Headline metric: pooling saves 31% at ρ = 0.3, not 50%

- Measure ρ on analog markets before quoting a saving.
- The forecast belongs to Planning.
- The toolkit is labeled.
- VERIFY-WITH-RAJESH client naming.

TRAP DRILLS NEW — THE THREE NASTIEST FOLLOW-UPS A STATISTICALLY RIGOROUS INTERVIEWER THROWS

T1 "Why the 99th percentile for the first buy — is that not wasteful?"

The numbers: CR = 300 ÷ 303 = 0.9901; z = 2.330; p99 buy 9,495: cost $12,009; p79 buy 7,200: $58,234; leftover 3,500 units × $3 = $10,500.

Say exactly: "Because a missed install costs a hundred unit-months of holding: at the ninety-ninth percentile the expected first-quarter cost is about $12 thousand against $58 thousand at the seventy-ninth, and the leftover — about thirty-five hundred units — is not waste, it is next quarter's demand."

T2 "Does the pooling benefit survive correlated markets?"

The numbers: k = 4: ρ = 0 → saves 50%; ρ = 0.3 → 31.1%; ρ = 0.6 → 16.3%; ρ = 1 → 0%.

Say exactly: "Only partly — half the buffer at zero correlation, about thirty-one percent at a correlation of point three, sixteen at point six; new markets share a product and a campaign, so I would measure it on analog markets before promising any number."

T3 "Your trigger is ±20% for two weeks. What are its false alarms and its power?"

The numbers: Weekly noise 10% (illustrative): no shift → 0.051 false triggers per series-year (both sides); true +25% → 77% within 4 weeks; +40% → 100%; +10% → 22% within 12 weeks.

Say exactly: "With ten percent weekly noise the trigger false-fires about once in 20 series-years, catches a true twenty-five percent shift within a month about 77 times in a hundred, and is nearly blind to a ten percent bias — which is why the bias sign publishes weekly beside it."

Question bank — likely questions for this interviewer, with a one-line answer plan

Behavioral and functional questions a Placement Analytics manager is likely to ask (rows 13–27 are new: the statistical questions the new primaries answer). Each plan points at a packet story or a named backup in the master packet; stories allocated to other interviewers are pointers only.

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
1	**Walk me through how you improved forecast accuracy.**	S6 pitch — honest supply definition first, MAPE/bias at the decision grain, waves by trust; 70 → 85% on scoped class-A SKUs; planners owned the forecast, I owned its measurement.	S6
2	**Tell me about a time you used data to change an inventory policy.**	S8 pitch — brochure → measured lanes, p99 promise, 99% bar, 580K units / ~$1.5M out with zero stockouts.	S8
3	**How do you decide where / how much safety stock to hold?**	Two error signals measured at the grain; cost asymmetry sets the percentile; release only by signed rule with a watch; fat tails keep buffer.	S8 → S6
4	**What is the softest assumption in your biggest result?**	S6: the expedite count (60 × $5K). S8: the late-arrival share. Say how each would be firmed from a ledger.	S6 / S8
5	**How do you measure lead-time performance?**	Lead-time accuracy as the master defines it = 1 − |actual − promised| / promised against a per-lane p99 promise; per shipment that cannot reach 99% against a p99 promise, so say which reading (coverage, or re-measured p99 within 1%) before quoting a tolerance VERIFY-WITH-RAJESH; drift flag and refresh cadence.	S8
6	**Tell me about a time a stakeholder wanted a number that looked better.**	Planning's request to count unfirmed MRP orders as supply — refused; released planned orders only.	S6
7	**How did you handle a dependency that slipped (MPN mapping)?**	Escalated to the sponsor as a re-sequenced date with the coverage number attached; made MPN a standing metric rather than a task.	S6
8	**How would you decompose a network placement problem (e.g., inbound placement cost vs. speed)?**	Master §3.99 opening: book of record and metric → stocks and flows → broken invariant → design → sequence; then lead-time and forecast error at the lane × node grain as the two levers.	Master §3.99
9	**How do you validate a model before the business acts on it?**	Parallel run against the source (daily reconciliation to SAP; manual sample per cohort); a published gate; the old process stays alive until the match holds.	S6 (S10 pointer — Josyula's)
10	**What is your quantitative background?**	MIT Supply Chain MicroMasters + Supply Chain Network Design certificate — acquired to design what I had been implementing; the formulas are that toolkit, labeled. Never "I build models for a living."	Master §4.1 toolkit line
11	**Tell me about a time you improved a process end to end (Plan → Deliver).**	S6 as the Plan redesign (definition → gate → alert → release rule) with S8 as the Source/Deliver parameter feeding it; the two programs stay separate in dollars.	S6 + S8
12	**Another example of Dive Deep?**	Pointers only: S10 (Austria — bar raiser's) and S7 (phantom math — unallocated; no BOM-explosion content). If pressed, S7 at the level of "six sites, three waves, five months, zero auto-loaded revision overlaps."	S7 pointer

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
13	**How do you know the lead-time distribution you measured is the right one — n, window, interval?**	S8a: ~400 shipments on the busy lane; the ranked p99 with its order-statistic interval; bell-curve p99 30.3 holds 96.5%, ranked p99 34 holds 99%; thin lanes print the interval or pool by cohort VERIFY-WITH-RAJESH NEW	S8a
14	**Planned vs actual lead time — how exactly was the plan wrong?**	S8a: plan 18 d vs mean 21 d (t = 15 on the 3-day bias); ERP σ 8 vs measured 4; tail 16 days short at p99 (34 vs 18) — wrong on location, spread and tail (illustrative lane) NEW	S8a
15	**How do you set safety stock and the service level?**	S8b: two terms — cover to the p99 lead time plus demand cover; critical ratio 0.99; lane A 12,937 → 8,102 units, lane B 5,238 → 6,621 (+1,383); 580K net is the record NEW	S8b
16	**How do you pick an alert threshold — precision, recall, and the cost of a false alarm?**	S6: alert at DoC ≤ lane p90; on an illustrative lane precision 71.2%, recall 98.6%; p99 trades twelve points of precision for about one of recall; cost $12,840 vs $8,913 at $60 / $5,000 NEW	S6
17	**How do you know it was your change and not seasonality?**	S8c: held-out quarter; paired old-vs-new plan on the same shipments (22 vs 175 late of 2,500); control lanes deferred a quarter if the record has them VERIFY-WITH-RAJESH; one draw, so rolling origin is the upgrade NEW	S8c
18	**What is the uncertainty on a result you are about to act on?**	S8c: 22 late of 2,500 → 99.1%, Wilson [98.7, 99.4]; z vs 96.5% = 7.1; 0 stockouts in 1,950 lane-weeks bounds the rate below 0.154% NEW	S8c
19	**Which of 1,000 lanes do you keep, and by what criteria?**	S9: spend Pareto + four-factor lane scoring + EDI survey; 597 lanes = 80% of spend first; ~400 deferred (~20%) on the same gate NEW	S9
20	**How would you design an estimated-delivery-date / promise policy?**	S19: promise at b = k·σ with k = 1.645 for a 95% floor; accuracy pinned on every cohort, speed floats; fast-promise share 7.6% → 39.6%; re-solve k per regime NEW	S19
21	**A headline metric is green — how do you check it?**	S19: headline 95.8% hid Φ(5.0) vs Φ(0.69) = 75.4% on the wide cohort; S20: raw +0.8 pt vs mix-adjusted −1.0 pt — publish the adjusted twin beside the raw NEW	S19 / S20
22	**Design a weekly business review for a delivery experience.**	S20: 7 outputs over 10 inputs and one joint headline; factor math names the owner in minutes; hard floors beside the headline; pre-wired reds NEW	S20
23	**How do you catch a degrading seller or carrier fast without false alarms?**	S20: Andon — n ≥ 30 floor plus a sustained signal; false-trip design bar < 5%; ~30 d → < 24 h (design, nothing in production) NEW	S20
24	**How would you test an incentive or contract change before launch?**	S21: replay a seller's orders under managed settings; c = 11 misses in 200 as the pre-agreed gate (illustrative); Wilson interval on the 85% handling adherence at n = 100: [76.7%, 90.7%] NEW	S21
25	**How would you forecast and place equipment for a new market with no history?**	E7: newsvendor with critical ratio 0.99 → z 2.33 → first buy 9,495 on a 6,000 mean; pooled central stock saves ~31% at ρ = 0.3; ±20% two-week re-forecast trigger; NACF INV-C for placement (illustrative; design, not run) NEW	E7

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
26	**Which number across your stories is the softest?**	S19's ≈$13.3M/yr and S21's ≈$3.7M/yr are modeled; S9's −10% freight has no control group VERIFY-WITH-RAJESH; S8's accuracy definition VERIFY-WITH-RAJESH. Name it first, then say how I would firm it. NEW	S19 / S21 / S9
27	**How do you separate your contribution from the client's?**	S21: I claim the contract design and wave-one mechanics; the adoption curve is the client's. S19–S20 are engagement designs — nothing in production moved. NEW	S21 / S19 / S20

Cheat-numbers box — canon values with basis labels

Canon numbers for all ten primaries with their basis labels, from the master packet's §4.1 ledger; the last ten rows are new (S8a–E7).

METRIC	VALUE TO SAY	BASIS	NEVER
S6 forecast accuracy	70% → 85% on scoped class-A SKUs	record	claiming I owned the forecast
S6 safety stock	$2M released under the both-errors-improved rule	record, one-time	"$2M per year"; adding to S8
S6 recurring	≈$0.8M/yr (carry 0.2 · expedites 0.3 · planner capacity 0.3); range $0.5–1.2M	modeled	presenting as record
S6 timing / scope	shortage alert live in 3 months; eight domains; six sites; adoption 8×; planner effort −50%; shipment data 60% complete at start	record	pasting "3 months" onto the six-site ingest (S7)
S6 alert rule	days-of-cover ≤ lane lead-time p90	record mechanism	—
S8 lanes / accuracy	200 lanes; 99% lead-time accuracy at the p99 promise	record	—
S8 release	580K units / ~$1.5M, zero stockouts in the watch window	record, one-time	"$1.5M to $2M"; annualizing it
S8 recurring	≈$0.6M/yr (carry 0.3 at 20% · expedites 0.3); range $0.3–1.4M	modeled	presenting as record
S8 unit value	≈$2.59/unit	derived from the record pair	—
Site count	"six sites"	record	"six NA sites"
Toolkit	MIT Supply Chain MicroMasters + Network Design certificate; formulas are illustrations	labeled	"I build supply chain models for a living"
S7 (pointer only)	six sites, three waves, 5 months; hardest site 6 weeks vs 3-week average	record	"500+ product families"; 60→95% inside S7; any BOM-explosion walk
S8a — example lane	plan 18 d · actual mean 21 d · p99 34 d · n ≈ 400 · 240 of 400 (60%) slower than the brochure · normal p99 30.3 d holds 96.5%	illustrative (secondary-source lane)	presenting the lane as the program's data

METRIC	VALUE TO SAY	BASIS	NEVER
S8b — safety-stock lanes	lane A 12,937 → 8,102 (SS 6,500 + 1,602 vs 9,437 + 3,500 pad); lane B 5,238 → 6,621 (+1,383); 150 × 4,500 − 50 × 1,900 = 580K	illustrative decomposition; the net 580K is record	saying the program "raised" fat-tail lanes as record VERIFY-WITH-RAJESH
S8c — held-out test	22 late of 2,500 → 99.12%, Wilson [98.67, 99.42]; 0 stockouts in 1,950 lane-weeks → < 0.154%	illustrative inputs	quoting window / quarter as fact VERIFY-WITH-RAJESH
S9 — EDI onboarding	1,000 asked → 597 = 80% of spend; completeness 60% → 95%; ≥95% valid-ASN gate held twice; freight −10% ≈ $2M in six months; ~200 next wave; ~400 lanes (~20%) deferred	record	annualizing the $2M; claiming −10% was causal without a control VERIFY-WITH-RAJESH
S9 — recurring	≈$0.4M/yr documentation labor	modeled	presenting as record
S19 — delivery promise	flat 1.10 d buffer → b = k·σ, k = 1.645; fast-promise share 7.6% → 39.6% at a held 95% floor; +$255K/wk ≈ $13.3M/yr	engagement model / modeled	the 97.8% variant and its $6.07M/wk — never claimed; any production result
S19 — regime drill	Q4 σ × 1.45 → accuracy 91.9% on a frozen dial; $12 breach cost vs $6.50 margin	model / drill	a live Q4 result
S20 — pre-purchase WBR	0.97 × 0.985 × 0.996 ≈ 95.2%; Promise Yield 41.5% = 43.6% × 95.2% (four-quarter target 50%); SE carrier 82% → accuracy 93.6%; ≈$0.58M per caught episode; false-trip bar < 5%; ~30 d → < 24 h	illustrative sample / modeled / design	any live Andon result — design engagement, nothing in production
S21 — managed shipping	enrolled ~38–42% stalled; 85% handling adherence; ≈$3.7M/yr per +10 pts of enrolled GMV; claim = contract design + wave-one mechanics	modeled	owning the adoption curve; presenting $3.7M as record
E7 — new-market buy	Cu ≈ $300, Co ≈ $3 → CR 0.99 → z 2.33; ±20% two-week trigger; $216K/yr per market + $100K stranding	illustrative; design, not run	saying it ran; naming the client unless confirmed VERIFY-WITH-RAJESH

Never-say, one glance before the call.

- Never add $2M (S6) and ~$1.5M (S8) into one number; never "$1.5M to $2M."
- Never "six NA sites" — "six sites." Never paste S6's "3 months" onto the six-site ingest.
- Never claim ownership of the demand forecast or the distributions — measurement, definitions, rule and policy were mine.
- Never present the safety-stock formula as a program artifact; MRP math at finished-goods grain only — no BOM explosion.
- Never present planner capacity as cash — time redeployed, not headcount removed.
- S10 (Austria) is the bar raiser's; S11/S12, S2/S18, S1/S3 belong to Tripathy, Goforth and Lan — pointers only here.

Backup bench — use only if a primary doesn't fit

Three fallbacks for an analytics manager: two master-packet backups plus one doc-derived Flexport micro-story — S4 (Sephora — the in-transit promise rule I got wrong; the failure / Are Right, A Lot question, promise math at lane σ) and S7 (Elementum — a contract manufacturer's planning data made safe to plan on, finished-goods grain only; Learn & Be Curious by math, not meetings); the doc-derived third is the FFA Asia latency investigation from the Flexport program record (Dive Deep on production evidence; record-from-docs labels, facts identical to the same bench in Josyula's packet). S9 (the 1,000 → 597 EDI story) was promoted to a full primary above. Tell the ten primaries first.

Bench discipline. The ten primaries above are the stories to tell first. Reach for a bench story only when the question genuinely does not fit a primary, or when the interviewer asks for "another example" on an LP already spent. Each entry is a 60-second pitch, a compressed STAR, one results line and two probes — enough to answer and stop; no full probing bank or derivation. Where a bench story is another interviewer's primary, the red note says so: use it only if the question demands it, and keep the facts identical to that packet.

BACKUP — ONLY IF ASKED S4 — Sephora S/4HANA — the in-transit promise rule I got wrong, fixed through the signed policy, not a side configuration

LPs this backup can serve: Are Right, A Lot (the failure question) · Customer Obsession (backup) · Earn Trust (owning the miss) · Learn and Be Curious — the designated policy-failure story

Basis. The v1 rule was mine; the indictment came from my own program's reports. Lane σ ≈ 0.3 day and the ~0.49-day window are the record's promise math; the record carries no lane counts and no dollar for the conversion lost while I was wrong — I will not invent the loss figure.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. My first ISM ruleset barred every channel from promising units in transit; the Aging In-Transit report showed dotcom starved of fast movers one day from the building. **Task.** Correct my own policy without weakening oversell protection, through the same three signatories. **Action.** Discovery: conversion-loss evidence from the aging and replenishment reports — the entry bar to reopen the policy; I stated that the v1 rule was mine. Plan & Design: promise math — lane σ ≈ 0.3 day, a ~0.49-day window ≈ 95% coverage; Promisable = in-transit × 1[ETA ≤ window] × 1[on-time ≥ floor]; Aging In-Transit as the kill switch; both blanket options rejected in writing. Build & Test: back-test against receiving history; four regression cells added to the node × channel matrix. Cutover: released on the normal monthly cadence as a signed FSD change, lane by lane — no hotfix, no side configuration. Hypercare: kill switch armed; never fired. Lesson: both sides of a rule get a number before it ships. **Result.** Primary metric: sellable units on qualifying in-transit lanes — starved → restored inside a per-lane promise window, oversell increase = zero, kill switch never fired. ≈$0.5M/yr modeled margin recovered — the record has no dollar for the conversion lost while I was wrong, and I will not invent one.

COMPRESSED STAR

- **S —** v1 of my own ISM ruleset: no channel could promise in-transit units — safe-feeling, and starving dotcom of fast movers reliably one day from the building.
- **T —** Fix my own signed policy without weakening oversell protection; take the miss back through the same three signatories.
- **A —** Evidence pack from Aging In-Transit and Replenishment; reframed as promise math (σ ≈ 0.3 d → ~0.49-d window ≈ 95%); rule with on-time floor and kill switch; back-test against receiving history; four regression cells; released lane by lane on the monthly cadence as an FSD change.

- **R —** Starved lanes restored inside the window; oversell increase = 0; kill switch never fired; re-signed by the same three signatories; lesson institutionalized — the Locus 98% gate (S2) and the 0.5% tolerance (S18) were set from data before the event.

RESULTS LINE

Sellable units on qualifying in-transit lanes starved → restored inside a back-tested ~0.49-day window (record) · oversell increase = 0; kill switch armed, never fired (record) · four regression cells added (record) · released as a signed FSD change on the normal cadence, no hotfix (record) · ≈$0.5M/yr modeled margin recovered (inputs illustrative).

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The v1 rule	`Promisable = 0 for ALL in-transit inventory`	equivalent to claiming σ → ∞ or on-time rate → 0 — infinite variability (record)	everything in transit hidden
2	One concrete lane	`Promisable = 120 × 0 = 0`	lane with InTransit = 120 units, ETA 0.4 day away (illustrative)	120 sellable units hidden
3	Across qualifying lanes	`≈300 fast-mover units/day × 0 promisable`	illustrative volume	≈300 units/day sat unsellable one day from the building
4	What starving costs (one SKU)	`100 × 60% × $20 = $1,200/day`	100 units one day out; 60% would convert; $20 margin (illustrative, not program data)	$1,200/day under-sell
5	The ignored other side	`100 × (1 − 0.97) = 3 late units; 3 × $15 = $45/day`	on-time rate 97%; $15 expedite (illustrative)	$45/day over-sell exposure — vs $1,200: the “safe” rule priced one side only

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Measure lane variability	`σ = 0.3 day`	trailing receiving history (record)	σ = 0.3 day
2	Choose coverage	`z at 95% = 1.645`	normal approximation on lane transit (toolkit label)	z = 1.645
3	Promise window	`w = z × σ = 1.645 × 0.3 = 0.4935`	rows 1–2	w ≈ 0.49 day
4	Coverage check	`Φ(w / σ) = Φ(0.49 / 0.3) = Φ(1.63) ≈ 0.95`	share of arrivals inside the window	≈95% covered
5	Corrected rule	`Promisable = InTransit × 1[ETA ≤ w] × 1[OTR ≥ floor]`	OTR = on-time rate over trailing receipts; floor = per-lane admission threshold, 95% here (floor illustrative)	lane by lane, not all-or-nothing
6	Lane that qualifies	`ETA 0.4 ≤ 0.49 → 1; OTR 97% ≥ 95% → 1; Promisable = 120 × 1 × 1 = 120`	the Table 1 lane, re-evaluated (ETA and OTR illustrative)	its 120 units become promisable
7	Lane that fails	`ETA 1.2 > 0.49 → 0; Promisable = 80 × 0 = 0`	a slower lane with 80 units in transit (illustrative)	stays at 0 — correctly out
8	Sample-size gate	`n = z²p(1−p)/e² = 3.84 × 0.95 × 0.05 / 0.05² = 3.84 × 0.0475 / 0.0025 ≈ 73`	OTR estimated within ±5% at 95% CI (illustrative)	a lane qualifies after ~73 receipts

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
9	Kill switch, values in	`trigger = 2 × p90 = 2 × 2.0 = 4.0 days; any unit in transit > 4.0 days → lane out`	lane p90 = 2.0 days (illustrative); armed, reviewed weekly by Inventory Control (record)	never fired (record)
10	Back-test	`rule replayed against receiving history; 4 regression cells`	before release (record)	no oversell in the back-test

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Sellable units on qualifying in-transit lanes	0	≈300 units/day restored inside the window	record mechanism; volume illustrative
2	Promise window	none — blanket exclusion	w ≈ 0.49 day at ~95% cover	record (promise math)
3	Oversell increase from the window	—	0	record
4	Kill switch	not armed	armed, never fired	record
5	Rule changes after	ad hoc	same 3 signatories, back-test + matrix regression	record — both sides priced from then on

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Would-be sales	`300 × 60% = 180 sales/day`	units restored (illustrative) × conversion (illustrative)	180 sales/day
2	Truly lost without the fix	`180 × 40% = 72 sales/day`	40% lost-sale share — the rest would have waited (illustrative)	72 lost sales/day
3	Margin recovered	`72 × $20 = $1,440/day`	$20 unit margin (illustrative)	≈$1,440/day
4	Annualized	`$1,440 × 365 = $525,600`	modeled; the mechanism is record	≈$0.5M/yr
5	Cost while v1 stood	`no dollar in the record — not claimed`	record	not claimed

TWO PROBES

Why not just hotfix it — you knew it was wrong?

Because the error was in the policy, so the fix had to be too — in the governed artifact, with regression, through the same signatures. A side configuration would have fixed the symptom and left the policy lying about what it allowed.

What did the miss cost?

The record carries no dollar for the conversion lost while v1 stood, and I will not invent one. What I can say is the mechanism: fast movers reliably one day from the building were unsellable on dotcom until the window existed; the ≈$0.5M/yr recovered is my model on illustrative inputs, labeled.

Canon numbers (basis labels): Lane σ ≈ 0.3 d; ~0.49-d window ≈ 95% coverage (record promise math) · oversell increase = 0 (record) · 4 regression cells (record) · kill switch never fired (record) · same three signatories re-signed (record) · ≈$0.5M/yr (modeled). Never a lane count; never a dollar for the loss while wrong.

BACKUP — ONLY IF ASKED S7 — Elementum — making a contract manufacturer's planning data safe to plan on: six sites, three waves, five months, zero auto-loaded revision overlaps

LPs this backup can serve: Learn and Be Curious (primary — learned the CM's data by math, not meetings) · Dive Deep · Earn Trust (external partner)

Basis. I owned the ingest requirements, the reconciliation gate, the wave plan and the weekly Flextronics call; engineers built the normalization layer and quarantine to the spec; the commercial relationship belonged to the customer's procurement — my leverage was evidence. The $1.5M / 580K belong to the p99 program this backbone enabled (S8). Finished-goods grain only here; no tolerance value — the record does not carry one.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. Flextronics supplied the planning data our component-demand visibility depended on, and it carried two traps documentation never confesses: pass-through assemblies that double-time demand if loaded as stocked, and overlapping revisions that rewrite history if you auto-load one. Six site systems carried six definitions of yield, lead time and order status. **Task.** I owned the ingest requirements, the reconciliation gate, the wave plan and the weekly Flextronics call — with no commercial leverage; evidence was my only lever. **Action.** Discovery: sampled finished goods and reconciled the CM's planning data against its own MRP — the trap signature found by math, named in a discovery memo. Plan & Design: an explicit marker for pass-through items; an effective-date quarantine with human disposition — never auto-load; one supply equation (on-hand + production plan + released planned orders) signed by customer planning. Build & Test: mapping tables owned by named site super-users; a per-site reconciliation report as the go-live gate. Cutover: three waves — two cleanest high-volume sites, then three on parameterized maps, then the hostile legacy ERP by snapshot-diffing at six weeks against a three-week average, priced up front. Hypercare: the weekly call left standing — it had become theirs. Lesson: reconcile to the source's math, not its documentation. **Result.** Primary metric: contract-manufacturer sites with planning-grade data — 0 → 6 in three waves and five months, zero auto-loaded revision overlaps. ≈$0.3M/yr modeled — the six sites and zero overlaps are measured; the dollars are my model; the $1.5M belongs to S8.

COMPRESSED STAR

- **S —** A contract manufacturer's planning feed with two silent traps; six sites, six definitions; one hostile legacy ERP (flat files, no change-data-capture); no commercial leverage.
- **T —** Own ingest requirements, the reconciliation gate, the 2/3/1 wave plan, and chair the weekly Flextronics call.
- **A —** Found the traps by sampling finished goods against the CM's own MRP; designed the two mechanisms that made the traps impossible rather than policed (explicit marker; effective-date quarantine, human disposition); reconciliation report as the per-site gate, not a date on a slide; wave 3's six weeks priced to leadership as the plan, not a slip; made the weekly call the CM's own protection against blame for misses its feed didn't cause.
- **R —** Planning-grade sites 0 → 6 in 3 waves / 5 months; zero auto-loaded overlaps ever; phantom double-timing eliminated at source; backbone handed to the 200-lane program (S8).

RESULTS LINE

CM sites with planning-grade data 0 → 6 in three waves / five months (record) · auto-loaded revision overlaps = 0, ever (record) · hardest site 6 weeks vs the 3-week average, priced up front and absorbed in plan (record) · ≈$0.3M/yr (modeled — reconciliation labor, expedites fired at false shortages, overlap rework); the record has no dollar for this story; the $1.5M belongs to S8.

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Planning-grade CM sites	`0 of 6; manual BOM-vs-MRP reconciliation at every site`	record	0 of 6
2	Manual recon burn	`6 h/wk × 6 sites = 36 h/wk`	modeled	36 h/wk of reconciliation
3	What MRP computes (finished-goods grain)	`NetReq(FG, t) = GrossReq − OnHand − SchedReceipts`	the netting identity; walk below at FG grain only (illustrative)	substituted in rows 4–5
4	Clean row	`NetReq = 500 − 150 − 200 = 150`	Gross 500; on-hand 150; receipts 200 (illustrative)	NetReq = 150; need-date = due − LT_FG
5	Distorted row — phantom FG record	`NetReq = 600 − 150 − 200 = 250; false shortage = 250 − 150 = 100`	a phantom/stale finished-goods record inflates gross requirements by 100 (illustrative)	an expedite fires on 100 units never needed
6	How often	`~5/site/mo × 6 sites × 12 mo = 360 false shortages/yr`	modeled count	360/yr
7	Timing corruption	`the same stale record pulls the need-date a lead-time early`	record mechanism	planners chase demand that was never real
8	Revision overlaps	`two revisions both partially true in [d0, d1]; ~24/yr`	auto-loading would rewrite demand for orders released under the prior revision (record rule; count modeled)	~24 history-rewrite risks/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Explicit marker	`distorted row reconciled: NetReq = 500 − 150 − 200 = 150`	the phantom record carries an explicit marker; it no longer inflates gross (record mechanism)	NetReq = 150 — matches the site's own MRP
2	Per-site gate	`count_model vs count_MRP on sampled finished goods must match`	the reconciliation report is the go-live gate, not a date on a slide (record)	a site opens on a matched report
3	Quarantine	`overlaps held for human disposition per order/effectivity`	record rule	auto-loaded overlaps = 0 — ever
4	Hostile flat-file site	`completeness = rows received / expected; a 40%-rows file would read as 60% deletions → blocked`	snapshot-diffing with a completeness flag (record mechanism)	partial files never consumed
5	Wave plan	`W1: 2 × 3 = 6 wk; W2: 3 × 3 = 9 wk; W3: 6 wk; 6 + 9 + 6 = 21 weeks`	hostile site at 6 wks vs the 3-wk average, priced up front (record)	≈5 months for six sites
6	Recon labor after	`6 → 1.5 h/wk per site — disposition queue only`	modeled	−4.5 h/wk per site

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Planning-grade CM sites	0 of 6	6 of 6, in 3 waves ≈ 5 months	record
2	Manual reconciliation	6 h/wk × 6 sites	1.5 h/wk × 6	modeled
3	Expedites on false shortages	~72/yr acted on	~0 — the marker removes the distortion	modeled
4	Auto-loaded revision overlaps	uncontrolled	0 — ever	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Reconciliation labor	`(6 − 1.5) = 4.5 h/wk; × 6 sites = 27 h/wk; × 48 wk = 1,296 ≈ 1,300 h; × $60 = $77,760`	modeled	≈$0.08M/yr
2	False-shortage expedites	`360 × 20% = 72 acted on; 72 × $3,000 = $216,000`	count from Table 1 row 6; premium benchmark $2–5K (modeled)	≈$0.2M/yr
3	Overlap rework avoided	`24 × 16 h = 384 h; 384 × $60 = $23,040`	the quarantine makes the error 0 by construction (modeled)	≈$0.02M/yr
4	Total	`0.08 + 0.2 + 0.02 = $0.3M`	the record has no dollar for S7; the $1.5M belongs to S8	≈$0.3M/yr

TWO PROBES

> **How did you get an external supplier with no obligation to care?**
>
> By making the weekly call theirs: it protected Flextronics from blame for misses their feed didn't cause. Procurement held the commercial relationship; my lever was the evidence in the discovery memo, confirmed by their own engineering.

> **Why three waves instead of one?**
>
> So one surprise could not spend six sites of credibility — the two cleanest high-volume sites first, three on parameterized maps second, the hostile ERP last with its six weeks priced up front. Speed was traded, never quality: a site went live off its matched reconciliation report, not a date.

Canon numbers (basis labels): 0 → 6 sites, 3 waves, 5 months (record) · zero auto-loaded overlaps (record) · 6 weeks vs 3-week average (record) · ≈$0.3M/yr (modeled). Never "six NA sites"; never "500+ product families"; 60 → 95% belongs to S9; the $1.5M / 580K belong to S8; no tolerance value.

BACKUP — ONLY IF ASKED Doc bench — Flexport FFA — the Asia latency investigation: a 21-second P95 traced to routing, decomposed to the layer, fixed in sequence (Dive Deep)

LPs this backup can serve: Dive Deep (primary — decomposition to the layer) · Deliver Results · Learn & Be Curious · Insist on the Highest Standards (a smoke-test protocol per change)

Basis. Doc-derived micro-story from the program record (the FFA Asia latency investigation, post-MVP 2022) — every figure is RECORD-FROM-DOCS: the investigation document, written when the FF App MVP had launched to 11 partners in 10 countries and 48 of 53 import partners (90.5%) had migrated. Program-lead voice: I owned the investigation, the decomposition and the fix sequence; infrastructure teams executed the WAF and CDN changes. Not in the master packet's S-numbering; supplements canon, never overrides it. The same bench appears in Josyula's packet — facts identical.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. Weeks after the FF App MVP launched to 11 partners in 10 countries — with 48 of 53 import partners already migrated — the monthly business review showed Asia page loads at a P95 of 21.19 seconds against 3.3 in the US. **Task.** Find the layer, not the anecdote, and fix it without a re-architecture in the middle of a live partner-facing launch. **Action.** I decomposed the problem before prescribing: static assets ran ~5s P95 in China against ~500ms in the US, and the app's domain had no geo-based DNS or reverse proxy — Asia traffic was riding the public internet across the Pacific; a double CDN sat on the static path. The fix was a sequence, not a project: enable the WAF, remove the double CDN (direct to S3), then a regional CDN proposal with a smoke-test protocol per change, then China tuning, then Asia tuning — each with a date and an owner. **Result.** Root cause isolated to routing — not the app, not the partners; two fixes shipped inside the investigation window and the regional plan scheduled with dates; and a standing rule imported into every later launch: the pilot's geography is a performance requirement, not a rollout detail.

RECORD-FROM-DOCS

COMPRESSED STAR

- **S —** Post-MVP, the MBR surfaced Asia P95 page load at 21.19s vs 3.3s in the US (17-day measurement window); 48 of 53 import partners (90.5%) already migrated — the app had to hold where the partners were.
- **T —** Isolate the layer causing Asia latency and sequence the fix set without destabilizing a live partner-facing launch.
- **A —** Split static from dynamic: static JS ~5s P95 in China vs ~500ms US; found no geo-based DNS / reverse proxy on the domain (public-internet routing) and a double CDN on static assets; sequenced WAF enablement → double-CDN removal (direct to S3) → regional CDN proposal → China tuning → Asia tuning, each gated by a smoke-test protocol with a named date and owner.
- **R —** Root cause isolated to routing; WAF and double-CDN fixes shipped inside the window; CDN / China / Asia work scheduled with owners; the investigation became the template for launch-geography checks.

RESULTS LINE

P95 page load 21.19s Asia vs 3.3s US (record-from-docs) · static ~5s China vs ~500ms US (record-from-docs) · 48 of 53 partners migrated = 90.5% (record-from-docs; canon's "48 of 53 within a month" — consistent) · two fixes shipped, three sequenced with dates (record-from-docs) · no post-fix latency figure exists in the record — never invent one.

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Page-load P95 by region	`Asia = 21.19s; US = 3.3s; Others = 4.45s`	Datadog @view.loading_time, 04/24–05/10 window (record-from-docs)	Asia is the outlier

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	The gap factor	`21.19 ÷ 3.3 = 6.42`	row 1 (record-from-docs)	6.42× slower than the US
3	Traffic behind the number	`6.31 ÷ 2.19 = 2.88`	Asia 6.31k vs US 2.19k page views, same window (record-from-docs)	≈2.9× the US traffic riding the worst latency
4	Static-asset leg	`JS P95 = 5s China vs 500ms US; 5 ÷ 0.5 = 10`	static.flexport.com CDN behavior (record-from-docs)	10× gap on the static leg
5	Dynamic leg	`no geo-DNS / reverse proxy for trade.flexport.com → Asia GraphQL hairpins over the public internet`	infra review (record-from-docs)	routing penalty on every data call
6	Known defect on the static path	`double CDN in China — Cloudflare stacked in front of Tencent`	infra review (record-from-docs)	one avoidable hop

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Decompose before touching	`page load = static resources + GraphQL requests + JS rendering`	the metric's own definition (record-from-docs)	three legs, measured and owned separately
2	Step 1 — security gate first	`enable WAF for trade.flexport.com: China, then global`	incident-207 follow-up; done 2022/06/30 (record-from-docs)	shipped — fix 1 of 2
3	Step 2 — remove the double CDN	`Tencent CDN → direct to the S3 bucket; Cloudflare hop removed for China`	done 2022/06/24 (record-from-docs)	shipped — fix 2 of 2
4	Step 3 — static proposal, CN/Asia	`alternatives written, reviewed, signed off`	ETA 2022/07/15 (record-from-docs)	sequenced with a date
5	Step 4 — China tuning	`e.g. replicate static assets in-region`	ETA 2022/07/30; depends on step 4 (record-from-docs)	sequenced with a date
6	Step 5 — Asia tuning	`the same playbook beyond China`	ETA 2022/08/15 (record-from-docs)	sequenced with a date
7	Held in reserve	`global CDN for trade.flexport.com — built only if the metrics still demand it`	2022 Q3 or later, metrics-gated (record-from-docs)	no speculative spend
8	Verification per change	`2 user types × 2 instances = 4 smoke passes before + 4 after = 8`	login → workview → logout → re-login; internal SSO + partner users; US + China hosts (record-from-docs; the multiplication counts the checklist)	8 workflow passes around every change

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Root cause	one number — "Asia is slow"	static leg (10×) and dynamic routing isolated, each with an owner and a sequence	record-from-docs
2	Fixes	none	2 shipped (WAF; double-CDN removal); 3 sequenced with dates (07/15, 07/30, 08/15)	record-from-docs
3	Post-fix P95	21.19s	no post-fix figure exists in the record — never claimed	record-from-docs

STEP	METRIC	BEFORE	AFTER	BASIS
4	Spend discipline	—	global CDN gated on metrics, not built on speculation	record-from-docs

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Partner base exposed	`48 ÷ 53 = 0.9057 ≈ 90.5% of import partners already migrated when the gap surfaced`	migration tracker (record-from-docs)	90.5% of partners on the affected app
2	Excess wait served per window	`gap vs US = 21.19 − 3.3 = 17.89s; 6,310 × 17.89 = 112,886 partner-seconds; ÷ 3,600 ≈ 31.4 hours`	6.31k Asia page views in the 17-day window (record-from-docs); gap arithmetic illustrative	≈31.4 partner-hours of excess wait every 17 days
3	Dollar value of the wait	`no dollar in the record — not claimed`	record-from-docs	not claimed
4	Post-fix saving	`no post-fix latency figure exists → no savings chain exists`	record-from-docs	never invent one

TWO PROBES

How did you know it was routing and not the application?

By decomposition before remediation: the static/dynamic split put ~5s of the China time on assets the app server never touched, and the same assets served in ~500ms from the US — the code was constant, the path was not. The domain's missing geo-DNS / reverse proxy and the double CDN explained both halves of the gap without a single application change.

Why not just stand up a China region?

A region is a re-architecture with a compliance tail; the evidence said the bottleneck was routing and CDN topology, which fixes in weeks. The sequenced plan — WAF, double-CDN removal, regional CDN, tuning — bought most of the win at a fraction of the risk, and the smoke-test protocol meant each step was reversible.

Canon numbers (basis labels): 21.19s vs 3.3s P95 (record-from-docs) · ~5s vs ~500ms static (record-from-docs) · 48/53 = 90.5% migrated (record-from-docs) · 11 partners / 10 countries MVP (record — matches S12 canon) · fix sequence WAF → double-CDN → regional CDN → China → Asia with smoke tests (record-from-docs). Never: a post-fix latency number (none in the record); never present scheduled items as delivered.

VERIFY BEFORE YOU SAY IT NEW

Each line is a detail that is **not in the record**, or a number resting on an illustrative input; it carries VERIFY-WITH-RAJESH where it appears in a story. Tick it, or leave it out of the room.

EVERYTHING

- ☐ Which client names I am comfortable saying aloud: “Walmart Marketplace” for S19–S21, “Google Fiber” for E7 — otherwise “a marketplace client” and “a broadband build-out client.”
- ☐ That none of S9, S19, S20, S21 or E7 is also planned as a primary for another interviewer — one question, one story across the loop.
- ☐ Every “illustrative” number is said with the word “illustrative” — the example lanes are a secondary source, not the program ledger.

S6

- ☐ How p90 was originally chosen for the alert (the precision/recall table is after-the-fact design arithmetic).
- ☐ How “demonstrably improved” was operationalized for the both-errors rule (the standard-error bar is my arithmetic, not the record).
- ☐ How large the MPN-mapping delay actually was.

S8

- ☐ Which definition of “99% lead-time accuracy” the program used — ratio, coverage, or p99-match — before quoting any tolerance.
- ☐ The thin-lane rule, the lane-cohort counts and the watch-window length (none is in the record).

S8A

- ☐ That the example-lane numbers (n ≈ 400, plan 18, mean 21, p99 34) are the ones I am comfortable quoting as illustrations.
- ☐ What the program actually did on thin lanes, and whether a quarterly cut of the +3-day bias was run.

S8B

- ☐ The 150 / 50 lane split and whether fat-tail lanes were “kept” (master) or “took stock in” (loop packet).
- ☐ The actual stakeholder dynamics behind the one-way-cut argument; whether a demand–supply correlation test was run; that the textbook form was my own starting point.

S8C

- ☐ Watch-window length; which quarter was held out, and whether it was a peak or a quiet quarter.
- ☐ Whether a control (deferred lanes) or a rolling-origin back-test existed; any pressure to release early.

S9

- ☐ The grain behind completeness 60% → 95% (event or shipment); the gate test sizes and whether the bar was an observed rate or a bound.
- ☐ Whether any control comparison was run for the −10% freight, and the base period behind “≈$2M in six months.” Cohort sizes are not in the record.

S19 / S20

- ☐ Any failure or live result beyond the stated drills — both are design engagements; nothing in production moved.
- ☐ The exact EWMA / CUSUM parameters of the Andon.

S21

- ☐ The order of discovery (funnel first, interviews second) and anything claimed beyond contract design plus wave-one mechanics.

E7

- ☐ Client naming; every CV, correlation and noise input is illustrative; the design was not run — say “I designed”, not “we ran”.

> **If you cannot tick it:** say “illustrative inputs, real outcomes”, give the record (200 lanes, 99%, 580K units / ~$1.5M, zero stockouts) and how you would firm the softer input. Never invent a window, a cohort count or a dollar.

NUMBERS I MUST NOT GET WRONG NEW

The canon numbers this packet uses, each with its basis word. Say the basis in the same breath as the number.

NUMBER	WHAT IT IS	BASIS	NEVER / CAREFUL
580K units / ~$1.5M	S8 — released with zero stockouts in the watch window; ≈$2.59/unit	record, one-time; $2.59 derived	never merged with the $2M into a range; never per year
≈$0.6M / yr	S8 — carry 0.3 at 20% + expedites 0.3; range $0.3–1.4M	modeled	not a result
200 lanes · 99% · p99	S8 — promise at p99; 99% lead-time accuracy gate	record	quote a tolerance only after naming the definition
70% → 85% · $2M	S6 — class-A forecast accuracy; safety stock released under the two-signal rule	record, one-time	never add to S8; never “I owned the forecast”
≈$0.8M / yr	S6 — carry 0.2 · expedites 0.3 · planner capacity 0.3; range $0.5–1.2M	modeled	planner capacity is time, not cash
3 months · 8× · −50%	S6 — shortage alert live; adoption; planner effort	record	never paste “3 months” onto the six-site ingest
1,000 → 597 = 80%	S9 — asked vs built first; ~400 lanes (~20%) deferred; ~200 next wave	record	not “1,000 lanes live”
60% → 95% · ≥95% gate	S9 — completeness; valid-ASN gate held twice	record	grain of the 95% VERIFY-WITH-RAJESH
−10% ≈ $2M in six months	S9 — freight spend	record, not annualized	no control group for causal claims VERIFY-WITH-RAJESH
1.10 d → b = k·σ, k = 1.645	S19 — flat buffer vs governed dial; fast-promise 7.6% → 39.6% at a 95% floor	engagement model	the 97.8% variant is never claimed
+$255K/wk ≈ $13.3M/yr	S19 — modeled value of the dial	modeled	never record; $12 breach vs $6.50 margin is the sensitivity
41.5% = 43.6% × 95.2%	S20 — Promise Yield on the docs' sample; target 50%	illustrative sample / design target	nothing in production moved
~30 d → < 24 h · < 5%	S20 — Andon detection-to-action; false-trip design bar	design	no live Andon result
~38–42% · 85% · ≈$3.7M/yr	S21 — stalled enrollment; handling adherence; per +10 pts of enrolled GMV	record / modeled	claim = contract + wave one only
CR 0.99 · z 2.33 · ±20%	E7 — newsvendor first buy and the two-week trigger; $216K/yr per market + $100K stranding	illustrative; design, not run	client name only if confirmed VERIFY-WITH-RAJESH
Six sites	every program	record	say it with no regional qualifier

Basis vocabulary. *record* — measured and signed in the program · *user anchor* — a number the candidate stated · *modeled* — my arithmetic on labeled inputs · *illustrative* — example inputs chosen to show the math · *design* — specified, not run · *engagement model* — a design-engagement calculation, not a production result.